Application Note

Basic design guide

MMSP™2

Application Note

MDMP2520FAN200503
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MP2520F
Basic design guide for MP2520F Application Processor

MAGIC EYES

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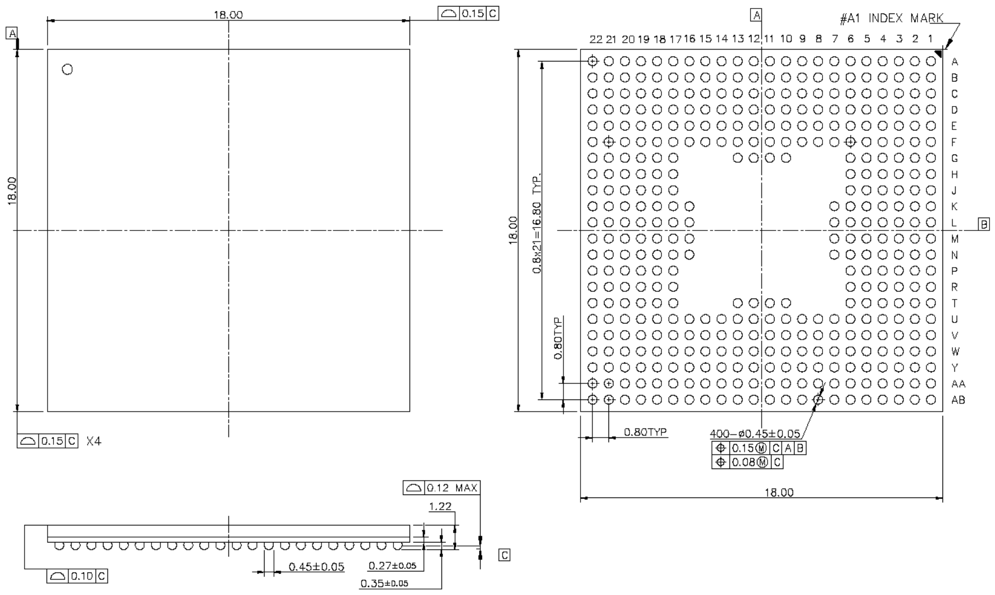
CONTENTS

1. IO Pin descriptions

1.1. Mechanical Dimensions



<Figure 1-1. MP2520F Package>

ITEM	DETAILS
UNIT	mm
Package Size	18.0TYP X 18.0TYP
Tolerance	±0.10
Ball Count	400 Balls
Ball Pitch	(x, y)=(0.8TYP, 0.8TYP)
Package Height	1.22TYP

<Table 1-1. Package Information>

1.2. MP2520F Signals Reference

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	
A	VDDA DC	VDD OP	DP1	GPIO O[4]	GPIO O[5]	GPIO D[5]	GPIO D[7]	nRESET	nPORSEL	XA[1]	XA[4]	XA[9]	XD[31]	XD[29]	XD[27]	XD[25]	XD[23]	XD[21]	XD[19]	XD[17]	GND	GND	A
B	VDDAPLL	ADCIN1	ADCIN3	ADCIN4	DP0	GPIO D[1]	GPIO D[6]	GPIO D[9]	GPIO D[10]	SDCLK	XA[0]	XA[5]	XD[30]	XD[28]	XD[26]	XD[24]	XD[22]	XD[20]	XD[18]	XD[16]	XD[15]	XD[14]	B
C	XTIRTC	XTORTC	VSSADC	ADCIN0	ADCIN2	DN0	DP2	VDDALIV E2	VDDOP	GPIO D[11]	nBATTF	XA[2]	XA[6]	XA[8]	XBA[1]	VDDOPA LIVE	GND	GND	GND	GND	XD[13]	XD[12]	C
D	XTI	XTO	VDDUPLL	VSSAPLL	ADCIN5	DN1	VDDOP	GPIO D[2]	VDDI	GPIO D[13]	EXTIN0	XA[3]	XA[7]	XA[10]	XBA[0]	VDDOPA LIVE	GND	GND	GND	GND	XD[11]	XD[10]	D
E	GPIO A[9]	GPIO A[14]	VDDFPLL	VSSUPLL	GND	VREF	DN2	VDDI	GPIO D[3]	GPIO D[8]	VDDI	EXTIN1	VDDALIV E1	XA[11]	XA[12]	GND	GND	GND	GND	VDDOPA LIVE	XD[9]	XD[8]	E
F	GPIO A[7]	GPIO A[11]	VDDRTC	VSSRTC	VSSFPLL	GND	GND	GPIO D[0]	GPIO D[4]	GPIO H[0]	VDDI	nPWRRG LTON	VDDOPA LIVE	GND	GND	GND	GND	GND	GND	VDDOPA LIVE	XD[7]	XD[6]	F
G	GPIO A[3]	GPIO A[8]	GPIO A[12]	GPIO A[13]	VDDOP	GPIO A[15]	NC	NC	NC	GND	GPIO D[12]	TEST_EN	VDDOPA LIVE	NC	NC	NC	GND	GND	SCKEX	VDDALIV E1	XD[5]	XD[4]	G
H	GPIO A[0]	GPIO A[4]	GPIO A[6]	VDDI	GPIO A[10]	GND	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	GND	nSCSX1	nSCSX0	VDDOPA LIVE	XD[3]	XD[2]	H
J	GPIO B[14]	GPIO B[15]	VDDOP	GPIO A[2]	GPIO A[5]	GND	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	GND	nSWEX	nSRASX	nSCASX	XD[1]	XD[0]	J
K	GPIO B[8]	GPIO B[11]	GPIO B[12]	GPIO B[13]	GPIO B[10]	GPIO A[1]	GND	NC	NC	NC	NC	NC	NC	NC	NC	GND	GND	SDQMX0	VDDI	SDQMX3	SDQMX1	SDQMX2	K
L	GPIO B[7]	GPIO B[4]	GPIO B[6]	GPIO B[5]	GPIO B[9]	VDDI	GND	NC	NC	NC	NC	NC	NC	NC	NC	GND	GND	GPIO K[5]	GPIO K[1]	GPIO K[2]	GPIO K[3]	GPIO K[0]	L
M	GPIO B[2]	GPIO B[0]	GPIO B[1]	VDDALIV E2	VDDOP	GPIO B[3]	GND	NC	NC	NC	NC	NC	NC	NC	NC	GND	GPIO J[0]	GPIO J[4]	GPIO J[1]	VDDOP	GPIO K[7]	GPIO K[4]	M
N	GPIO L[14]	GPIO L[13]	GPIO L[10]	GPIO L[11]	GPIO L[12]	GPIO L[7]	GND	NC	NC	NC	NC	NC	NC	NC	NC	GND	GPIO J[10]	GPIO J[8]	GPIO J[5]	VDDI	GPIO J[2]	GPIO K[6]	N
P	GPIO L[9]	GPIO L[8]	GPIO L[5]	VDDI	GPIO L[3]	GND	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	VDDI	GPIO J[12]	GPIO J[13]	GPIO J[14]	GPIO J[3]	VDDALIE2	P
R	GPIO L[6]	GPIO L[4]	GPIO L[0]	GPIO L[1]	GPIO M[4]	VDDA920	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	GND	GPIO I[4]	GPIO I[5]	GPIO I[2]	GPIO J[9]	GPIO J[6]	R
T	GPIO L[2]	GPIO M[8]	GPIO M[1	GPIO M[2]	VDDA920	GPIO N[6]	NC	NC	NC	GND	GND	GND	GND	NC	NC	NC	GND	VDDA940	GPIO I[7]	GPIO I[6]	GPIO J[15]	GPIO J[7]	T
U	GPIO M[7]	GPIO M[5]	GPIO N[6]	GPIO N[4]	GPIO N[5]	GND	GND	GND	VDDALIV E2	GND	GND	GPIO E[8]	GPIO E[0]	GND	GND	GND	GND	VDDOP	GPIO I[13]	GPIO I[12]	GPIO I[0]	GPIO J[11]	U
V	GPIO M[3]	GPIO M[0]	GPIO N[3]	nGRESET O	GND	GND	GPIO H[3]	SDA	GPIO C[14]	GPIO C[10]	GPIO C[2]	GPIO E[12]	GPIO E[4]	GPIO F[7]	VDDA940	VDDI	GPIO G[11]	GND	GPIO G[2]	GPIO I[15]	GPIO I[3]	VDDOP	V
W	GPIO N[7]	GPIO N[2]	GPIO N[0]	GND	GND	GND	GPIO H[5]	VDDA920	VDDA920	GPIO C[5]	VDDI	GPIO E[13]	VDDI	GPIO E[1]	GPIO F[5]	GPIO G[14]	GPIO G[10]	GND	GPIO G[5]	GPIO I[14]	GPIO I[8]	GPIO I[1]	W
Y	GPIO N[1]	OM0	TDI	TCK	TMS	GPIO O[2]	GPIO H[4]	SCL	GPIO C[13]	VDDOP	GPIO E[15]	VDDOP	GPIO E[6]	VDDALIV E2	VDDOP	GPIO G[15]	GPIO G[9]	GPIO G[8]	GPIO G[4]	GPIO G[3]	GPIO I[11]	GPIO I[9]	Y
AA	TDO	VDDOP	TRST	GPIO O[0]	GPIO H[2]	EXTCLKI N	GPIO C[12]	GPIO C[9]	GPIO C[6]	GPIO C[4]	GPIO C[0]	GPIO E[10]	GPIO E[7]	GPIO E[3]	GPIO F[9]	GPIO F[4]	GPIO F[1]	GPIO F[0]	GPIO G[7]	GPIO G[6]	GPIO G[1]	GPIO I[10]	AA
AB	GPIO O[3]	GPIO O[1]	GPIO H[6]	GPIO H[1]	GPIO C[15]	GPIO C[11]	GPIO C[8]	GPIO C[7]	GPIO C[3]	GPIO C[1]	GPIO E[14]	GPIO E[11]	GPIO E[9]	GPIO E[5]	GPIO E[2]	GPIO F[8]	GPIO F[6]	GPIO F[3]	GPIO F[2]	GPIO G[13]	GPIO G[12]	GPIO G[0]	AB
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	

<Figure 1-2. MP2520F Package FBGA Ball Map>

Signal Name	Type	GPIO	Descriptions
Memory Controller Signals			
SDCLK_X	O	-	SDRAM clock
XA[14:0]	O	-	SDRAM address bus. This bus signals the address requested for memory accesses.
XD[15:0]	B	-	SDRAM data bus. XD[15:0] is used for 16-bit data mode.
XD[31:16]	B	-	SDRAM data bus. These data bits are used for 32-bit memory.
SCKEx	O	-	SDRAM clock enable.
nSCSx0	O	-	SDRAM CS for bank 0. This signal should be connected to the chip select (CS) pins for SDRAM.
nSCSx1	O	-	SDRAM CS for bank 1. This signal should be connected to the chip select (CS) pins for SDRAM.
NSRASx	O	-	SDRAM RAS. This signal should be connected to the row address strobe (RAS) pins for all banks of SDRAM.
NSCASx	O	-	SDRAM CAS. This signal should be connected to the column address strobe (CAS) pins for all banks of SDRAM.
NSWEx	O	-	SDRAM write enable. This signal should be connected to the write enables of SDRAM.
SDQMx[3:0]	O	-	SDRAM DQM for data bytes 0 through 3. These signals should be connected to the data output mask enables (DQM) for SDRAM
Display Controller Pins			
VD[23:8]	O	GPIOA[15:0]	Video output data (RGB, Multiplexed-RGB, YCbCr)
VD[7:0]	O	GPIOB[15:8]	Video output data (RGB, Multiplexed-RGB, YCbCr)
CLKH	O	GPIOB[7]	Pixel Clock Output
DE	O	GPIOB[6]	Data Enable
HSYNC	O	GPIOB[5]	Horizontal Sync
VSYNC	O	GPIOB[4]	Vertical Sync
POL	O	GPIOB[3]	Data reverse signal of source driver
CLKV	O	GPIOB[2]	Vertical Clock output
PS	O	GPIOB[1]	Power Save
FG	O	GPIOB[0]	Clock signal of gate driver
XDOFF	O	GPIOH[6]	Gate-OFF Control signal for gate driver
VCLKIN	I	GPIOH[5]	Video Clock Input
STVD	B	GPIOK[7]	Shift clock output for Gate Driver IC

STHR	B	GPIOK[3]	Start Pulse of source Driver IC
Image Signal Processor Pins			
ID[15:0]	I	GPIOC[15:0]	Image Capture Data Input
ICLKIN	I	GPIOH[4]	Image Clock In
IHSYNC	B	GPIOH[3]	Image HSYNC
IVSYNC	B	GPIOH[2]	Image VSYNC
ICLKOUT	O	GPIOH[1]	Image Clock Out
UART Pins			
TX0	O	GPIOD[0]	UART0 Transmit Pin
RX0	I	GPIOD[1]	UART0 Receive Pin
nRIO	I	GPIOD[2]	UART1, UART2 Ring Indicator Pin
nDCD	I	GPIOD[3]	UART1, UART2 Data-Carrier-Detect Pin
nDSR	I	GPIOD[4]	UART1, UART2 Data-Set-Ready Pin
nDTR	O	GPIOD[5]	UART1, UART2 Data-Terminal-Ready Pin
nCTS	I	GPIOD[6]	UART1, UART2 Clear-To-Send Pin
nRTS	O	GPIOD[7]	UART1, UART2 Ready-To-Send Pin
TX1	O	GPIOD[8]	UART1 Transmit Pin
RX1	I	GPIOD[9]	UART1 Receive Pin
RX2	I	GPIOD[10]	UART2 Receive Pin
TX2	O	GPIOD[11]	UART2 Transmit Pin
RX3	I	GPIOD[12]	UART3 Receive Pin
TX3	O	GPIOD[13]	UART3 Transmit Pin
SIR **Pins**			
SIR **RX**	I	GPIOD[10]	SIR Receive Pin (SIR Mode)
SIR **TX**	O	GPIOD[11]	SIR Transmit Pin (SIR Mode)
PPM Pin			
PPM	I	GPIOL[13]	PPM Interface Pin

Non DRAM Controller Pins			
CD[15:0]	B	GPIOE[15:0]	Non DRAM Data Bus
CA[25:23](OM[3:1])	O	GPIOF[9:7]	Non DRAM Address Bus
CA[22:16]	O	GPIOF[6:0]	Non DRAM Address Bus
CA[15:1]	O	GPIOG[15:1]	Non DRAM Address Bus
CA[0](DQMz[0],BTENB0)	O	GPIOG[0]	Non DRAM Address Bus, Negative byte enable. This signal should be connected to the low byte enable for 16bit SRAM.
nPWAIT	I	GPIOI[13]	Wait Control for PCMCIA /CF/IDE/Static Memory. This signal is an input and is driven low by the PCMCIA/CF/IDE/ Static Memory to extend the length of the transfers to/from applications processor.
nPCS[1]	O	GPIOI[7]	PCMCIA/CF Chip Select, This is used to select PCMCIA card, and enable the low byte lane.
nPCS[0]	O	GPIOI[6]	PCMCIA/CF Chip Select. This is used to select PCMCIA card, and enable the High byte lane.
POE	O	GPIOJ[12]	PCMCIA/CF/SM Output Enable. This signal is an output and performs reads from memory and attribute space.
nPWE	O	GPIOJ[11]	PCMCIA/CF/SM Write Enable. This signal is an output and performs writes to memory and attribute space.
nPIOR	O	GPIOJ[10]	PCMCIA/CF/IDE I/O Read Enable. This signal is an output and performs read transactions from the PCMCIA I/O space.
nPIOW	O	GPIOJ[9]	PCMCIA/CF/IDE I/O Write Enable. This signal is an output and performs write transactions to the PCMCIA I/O space.
PSKSEL	O	GPIOJ[8]	PCMCIA/CF Socket Select. This signal is an output and is used by external steering logic to route control, address and data signals to one of the two PCMCIA sockets.
nPREG	O	GPIOJ[7]	PCMCIA/CF Register Select. This signal is an output and indicates that, on a memory transaction, the target address is attribute space. This signal has the same timing as address.
nPIOIS16	I	GPIOJ[6]	PCMCIA/CF/IDE Select 16bit Access. This signal is an input and is an acknowledgement from the PCMCIA card that the current address is a valid 16 bit wide I/O address.
RDnWR	O	GPIOJ[5]	Buffer Direction Control. Read/Write for static interface. Intended for use as a steering signal for buffering logic.
nBUFOE	O	GPIOJ[4]	Buffer Output Enable Intended for use as a buffer direction control for data bus buffering logic.
nBUFENB	O	GPIOJ[3]	Buffer Enable. Intended for use as a buffer enable for data bus buffering logic.
Static Memory Pins			
DQMz[1] (BTENB1)	O	GPIOI[12]	Negative Byte Enable, This signal should be connected to the high byte enable for 16bit SRAM.
nSCS[3]	O	GPIOI[11]	Static Memory Chip Select. This signal is chip selects to static memory devices such as ROM and Flash
nSCS[2]	O	GPIOI[10]	Static Memory Chip Select. This signal is chip selects to static memory devices such as ROM and Flash
nSCS[1]	O	GPIOI[9]	Static Memory Chip Select. This signal is chip selects to static memory devices such as ROM and Flash

nSCS[0]	O	GPIOI[8]	Static Memory Chip Select. This is the chip select for the boot memory.
IDE Pins			
nICS[1]	O	GPIOI[5]	IDE Chip Select. This is used to select PCMCIA card, and enable the High byte lane.
nICS[0]	O	GPIOI[4]	IDE Chip Select. This is used to select PCMCIA card, and enable the low byte lane.
Nand Flash Pins			
nNFCE[3]	O	GPIOI[3]	NAND Chip Select. This is chip selects to NAND Flash memory.
nNFCE[2]	O	GPIOI[2]	NAND Chip Select This is chip selects to NAND Flash memory.
nNFCE[1]	O	GPIOI[1]	NAND Chip Select This is chip selects to NAND Flash memory.
nNFCE[0]	O	GPIOI[0]	NAND Chip Select This is chip selects to NAND Flash memory for NAND boot memory.
nPIOW/ALE	O	GPIOJ[9]	NAND ALE. This is address latch enable to NAND Flash memory
nPIOR/CLE	O	GPIOJ[10]	NAND CLE. This is command latch enable to NAND Flash memory
RnB	I	GPIOJ[2]	NAND Ready & Busy. This is Ready/Busy signal of NAND Flash memory.
nNFOE	O	GPIOJ[1]	NAND Output Enable
nNFWE	O	GPIOJ[0]	NAND Write Enable
External DMA Pins			
DREQ[3:0]	I	GPIOK[7:4]	DMA Request. This should be used as IDE DMA Request and Flyby DMA request.
DVAL[3:0]	O	GPIOK[3:0]	DMA Validate. This should be used as Flyby DMA data valid strobes and IDE DMA Acknowledge signal.
PWM Pins			
PWMOUT[3:0]	O	GPIOL[14:11]	Pulse Width Modulation channel
AC97 **Controller Pins**			
ABITCLK	I	GPIOL[10]	AC97 Audio Port bit clock
ADATAIN	I	GPIOL[9]	AC97 Audio Port data in
ASYNC	O	GPIOL[8]	AC97 Audio Port sync signal
nARST	O	GPIOL[7]	AC97 Audio Port reset signal
ADATOUT	O	GPIOL[6]	AC97 Audio Port data out
I^2S Controller Pins			
I^2S CLK	O	GPIOL[10]	I^2S Clock(QCLK_ I^2S)

I^2S DATAIN	I	GPIOL[9]	I^2S SDATA IN
I^2S SYNC	O	GPIOL[8]	I^2S SYNC
I^2S ACLK	O	GPIOL[7]	I^2S System clock(ACLK_ I^2S)
I^2S DATAOUT	O	GPIOL[6]	I^2S DATA OUT
MMC/SD Controller Pins			
SDICLK	O	GPIOL[5]	MMC/SD Clock
SDICMD	B	GPIOL[4]	MMC/SD Command
SDIDAT[3:0]	B	GPIOL[3:0]	MMC/SD Data
656 Type Video Pins			
VD656[7:0]	B	GPIOM[8:1]	656 Type Video Data
VDCLKIN	B	GPIOM[0]	656 Type Video Clock
Transport Stream I/F Pins			
TSISYNC	I	GPIOM[8]	MPEG I/F Sync
TSIDP	I	GPIOM[7]	MPEG I/F Data Valid
TSIERR	I	GPIOM[6]	MPEG I/F Error
TSIDI0	I	GPIOM[5]	MPEG I/F Data bus (Serial Mode)
TSIDI[7:0]	I	GPIOM[7:0]	MPEG I/F Data bus (Parallel Mode)
TSICLK	I	GPIOM[4]	MPEG I/F Clock
Memory Stick Controller Pins			
MSBS	O	GPIOL[3]	Memory Stick BS
MSDAT	B	GPIOL[2]	Memory Stick SDAT
MSINS	I	GPION[2]	Memory Stick INS
MSCLK	O	GPION[1]	Memory Stick Clock
CDROM Controller Pins			
BCLK	I	GPIOM[7]	CD I/F BCLK
LRCK	I	GPIOM[8]	CD I/F LRCK
SDAT	I	GPIOM[6]	CD I/F SDAT

SPDIF In/Out Pins			
SPDIFIN	I	GPIOO[5]	SPDIF In
SPDIFOUT	O	GPIOO[4]	SPDIF Out
SSP Pins			
SSPCLK	O	GPIOO[3]	Synchronous Serial Port Clock
SSPFRM	O	GPIOO[2]	Synchronous Serial Port Frame Signal
SSPTXD	O	GPIOO[1]	Synchronous Serial Port Data
SSPRXD	I	GPIOL[14]	Synchronous Serial Transmit Data
SSPEXTCLKIN	I	GPIOD[12]	Synchronous Serial Port external clock input
One Wire Master Pin			
1WIRE	O	GPIOO[0]	1-wire Master Output
Miscellaneous Pins			
EXTCLKO	O	GPIOH[0]	External Clock Out
OM[3:0]	I	GPIOF[9:7], OM0	Test Mode (External Clock selection)
TEST_EN	I	-	Test Mode Enable
nPORSEL	I	-	POWER On Reset Selection. This signal determines whether the internal POR circuit or nRESET isto be used. If nPORSEL is 0, the internal POR circuit would be used.
nRESET	I	-	Global Reset In. Active low input. nRESET is a level-sensitive input which is used to start the processor from a known address. A LOW level will cause the current instruction to terminate abnormally, and all on-chip state to be reset. When nRESET is driven HIGH, the processor will re-start from address 0.
nGRESETOut	O	-	Global Reset Out
nPWRRGLTOn	O	-	Active Low Power Control stop : 1 run : 0
EXTCLK	I	-	External Clock In
EXTIN0	I	-	Wake-up Control input #0
EXTIN1	I	-	Wake-up Control Input #1
nBATTF	I	-	Battery Fault. Active low input. Signals MP2520F that the main power source is failing (battery is low or is removed from the system.)
Crystal Pins			
XTI	I	-	7.3728Mhz Crystal In

XTO	O	-	7.3728Mhz Crystal Out
XTIRTC	I	-	32.768Khz RTC Crystal input
XTORTC	O	-	32.768Khz RTC Crystal output
JTAG Pins			
TCK	I	-	Clock input for the JTAG Logic
nTRST	I	-	Reset input for the JTAG Logic
TDO	O	-	Serial output for test instructions and data for JTAG
TMS	I	-	TMS controls the sequence of the TAP controller's state
TDI	I	-	Serial Data input for JTAG
Dedicated GPIO			
GPIOI[15]	B		GPIO : Dedicated GPIO(CBBUSREQ) ALT2 : 656IN[7]
GPIOI[14]	B		GPIO : Dedicated GPIO(CBBUSGNT) ALT2 : 656IN[6]
GPIOJ[15]	B		GPIO : Dedicated GPIO(CBREQ) ALT2 : 656IN[2]
GPIOJ[14]	B		GPIO : Dedicated GPIO(CBACK) ALT2 : 656IN[1]
GPIOJ[13]	B		GPIO : Dedicated GPIO(CBRDY) ALT2 : 656IN[0]
Power and Ground Pins			
VDDA920	SUP		ARM920T power supply
VDDA940	SUP		ARM940T power supply
VDDADC	SUP		ADC power supply
VDDALIVE1	SUP		Alive1 block power supply
VDDALIVE2	SUP		Alive2 block power supply
VDDAPLL	SUP		APLL power supply
VDDFPLL	SUP		FPLL power supply
VDDUPLL	SUP		UPLL power supply
VDDI	SUP		Internal logic power supply
VDDOP	SUP		I/O power supply
VDDOPALIVE	SUP		SDRAM I/O power supply

VDDRTC	SUP	RTC power supply
VSSADC	SUP	ADC ground
VSSAPLL	SUP	APLL ground
VSSFPLL	SUP	FPLL ground
VSSUPLL	SUP	UPLL ground
VSSRTC	SUP	RTC ground

<Table 1-2. Signal Description of MP2520F>

1.3. MP2520F I/O Pin Listing

Ball#	Name	Type	Func. After Reset	Primary Function	Alternate Function 1	Alternate Function 2
A1	**VDDADC**	SUP				
A2	**VDDOP**	SUP				
A3	**DP1**	IAOA	DP1	DP1	-	-
A4	**GPIOO[4]/SPDIFOUT**	ISOSZP	GPIOO[4]	GPIOO[4]	SPDIFOUT	-
A5	**GPIOO[5]/SPDIFIN**	ISOSZP	GPIOO[5]	GPIOO[5]	SPDIFIN	
A6	**GPIOD[5]/nDTR**	ISOSZP	GPIOD[5]	GPIOD[5]	nDTR	-
A7	**GPIOD[7]/nRTS**	ISOSZP	GPIOD[7]	GPIOD[7]	nRTS	-
A8	**nRESET**	IS	nRESET	nRESET	-	-
A9	**nPORSEL**	IS	nPORSEL	nPORSEL	-	-
A10	**XA[1]**	OS	XA[1]	XA[1]	-	-
A11	**XA[4]**	OS	XA[4]	XA[4]	-	-
A12	**XA[9]**	OS	XA[9]	XA[9]	-	-
A13	**XD[31]**	ISOS	XD[31]	XD[31]	-	-
A14	**XD[29]**	ISOS	XD[29]	XD[29]	-	-
A15	**XD[27]**	ISOS	XD[27]	XD[27]	-	-
A16	**XD[25]**	ISOS	XD[25]	XD[25]	-	-
A17	**XD[23]**	ISOS	XD[23]	XD[23]	-	-

A18	**XD[21]**	ISOS	XD[21]	XD[21]	-	-
A19	**XD[19]**	ISOS	XD[19]	XD[19]	-	-
A20	**XD[17]**	ISOS	XD[17]	XD[17]	-	-
A21	**GND**	SUP				
A22	**GND**	SUP				
B1	**VDDAPLL**	SUP				
B2	**ADCIN1**	IA	ADCIN1	ADCIN1	-	-
B3	**ADCIN3**	IA	ADCIN3	ADCIN3	-	-
B4	**ADCIN4**	IA	ADCIN4	ADCIN4	-	-
B5	**DP0**	IAOA	DP0	DP0	-	-
B6	**GPIOD[1]/RX0**	ISOSZP	GPIOD[1]	GPIOD[1]	RX0	-
B7	**GPIOD[6]/nCTS**	ISOSZP	GPIOD[6]	GPIOD[6]	nCTS	-
B8	**GPIOD[9]/RX1**	ISOSZP	GPIOD[9]	GPIOD[9]	RX1	-
B9	**GPIOD[10]/RX2**	ISOSZP	GPIOD[10]	GPIOD[10]	RX2	-
B10	**SDCLK**	OS	SDCLK	SDCLK	-	-
B11	**XA[0]**	OS	XA[0]	XA[0]	-	-
B12	**XA[5]**	OS	XA[5]	XA[5]	-	-
B13	**XD[30]**	ISOS	XD[30]	XD[30]	-	-
B14	**XD[28]**	ISOS	XD[28]	XD[28]	-	-
B15	**XD[26]**	ISOS	XD[26]	XD[26]	-	-
B16	**XD[24]**	ISOS	XD[24]	XD[24]	-	-
B17	**XD[22]**	ISOS	XD[22]	XD[22]	-	-
B18	**XD[20]**	ISOS	XD[20]	XD[20]	-	-
B19	**XD[18]**	ISOS	XD[18]	XD[18]	-	-
B20	**XD[16]**	ISOS	XD[16]	XD[16]	-	-
B21	**XD[15]**	ISOS	XD[15]	XD[15]	-	-
B22	**XD[14]**	ISOS	XD[14]	XD[14]	-	-
C1	**XTIRTC**	IS	XTIRTC	XTIRTC	-	-
C2	**XTORTC**	OS	XTORTC	XTORTC	-	-

C3	**VSSADC**	SUP				
C4	**ADCIN0**	IA	ADCIN0	ADCIN0	-	-
C5	**ADCIN2**	IA	ADCIN2	ADCIN2	-	-
C6	**DN0**	IAOA	DN0	DN0	-	-
C7	**DP2**	IAOA	DP2	DP2	-	-
C8	**VDDALIVE2**	SUP				
C9	**VDDOP**	SUP				
C10	**GPIOD[11]/TX2**	ISOSZP	GPIOD[11]	GPIOD[11]	TX2	-
C11	**nBATTF**	IS	nBATTF	nBATTF	-	-
C12	**XA[2]**	OS	XA[2]	XA[2]	-	-
C13	**XA[6]**	OS	XA[6]	XA[6]	-	-
C14	**XA[8]**	OS	XA[8]	XA[8]	-	-
C15	**XBA[1]**	OS	XBA[1]	XBA[1]	-	-
C16	**VDDOPALIVE**	SUP				
C17	**GND**	SUP				
C18	**GND**	SUP				
C19	**GND**	SUP				
C20	**GND**	SUP				
C21	**XD[13]**	ISOS	XD[13]	XD[13]	-	-
C22	**XD[12]**	ISOS	XD[12]	XD[12]	-	-
D1	**XTI**	IS	XTI	XTI	-	-
D2	**XTO**	OS	XTO	XTO	-	-
D3	**VDDUPLL**	SUP				
D4	**VSSAPLL**					
D5	**ADCIN5**	IA	ADCIN5	ADCIN5	-	-
D6	**DN1**	IAOA	DN1	DN1	-	-
D7	**VDDOP**	SUP				
D8	**GPIOD[2]/nRIO**	ISOSZP	GPIOD[2]	GPIOD[2]	nRIO	-

D9	**VDDI**	SUP				
D10	**GPIOD[13]/TX3**	ISOSZP	GPIOD[13]	GPIOD[13]	TX3	-
D11	**EXTIN0**	IS	EXTIN0	EXTIN0	-	-
D12	**XA[3]**	OS	XA[3]	XA[3]	-	-
D13	**XA[7]**	OS	XA[7]	XA[7]	-	-
D14	**XA[10]**	OS	XA[10]	XA[10]	-	-
D15	**XBA[0]**	OS	XBA[0]	XBA[0]	-	-
D16	**VDDOPALIVE**	SUP				
D17	**GND**	SUP				
D18	**GND**	SUP				
D19	**GND**	SUP				
D20	**GND**	SUP				
D21	**XD[11]**	ISOS	XD[11]	XD[11]	-	-
D22	**XD[10]**	ISOS	XD[10]	XD[10]	-	-
E1	**GPIOA[9]/VD[17]**	ISOSZP	GPIOA[9]	GPIOA[9]	VD[17]	-
E2	**GPIOA[14]/VD[22]**	ISOSZP	GPIOA[14]	GPIOA[14]	VD[17]	-
E3	**VDDFPLL**	SUP				
E4	**VSSUPLL**	SUP				
E5	**GND**	SUP				
E6	**VREF**	IA	VREF	VREF	-	-
E7	**DN2**	IAOA	DN2	DN2	-	-
E8	**VDDI**	SUP				
E9	**GPIOD[3]/nDCD**	ISOSZP	GPIOD[3]	GPIOD[3]	nDCD	-
E10	**GPIOD[8]/TX1**	ISOSZP	GPIOD[8]	GPIOD[8]	TX1	-
E11	**VDDI**	SUP				
E12	**EXTIN1**	IS	EXTIN1	EXTIN1	-	-
E13	**VDDALIVE1**	SUP				
E14	**XA[11]**	OS	XA[11]	XA[11]	-	-

E15	**XA[12]**	OS	XA[12]	XA[12]	-	-
E16	**GND**	SUP				
E17	**GND**	SUP				
E18	**GND**	SUP				
E19	**GND**	SUP				
E20	**VDDOPALIVE**	SUP				
E21	**XD[9]**	ISOS	XD[9]	XD[9]	-	-
E22	**XD[8]**	ISOS	XD[8]	XD[8]	-	-
F1	**GPIOA[7]/VD[15]**	ISOSZP	GPIOA[7]	GPIOA[7]	VD[15]	-
F2	**GPIOA[11]/VD[19]**	ISOSZP	GPIOA[11]	GPIOA[11]	VD[19]	-
F3	**VDDRTC**	SUP				
F4	**VSSRTC**					
F5	**VSSFPLL**	SUP				
F6	**GND**	SUP				
F7	**GND**	SUP				
F8	**GPIOD[0]/TX0**	ISOSZP	GPIOD[0]	GPIOD[0]	TX0	-
F9	**GPIOD[4]/nDSR**	ISOSZP	GPIOD[4]	GPIOD[4]	nDSR	-
F10	**GPIOH[0]/EXTCLKO**	ISOSZP	GPIOH[0]	GPIOH[0]	EXTCLKO	-
F11	**VDDI**	SUP				
F12	**nPWRRGLTON**	OS	nPWRRGLTOn	nPWRRGLTOn	-	-
F13	**VDDOPALIVE**	SUP				
F14	**GND**	SUP				
F15	**GND**	SUP				
F16	**GND**	SUP				
F17	**GND**	SUP				
F18	**GND**	SUP				
F19	**GND**	SUP				
F20	**VDDOPALIVE**	SUP				

F21	**XD[7]**	ISOS	XD[7]	XD[7]	-	-
F22	**XD[6]**	ISOS	XD[6]	XD[6]	-	-
G1	**GPIOA[3]/VD[11]**	ISOSZP	GPIOA[3]	GPIOA[3]	VD[11]	-
G2	**GPIOA[8]/VD[16]**	ISOSZP	GPIOA[8]	GPIOA[8]	VD[16]	-
G3	**GPIOA[12]/VD[20]**	ISOSZP	GPIOA[12]	GPIOA[12]	VD[20]	-
G4	**GPIOA[13]/VD[21]**	ISOSZP	GPIOA[13]	GPIOA[13]	VD[21]	-
G5	**VDDOP**	SUP				
G6	**GPIOA[15]/VD[23]**	ISOSZP	GPIOA[15]	GPIOA[15]	VD[23]	-
G10	**GND**	SUP				
G11	**GPIOD[12]/RX3/SSPEXTCLKIN**	ISOSZP	GPIOD[12]	GPIOD[12]	RX3	SSPEXTCLKIN
G12	**TEST_EN**	IS	TEST_EN	TEST_EN	-	-
G13	**VDDOPALIVE**	SUP				
G17	**GND**	SUP				
G18	**GND**	SUP				
G19	**SCKEX**	OS	SCKEx	SCKEx	-	-
G20	**VDDALIVE1**	SUP				
G21	**XD[5]**	ISOS	XD[5]	XD[5]	-	-
G22	**XD[4]**	ISOS	XD[4]	XD[4]	-	-
H1	**GPIOA[0]/VD[8]**	ISOSZP	GPIOA[0]	GPIOA[0]	VD[8]	-
H2	**GPIOA[4]/VD[12]**	ISOSZP	GPIOA[4]	GPIOA[4]	VD[12]	-
H3	**GPIOA[6]/VD[14]**	ISOSZP	GPIOA[6]	GPIOA[6]	VD[14]	-
H4	**VDDI**	SUP				
H5	**GPIOA[10]/VD[18]**	ISOSZP	GPIOA[10]	GPIOA[10]	VD[18]	-
H6	**GND**	SUP				
H17	**GND**	SUP				
H18	**nSCSX1**	OS	nSCSX1	nSCSX1	-	-
H19	**nSCSX0**	OS	nSCSX0	nSCSX0	-	-
H20	**VDDOPALIVE**	SUP				

H21	**XD[3]**	ISOS	XD[3]	XD[3]	-	-
H22	**XD[2]**	ISOS	XD[2]	XD[2]	-	-
J1	**GPIOB[14]/VD[6]**	ISOSZP	GPIOB[14]	GPIOB[14]	VD[6]	-
J2	**GPIOB[15]/VD[7]**	ISOSZP	GPIOB[15]	GPIOB[15]	VD[7]	-
J3	**VDDOP**	SUP				
J4	**GPIOA[2]/VD[10]**	ISOSZP	GPIOA[2]	GPIOA[2]	VD[10]	-
J5	**GPIOA[5]/VD[13]**	ISOSZP	GPIOA[5]	GPIOA[5]	VD[13]	-
J6	**GND**	SUP				
J17	**GND**	SUP				
J18	**nSWEX**	OS	nSWEx	nSWEx	-	-
J19	**nSRASX**	OS	nSRASx	nSRASx	-	-
J20	**nSCASX**	OS	nSCASx	nSCASx	-	-
J21	**XD[1]**	ISOS	XD[1]	XD[1]	-	-
J22	**XD[0]**	ISOS	XD[0]	XD[0]	-	-
K1	**GPIOB[8]/VD[0]**	ISOSZP	GPIOB[8]	GPIOB[8]	VD[0]	-
K2	**GPIOB[11]/VD[3]**	ISOSZP	GPIOB[11]	GPIOB[11]	VD[3]	-
K3	**GPIOB[12]/VD[4]**	ISOSZP	GPIOB[12]	GPIOB[12]	VD[4]	-
K4	**GPIOB[13]/VD[5]**	ISOSZP	GPIOB[13]	GPIOB[13]	VD[5]	-
K5	**GPIOB[10]/VD[2]**	ISOSZP	GPIOB[10]	GPIOB[10]	VD[2]	-
K6	**GPIOA[1]/VD[9]**	ISOSZP	GPIOA[1]	GPIOA[1]	VD[9]	-
K7	**GND**	SUP				
K16	**GND**	SUP				
K17	**GND**	SUP				
K18	**SDQMX0**	OS	SDQMx0	SDQMx0	-	-
K19	**VDDI**	SUP				
K20	**SDQMX3**	OS	SDQMx3	SDQMx3	-	-
K21	**SDQMX1**	OS	SDQMx1	SDQMx1	-	-
K22	**SDQMX2**	OS	SDQMx2	SDQMx2	-	-

L1	**GPIOB[7]/CLKH**	ISOSZP	GPIOB[7]	GPIOB[7]	CLKH	-
L2	**GPIOB[4]/VSYNC**	ISOSZP	GPIOB[4]	GPIOB[4]	VSYNC	-
L3	**GPIOB[6]/DE**	ISOSZP	GPIOB[6]	GPIOB[6]	DE	-
L4	**GPIOB[5]/HSYNC**	ISOSZP	GPIOB[5]	GPIOB[5]	HSYNC	-
L5	**GPIOB[9]/VD[1]**	ISOSZP	GPIOB[9]	GPIOB[9]	VD[1]	-
L6	**VDDI**	SUP				
L7	**GND**	SUP				
L16	**GND**	SUP				
L17	**GND**	SUP				
L18	**GPIOK[5]/DREQ1**	ISOSZP	GPIOK[5]	GPIOK[5]	DREQ1	-
L19	**GPIOK[1]/DVAL1**	ISOSZP	GPIOK[1]	GPIOK[1]	DVAL1	-
L20	**GPIOK[2]/DVAL2/V656CLK**	ISOSZP	GPIOK[2]	GPIOK[2]	DVAL2	V656CLK
L21	**GPIOK[3]/ DVAL3/STHR**	ISOSZP	GPIOK[3]	GPIOK[3]	DVAL3	STHR
L22	**GPIOK[0]/ DVAL0**	ISOSZP	GPIOK[0]	GPIOK[0]	DVAL0	-
M1	**GPIOB[2]/CLKV**	ISOSZP	GPIOB[2]	GPIOB[2]	CLKV	-
M2	**GPIOB[0]/FG**	ISOSZP	GPIOB[0]	GPIOB[0]	FG	-
M3	**GPIOB[1]/PS**	ISOSZP	GPIOB[1]	GPIOB[1]	PS	-
M4	**VDDALIVE2**	SUP				
M5	**VDDOP**	SUP				
M6	**GPIOB[3]/POL**	ISOSZP	GPIOB[3]	GPIOB[3]	POL	-
M7	**GND**	SUP				
M16	**GND**	SUP				
M17	**GPIOJ[0]/nNFWE**	ISOSZP	GPIOJ[0]	GPIOJ[0]	nNFWE	-
M18	**GPIOJ[4]/nBUFOE**	ISOSZP	GPIOJ[4]	GPIOJ[4]	nBUFOE	-
M19	**GPIOJ[1]/nNFOE**	ISOSZP	GPIOJ[1]	GPIOJ[1]	nNFOE	-
M20	**VDDOP**	SUP				
M21	**GPIOK[7]/DREQ3/STVD**	ISOSZP	GPIOK[7]	GPIOK[7]	DREQ3	STVD
M22	**GPIOK[4]/DREQ0**	ISOSZP	GPIOK[4]	GPIOK[4]	DREQ0	-

N1	**GPIOL[14]/PWMOUT3/SSPRXD**	ISOSZP	GPIOL[14]	GPIOL[14]	PWMOUT3	SSPRXD
N2	**GPIOL[13]/PWMOUT2/PPM**	ISOSZP	GPIOL[13]	GPIOL[13]	PWMOUT2	PPM
N3	**GPIOL[10]/ABITCLK/ I^2S CLK**	ISOSZP	GPIOL[10]	GPIOL[10]	ABITCLK	-
N4	**GPIOL[11]/PWMOUT0**	ISOSZP	GPIOL[11]	GPIOL[11]	PWMOUT0	-
N5	**GPIOL[12]/PWMOUT1**	ISOSZP	GPIOL[12]	GPIOL[12]	PWMOUT1	-
N6	**GPIOL[7]/nARST/ I^2S ACLK**	ISOSZP	GPIOL[7]	GPIOL[7]	nARST	-
N7	**GND**	SUP				
N16	**GND**	SUP				
N17	**GPIOJ[10]/nPIOR**	ISOSZP	GPIOJ[10]	GPIOJ[10]	nPIOR	-
N18	**GPIOJ[8]/PSKSEL**	ISOSZP	GPIOJ[8]	GPIOJ[8]	PSKSEL	-
N19	**GPIOJ[5]/RDNWR**	ISOSZP	GPIOJ[5]	GPIOJ[5]	RDNWR	-
N20	**VDDI**	SU				
N21	**GPIOJ[2]/RNB**	ISOSZP	GPIOJ[2]	GPIOJ[2]	RnB	-
N22	**GPIOK[6]/DREQ2**	ISOSZP	GPIOK[6]	GPIOK[6]	DREQ2	-
P1	**GPIOL[9]/ADATAIN/ I^2S DATAIN**	ISOSZP	GPIOL[9]	GPIOL[9]	ADATAIN	-
P2	**GPIOL[8]/ASYNC/ I^2S SYNC**	ISOSZP	GPIOL[8]	GPIOL[8]	ASYNC	-
P3	**GPIOL[5]/SDICLK**	ISOSZP	GPIOL[5]	GPIOL[5]	SDICLK	-
P4	**VDDI**	SUP				
P5	**GPIOL[3]/SDIDAT[3]/MSBS**	ISOSZP	GPIOL[3]	GPIOL[3]	SDIDAT[3]	MSBS
P6	**GND**	SUP				
P17	**VDDI**	SUP				
P18	**GPIOJ[12]/nPOE**	ISOSZP	GPIOJ[12]	GPIOJ[12]	nPOE	-
P19	**GPIOJ[13](CBRDY)/ /656IN[0]**	ISOSZP	GPIOJ[13]	GPIOJ[13] (CBRDY)		656IN[0]
P20	**GPIOJ[14](CBACK)/ /656IN[1]**	ISOSZP	GPIOJ[14]	GPIOJ[14] (CBACK)		656IN[1]
P21	**GPIOJ[3]/nBUFENB**	ISOSZP	GPIOJ[3]	GPIOJ[3]	nBUFENB	-
P22	**VDDALIE2**	SUP				
R1	**GPIOL[6]/ADATOUT/ I^2S DATAOUT**	ISOSZP	GPIOL[6]	GPIOL[6]	ADATOUT	-
R2	**GPIOL[4]/SDICMD**	ISOSZP	GPIOL[4]	GPIOL[4]	SDICMD	-

R3	**GPIOL[0]/SDIDAT[0]**	ISOSZP	GPIOL[0]	GPIOL[0]	SDIDAT[0]	-
R4	**GPIOL[1]/SDIDAT[1]**	ISOSZP	GPIOL[1]	GPIOL[1]	SDIDAT[1]	-
R5	**GPIOM[4]/VD656[3]/TSICLK**	ISOSZP	GPIOM[4]	GPIOM[4]	VD656[3]	TSICLK
R6	**VDDA920**	SUP				
R17	**GND**	SUP				
R18	**GPIOI[4]/nICS[0]**	ISOSZP	GPIOI[4]	GPIOI[4]	nICS[0]	-
R19	**GPIOI[5]/nICS[1]**	ISOSZP	GPIOI[5]	GPIOI[5]	nICS[1]	-
R20	**GPIOI[2]/nNFCE[2]/656IN[4]**	ISOSZP	GPIOI[2]	GPIOI[2]	nNFCE[2]	656IN[4]
R21	**GPIOJ[9]/nPIOW**	ISOSZP	GPIOJ[9]	GPIOJ[9]	nPIOW	-
R22	**GPIOJ[6]/nPIOIS16**	ISOSZP	GPIOJ[6]	GPIOJ[6]	nPIOIS16	-
T1	**GPIOL[2]/SDIDAT[2]/MSDAT**	ISOSZP	GPIOL[2]	GPIOL[2]	SDIDAT[2]	MSDAT
T2	**GPIOM[8]/VD656[7]/TSISYNC(LRCK)**	ISOSZP	GPIOM[8]	GPIOM[8]	VD656[7]	TSISYNC (LRCK)
T3	**GPIOM[1]/VD656[0]**	ISOSZP	GPIOM[1]	GPIOM[1]	VD656[0]	-
T4	**GPIOM[2]/VD656[1]**	ISOSZP	GPIOM[2]	GPIOM[2]	VD656[1]	-
T5	**VDDA920**	SUP				
T6	**GPIOM[6]/VD656[5]/TSIERR(SDAT)**	ISOSZP	GPIOM[6]	GPIOM[6]	VD656[5]	TSIERR (SDAT)
T10	**GND**	SUP				
T11	**GND**	SUP				
T12	**GND**	SUP				
T13	**GND**	SUP				
T17	**GND**	SUP				
T18	**VDDA940**	SUP				
T19	**GPIOI[7]/nPCS[1]**	ISOSZP	GPIOI[7]	GPIOI[7]	nPCS[1]	-
T20	**GPIOI[6]/nPCS[0]**	ISOSZP	GPIOI[6]	GPIOI[6]	nPCS[0]	-
T21	**GPIOJ[15] (CBREQ)/ /656IN[2]**	ISOSZP	GPIOJ[15]	GPIOJ[15] (CBREQ)		656IN[2]
T22	**GPIOJ[7]/nPREG**	ISOSZP	GPIOJ[7]	GPIOJ[7]	nPREG	-
U1	**GPIOM[7]/VD656[6]/TSIDP (BCLK)**	ISOSZP	GPIOM[7]	GPIOM[7]	VD656[6]	TSIDP (BCLK)
U2	**GPIOM[5]/VD656[4]/TSIDI0**	ISOSZP	GPIOM[5]	GPIOM[5]	VD656[4]	TSIDI0

U3	**GPION[6]/TSIDI[6]/TSIDP**	ISOSZP	GPION[6]	GPION[6]	TSIDI[6]	TSIDP
U4	**GPION[4]/TSIDI[4]/TSIDI0 (BCLK)**	ISOSZP	GPION[4]	GPION[4]	TSIDI[4]	TSIDI0 (BCLK)
U5	**GPION[5]/TSIDI[5]/TSIERR (LRCK)**	ISOSZP	GPION[5]	GPION[5]	TSIDI[5]	TSIERR (LRCK)
U6	**GND**	SUP				
U7	**GND**	SUP				
U8	**GND**	SUP				
U9	**VDDALIVE2**	SUP				
U10	**GND**	SUP				
U11	**GND**	SUP				
U12	**GPIOE[8]/ZD[8]**	ISOSZP	GPIOE[8]	GPIOE[8]	CD[8]	-
U13	**GPIOE[0]/ZD[0]**	ISOSZP	GPIOE[0]	GPIOE[0]	CD[0]	-
U14	**GND**	SUP				
U15	**GND**	SUP				
U16	**GND**	SUP				
U17	**GND**	SUP				
U18	**VDDOP**	SUP				
U19	**GPIOI[13]/nPWAIT**	ISOSZP	GPIOI[13]	GPIOI[13]	nPWAIT	-
U20	**GPIOI[12]/DQMZ1(BTENB1)**	ISOSZP	GPIOI[12]	GPIOI[12]	DQMZ1 (BTENB1)	-
U21	**GPIOI[0]/nNFCE[0]**	ISOSZP	GPIOI[0]	GPIOI[0]	nNFCE[0]	-
U22	**GPIOJ[11]/nPWE**	ISOSZP	GPIOJ[11]	GPIOJ[11]	nPWE	-
V1	**GPIOM[3]/VD656[2]**	ISOSZP	GPIOM[3]	GPIOM[3]	VD656[2]	-
V2	**GPIOM[0]/VDCLKIN**	ISOSZP	GPIOM[0]	GPIOM[0]	VDCLKIN	-
V3	**GPION[3]/TSIDI[3]/TSICLK (SDAT)**	ISOSZP	GPION[3]	GPION[3]	TSIDI[3]	TSICLK (SDAT)
V4	**nGRESETOut**	OS	nGRESETOut	nGRESETOut	-	-
V5	**GND**	SUP				
V6	**GND**	SUP				
V18	**GND**	SUP				

V7	**GPIOH[3]/IHSYNC**	ISOSZP	GPIOH[3]	GPIOH[3]	IHSYNC	-
V8	**SDA**	OD	SDA	SDA	-	-
V9	**GPIOC[14]/ID[14]**	ISOSZP	GPIOC[14]	GPIOC[14]	ID[14]	-
V10	**GPIOC[10]/ID[10]**	ISOSZP	GPIOC[10]	GPIOC[10]	ID[10]	-
V11	**GPIOC[2]/ID[2]**	ISOSZP	GPIOC[2]	GPIOC[2]	ID[2]	-
V12	**GPIOE[12]/ZD[12]**	ISOSZP	GPIOE[12]	GPIOE[12]	CD[12]	-
V13	**GPIOE[4]/ZD[4]**	ISOSZP	GPIOE[4]	GPIOE[4]	CD[4]	-
V14	**GPIOF[7]/ZA[23]/OM[1]**	ISOSZP	GPIOF[7]	GPIOF[7]	CA[23]	OM[1]
V15	**VDDA940**	SUP				
V16	**VDDI**	SUP				
V17	**GPIOG[11]/ZA[11]**	ISOSZP	GPIOG[11]	GPIOG[11]	CA[11]	-
V19	**GPIOG[2]/ZA[2]**	ISOSZP	GPIOG[2]	GPIOG[2]	CA[2]	-
V20	**GPIOI[15] (CBBUSREQ)/ /656IN[7]**	ISOSZP	GPIOI[15]	GPIOI[15] (CBBUSREQ)		656IN[7]
V21	**GPIOI[3]/nNFCE[3]/656IN[5]**	ISOSZP	GPIOI[3]	GPIOI[3]	nNFCE[3]	656IN[5]
V22	**VDDOP**	SUP				
W1	**GPION[7]/TSIDI[7]/TSISYNC**	ISOSZP	GPION[7]	GPION[7]	TSIDI[7]	TSISYNC
W2	**GPION[2]/TSIDI[2]/MSINS**	ISOSZP	GPION[2]	GPION[2]	TSIDI[2]	MSINS
W3	**GPION[0]/TSIDI[0]**	ISOSZP	GPION[0]	GPION[0]	TSIDI[0]	-
W4	**GND**	SUP				
W5	**GND**	SUP				
W6	**GND**	SUP				
W7	**GPIOH[5]/VCLKIN**	ISOSZP	GPIOH[5]	GPIOH[5]	VCLKIN	-
W8	**VDDA920**	SUP				
W9	**VDDA920**	SUP				
W10	**GPIOC[5]/ID[5]**	ISOSZP	GPIOC[5]	GPIOC[5]	ID[5]	-
W11	**VDDI**	SUP				
W12	**GPIOE[13]/ZD[13]**	ISOSZP	GPIOE[13]	GPIOE[13]	CD[13]	-

W13	**VDDI**	SUP				
W14	**GPIOE[1]/ZD[1]**	ISOSZP	GPIOE[1]	GPIOE[1]	CD[1]	-
W15	**GPIOF[5]/ZA[21]**	ISOSZP	GPIOF[5]	GPIOF[5]	CA[21]	-
W16	**GPIOG[14]/ZA[14]**	ISOSZP	GPIOG[14]	GPIOG[14]	CA[14]	-
W17	**GPIOG[10]/ZA[10]**	ISOSZP	GPIOG[10]	GPIOG[10]	CA[10]	-
W18	**GND**	SUP				
W19	**GPIOG[5]/ZA[5]**	ISOSZP	GPIOG[5]	GPIOG[5]	CA[5]	-
W20	**GPIOI[14] (CBBUSGNT)/ /656IN[6]**	ISOSZP	GPIOI[14]	GPIOI[14] (CBBUSGNT)		656IN[6]
W21	**GPIOI[8]/nSCS[0]**	ISOSZP	GPIOI[8]	GPIOI[8]	nSCS[0]	-
W22	**GPIOI[1]/nNFCE[1]/656IN[3]**	ISOSZP	GPIOI[1]	GPIOI[1]	nNFCE[1]	656IN[3]
Y1	**GPION[1]/TSIDI[1]/MSCLK**	ISOSZP	GPION[1]	GPION[1]	TSIDI[1]	MSCLK
Y2	**OM0**	OS	OM0	OM0	-	-
Y3	**TDI**	IS	TDI	TDI	-	-
Y4	**TCK**	IS	TCK	TCK	-	-
Y5	**TMS**	IS	TMS	TMS	-	-
Y6	**GPIOO[2]/SSPFRM**	ISOSZP	GPIOO[2]	GPIOO[2]	SSPFRM	-
Y7	**GPIOH[4]/ICLKIN**	ISOSZP	GPIOH[4]	GPIOH[4]	ICLKIN	
Y8	**SCL**	OD	SCL	SCL	-	-
Y9	**GPIOC[13]/ID[13]**	ISOSZP	GPIOC[13]	GPIOC[13]	ID[13]	-
Y10	**VDDOP**	SUP				
Y11	**GPIOE[15]/ZD[15]**	ISOSZP	GPIOE[15]	GPIOE[15]	CD[15]	-
Y12	**VDDOP**	SUP				
Y13	**GPIOE[6]/ZD[6]**	ISOSZP	GPIOE[6]	GPIOE[6]	CD[6]	-
Y14	**VDDALIVE2**	SUP				
Y15	**VDDOP**	SUP				
Y16	**GPIOG[15]/ZA[15]**	ISOSZP	GPIOG[15]	GPIOG[15]	CA[15]	-
Y17	**GPIOG[9]/ZA[9]**	ISOSZP	GPIOG[9]	GPIOG[9]	CA[9]	-

Y18	**GPIOG[8]/ZA[8]**	ISOSZP	GPIOG[8]	GPIOG[8]	CA[8]	-
Y19	**GPIOG[4]/ZA[4]**	ISOSZP	GPIOG[4]	GPIOG[4]	CA[4]	-
Y20	**GPIOG[3]/ZA[3]**	ISOSZP	GPIOG[3]	GPIOG[3]	CA[3]	-
Y21	**GPIOI[11]/nSCS[3]**	ISOSZP	GPIOI[11]	GPIOI[11]	nSCS[3]	-
Y22	**GPIOI[9]/nSCS[1]**	ISOSZP	GPIOI[9]	GPIOI[9]	nSCS[1]	-
AA1	**TDO**	OSZ	TDO	TDO	-	-
AA2	**VDDOP**	SUP				
AA3	**NTRST**	IS	nTRST	nTRST	-	-
AA4	**GPIOO[0]/1WIRE**	OD	GPIOO[0]	GPIOO[0]	1WIRE	-
AA5	**GPIOH[2]/IVSYNC**	ISOSZP	GPIOH[2]	GPIOH[2]	IVSYNC	-
AA6	**EXTCLKIN**	IS	EXTCLKIN	EXTCLKIN		-
AA7	**GPIOC[12]/ID[12]**	ISOSZP	GPIOC[12]	GPIOC[12]	ID[12]	-
AA8	**GPIOC[9]/ID[9]**	ISOSZP	GPIOC[9]	GPIOC[9]	ID[9]	-
AA9	**GPIOC[6]/ID[6]**	ISOSZP	GPIOC[6]	GPIOC[6]	ID[6]	-
AA10	**GPIOC[4]/ID[4]**	ISOSZP	GPIOC[4]	GPIOC[4]	ID[4]	-
AA11	**GPIOC[0]/ID[0]**	ISOSZP	GPIOC[0]	GPIOC[0]	ID[0]	-
AA12	**GPIOE[10]/ZD[10]**	ISOSZP	GPIOE[10]	GPIOE[10]	CD[10]	-
AA13	**GPIOE[7]/ZD[7]**	ISOSZP	GPIOE[7]	GPIOE[7]	CD[7]	-
AA14	**GPIOE[3]/ZD[3]**	ISOSZP	GPIOE[3]	GPIOE[3]	CD[3]	-
AA15	**GPIOF[9]/ZA[25]/OM[3]**	ISOSZP	GPIOF[9]	GPIOF[9]	CA[25]	OM[3]
AA16	**GPIOF[4]/ZA[20]**	ISOSZP	GPIOF[4]	GPIOF[4]	CA[20]	-
AA17	**GPIOF[1]/ZA[17]**	ISOSZP	GPIOF[1]	GPIOF[1]	CA[17]	-
AA18	**GPIOF[0]/ZA[16]**	ISOSZP	GPIOF[0]	GPIOF[0]	CA[16]	-
AA19	**GPIOG[7]/ZA[7]**	ISOSZP	GPIOG[7]	GPIOG[7]	CA[7]	-
AA20	**GPIOG[6]/ZA[6]**	ISOSZP	GPIOG[6]	GPIOG[6]	CA[6]	-
AA21	**GPIOG[1]/ZA[1]**	ISOSZP	GPIOG[1]	GPIOG[1]	CA[1]	-
AA22	**GPIOI[10]/nSCS[2]**	ISOSZP	GPIOI[10]	GPIOI[10]	nSCS[2]	-
AB1	**GPIOO[3]/SSPCLK**	ISOSZP	GPIOO[3]	GPIOO[3]	SSPCLK	-

AB2	**GPIOO[1]/SSPTXD**	ISOSZP	GPIOO[1]	GPIOO[1]	SSPTXD	-
AB3	**GPIOH[6]/XDOFF**	ISOSZP	GPIOH[6]	GPIOH[6]	XDOFF	-
AB4	**GPIOH[1]/ICLKOUT**	ISOSZP	GPIOH[1]	GPIOH[1]	ICLKOUT	-
AB5	**GPIOC[15]/ID[15]**	ISOSZP	GPIOC[15]	GPIOC[15]	ID[15]	-
AB6	**GPIOC[11]/ID[11]**	ISOSZP	GPIOC[11]	GPIOC[11]	ID[11]	-
AB7	**GPIOC[8]/ID[8]**	ISOSZP	GPIOC[8]	GPIOC[8]	ID[8]	-
AB8	**GPIOC[7]/ID[7]**	ISOSZP	GPIOC[7]	GPIOC[7]	ID[7]	-
AB9	**GPIOC[3]/ID[3]**	ISOSZP	GPIOC[3]	GPIOC[3]	ID[3]	-
AB10	**GPIOC[1]/ID[1]**	ISOSZP	GPIOC[1]	GPIOC[1]	ID[1]	-
AB11	**GPIOE[14]/ZD[14]**	ISOSZP	GPIOE[14]	GPIOE[14]	CD[14]	-
AB12	**GPIOE[11]/ZD[11]**	ISOSZP	GPIOE[11]	GPIOE[11]	CD[11]	-
AB13	**GPIOE[9]/ZD[9]**	ISOSZP	GPIOE[9]	GPIOE[9]	CD[9]	-
AB14	**GPIOE[5]/ZD[5]**	ISOSZP	GPIOE[5]	GPIOE[5]	CD[5]	-
AB15	**GPIOE[2]/ZD[2]**	ISOSZP	GPIOE[2]	GPIOE[2]	CA[24]	-
AB16	**GPIOF[8]/ZA[24]/OM[2]**	ISOSZP	GPIOF[8]	GPIOF[8]	CA[24]	OM[2]
AB17	**GPIOF[6]/ZA[22]**	ISOSZP	GPIOF[6]	GPIOF[6]	CA[22]	-
AB18	**GPIOF[3]/ZA[19]**	ISOSZP	GPIOF[3]	GPIOF[3]	CA[19]	-
AB19	**GPIOF[2]/ZA[18]**	ISOSZP	GPIOF[2]	GPIOF[2]	CA[18]	-
AB20	**GPIOG[13]/ZA[13]**	ISOSZP	GPIOG[13]	GPIOG[13]	CA[13]	-
AB21	**GPIOG[12]/ZA[12]**	ISOSZP	GPIOG[12]	GPIOG[12]	CA[12]	-
AB22	**GPIOG[0]/ZA[0] (DQMz[0], BTENB0)**	ISOSZP	GPIOG[0]	GPIOG[0]	CA[0] (DQMz[0],BTENB0)	-

<Table 1-3. Pin Listing of MP2520F>

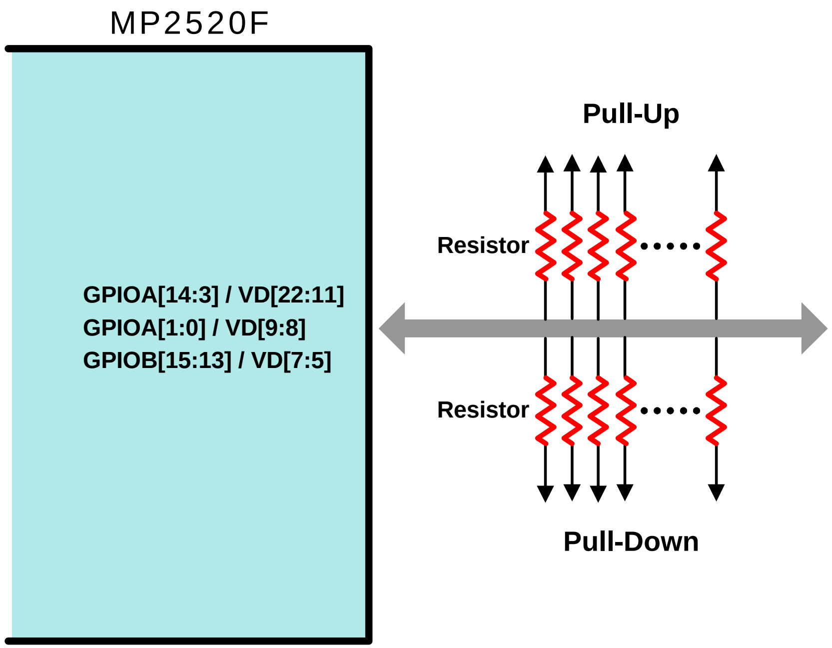
1.4. Pin Functions

1.4.1. System Configuration Pins

Overview

MP2520F has System Configuration Pins that configure the system initial value like as NAND boot, NOR boot, SDRAM bus width, SDRAM capacity, and so forth. If a reset signal is asserted to MP2520F, MP2520F will read the value of Configuration Pins and it will be initialized according to the setting value of Configuration Pins. Therefore they must be pulled up(or pulled down) correctly. <Table 1-4> shows these pins

<Caution 1-1>
These pins work as system configuration function pins when reset occurring and work as GPIO function pins or ALTERNATE(hereafter ALT) function pins after reset according to the setting value of GPIO Alternate Function Registers.



<Figure 1-1. The concept of System Configuration Pin Setting>

Pin Name	Ball	Function Name	Description
GPIOA[14]/VD[22]	E2	RstCfgNFBoot	**NAND Boot Select** 1 : NAND Boot Enable 0 : NAND Boot Disable
GPIOA[13]/VD[21]	G4	RstCfgNFBSize	**NAND Boot Size** 1 : Not Support 0 : 512byte
GPIOA[12]/VD[20]	G3	RstCfgNFBUSWidth	**NAND flash Bus Width** 1 : 16bit 0 : 8bit
GPIOA[11]/VD[19]	F2	RstCfgNFType	**NAND flash Address Type** 1 : 4 Address 0 : 3 Address
GPIOA[10]/VD[18]	H5	RstCfgSR0BUSWidth	**SRAM Bus Width** 1 : 16 bit 0 : 8bit
GPIOA[9]/VD[17]	E1	RstCfgSDRTYPE	**DRAM Type** 1 : Not Support 0 : SDRAM
GPIOA[8]/VD[16]	G2	RstCfgBUSCBW	**SDRAM Bank Total Bus Width** 1 : 32 bit 0 : 16 bit
GPIOA[7]/VD[15]	F1	RstCfgSDRBW[1]	**SDRAM Bus Width** 10 : 16bit 01 : 8bit 00 : 4bit
GPIOA[6]/VD[14]	H3	RstCfgSDRBW[0]	
GPIOA[5]/VD[13]	J5	RstCfgSDRCAP[1]	**SDRAM Capacity** 11 : Reserved 10 : 256 Mbit 01 : 128Mbit 00 : 64 Mbit
GPIOA[4]/VD[12]	H2	RstCfgSDRCAP[0]	
GPIOA[3]/VD[11]	G1	RstCfgShadow	**Shadow Mode Setting** 1 : Shadow Enable 0 : Shadow Disable
GPIOA[1]/VD[9]	K6	RstCfgUARTBootClk	**UART Boot Clock** 1 : ABITCLK 0 : PLL clock
GPIOA[0]/VD[8]	H1	RstCfgBootDown	**UART Boot Setting** 1 : UART Boot Enable 0 : UART Boot Disable
GPIOB[15]/VD[7]	J2	ARM940T DEBUG Enable	**ARM940 ICE Debug Setting** 1 : ARM940 ICE Debug mode Enable

			0 : ARM940 ICE Debug mode Disable
GPIOB[14]/VD[6]	J1	RstCfgSRBuf	**SRAM Buffer Setting** 1 : SRAM Buffer Enable 0 : SRAM Buffer Disable
GPIOB[13]/VD[5]	K4	RstCfgNFBuf	**Nand flash Buffer Setting** 1 : NAND flash Buffer Enable 0 : NAND flash Buffer Disable

<Table 1-4. Pin Description of MP2520F System Configuration>

<Caution 1-2> The meaning of bank

RstCfgBUSCBW means a total bus width of SDRAM bank. This 'bank' means SDRAM bank of MP2520F(different from the meaning of bank that is mentioned at SDRAM specification). For example, Samsung K4S561632D is a 256Mbit SDRAM IC and its structure is 4M x 16bit x 4bank. This 'bank' means an inner bank of SDRAM IC.
Please do not confuse 'bank' of MP2520F with 'bank' of SDRAM IC.

<Caution 1-3> The setting guide of RstCfgBUSCBW, RstCfgSDRBW, and RstCfgSDRCAP in case of Mobile SDRAM with 32bit data bus width

RstCfgBUSCBW, RstCfgSDRBW, and RstCfgSDRCAP of <Table 1-4> describe the setting guide related to SDRAM with 4bit, 8bit, and 16bit data bus width. If a system developer wants to use Mobile SDRAM with 32bit data bus width, he should set RstCfgBUSCBW, RstCfgSDRBW, and RstCfgSDRCAP pins as if two SDRAMs with 16bit data bus width would be used.

Ex 1) For using SAMSUNG Mobile SDRAM K4S513233F (64MB[512Mbit] capacity, 32bit data bus width)
RstCfgBUSCBW : 1 (32bit bus SDRAM bank)
RstCfgSDRBW[1:0] : 10_2 (16bit bus SDRAM)
RstCfgSDRCAP[1:0] : 10_2 (32MB[256Mbit] capacity)

Ex 2) For using SAMSUNG Mobile SDRAM K4S563233F (32MB[256Mbit] capacity, 32bit data bus width)
RstCfgBUSCBW : 1 (32bit bus SDRAM bank)
RstCfgSDRBW[1:0] : 10_2 (16bit bus SDRAM)
RstCfgSDRCAP[1:0] : 01_2 (16MB[128Mbit] capacity)

Setting Examples of System Configuration Pin

Case 1)

<Figure 1-2> shows the design example related to System Configuration Pin settings. The preconditions of <Figure 1-2> are as follows.

Num	System specifications	Related Configuration Pin	Configuration Pin setting
1	**Use NAND boot** - NAND type : Small block size NAND flash (block size : 16KB)	RstCfgNFBoot	1
2	NAND boot size : 512byte	RstCfgNFBSize	0
3	NAND data bus width : 8bit	RstCfgNFBUSWidth	0
4	NAND address type : 4 address type - NAND size : 64MB	RstCfgNFType	1
5	NOR flash(or ROM or SRAM) bus width : Not use NOR boot	RstCfgSR0BUSWidth	meaningless
6	SDRAM Type : SDR-SDRAM	RstCfgSDRTYPE	0
7	SDRAM bank data bus width : 32bit	RstCfgBUSCBW	1
8	SDRAM data bus width : 16bit	RstCfgSDRBW[1]	1
		RstCfgSDRBW[0]	0
9	SDRAM capacity : 32MB(256Mbit)	RstCfgSDRCAP[1]	1
		RstCfgSDRCAP[0]	0
10	Shadow mode : ON	RstCfgShadow	1
11	(if use UART boot) UART boot clock source : Not use UART boot	RstCfgUARTBootClk	meaningless
12	Not use UART boot	RstCfgBootDown	0
13	Use ARM940 ICE debug option	ARM940TDEBUGEnable	1
14	Not use buffer IC halfway between MMSP2 and NOR flash	RstCfgSR0Buf	0
15	Not use buffer IC halfway between MMSP2 and NAND flash	RstCfgNFBuf	0

Case 2)

<Figure 1-3> shows the design example related to System Configuration Pin settings. The preconditions of <Figure 1-3> are as follows.

Num	System specifications	Related Configuration Pin	Configuration Pin setting
1	**Use NAND boot** - NAND type : Small block size NAND flash (block size : 16KB)	RstCfgNFBoot	1
2	NAND boot size : 512byte	RstCfgNFBSize	0
3	NAND data bus width : 8bit	RstCfgNFBUSWidth	0
4	NAND address type : 3 address type - NAND size : 32MB	RstCfgNFType	0
5	NOR flash(or ROM or SRAM) bus width : Not use NOR boot	RstCfgSR0BUSWidth	meaningless
6	SDRAM Type : SDR-SDRAM	RstCfgSDRTYPE	0
7	SDRAM bank data bus width : 32bit	RstCfgBUSCBW	1
8	SDRAM data bus width : 32bit	RstCfgSDRBW[1]	1
		RstCfgSDRBW[0]	0

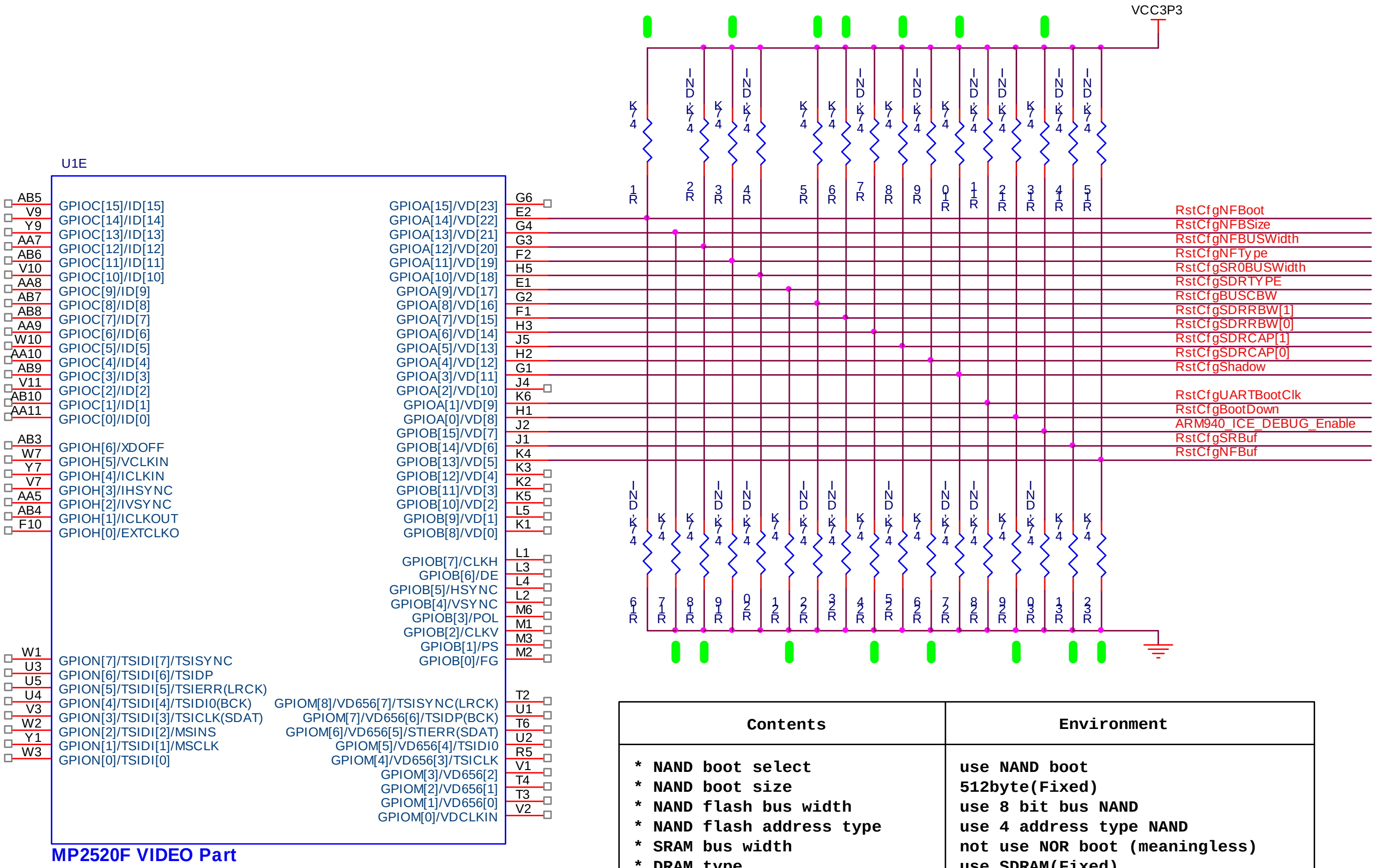
9	SDRAM capacity : 64MB(512Mbit)	RstCfgSDRCAP[1]	1
		RstCfgSDRCAP[0]	0
10	Shadow mode : ON	RstCfgShadow	1
11	(if use UART boot) UART boot clock source : Not use UART boot	RstCfgUARTBootClk	meaningless
12	Not use UART boot	RstCfgBootDown	0
13	Not use ARM940 ICE debug option	ARM940TDEBUGEnable	0
14	Not use buffer IC halfway between MMSP2 and NOR flash	RstCfgSR0Buf	0
15	Use buffer IC halfway between MMSP2 and NAND flash	RstCfgNFBuf	1

Case 3)

<Figure 1-4> shows the design example related to System Configuration Pin settings. The preconditions of <Figure 1-4> are as follows.

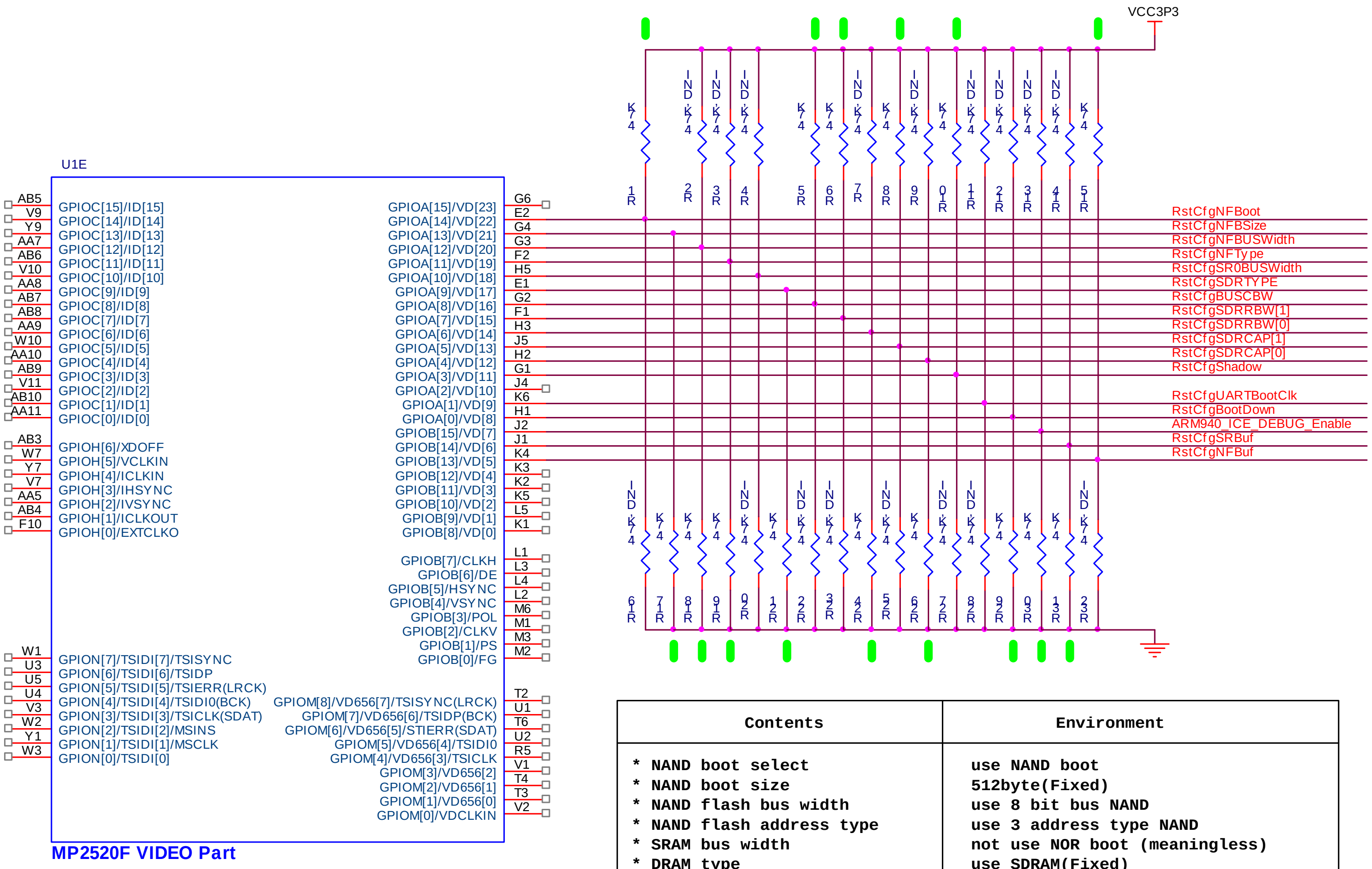
Num	System specifications	Related Configuration Pin	Configuration Pin setting
1	**Use NOR boot** - In case of using 16bit bus width NOR flash	RstCfgNFBoot	0
2	NAND boot size : Not use NAND boot	RstCfgNFBSize	meaningless
3	NAND data bus width : Not use NAND boot	RstCfgNFBUSWidth	meaningless
4	NAND address type : Not use NAND boot	RstCfgNFType	meaningless
5	NOR flash(or ROM or SRAM) bus width : 16bit	RstCfgSR0BUSWidth	1
6	SDRAM Type : SDR-SDRAM	RstCfgSDRTYPE	0
7	SDRAM bank data bus width : 32bit	RstCfgBUSCBW	1
8	SDRAM data bus width : 16bit	RstCfgSDRBW[1]	1
		RstCfgSDRBW[0]	0
9	SDRAM capacity : 32MB(256Mbit)	RstCfgSDRCAP[1]	1
		RstCfgSDRCAP[0]	0
10	Shadow mode : ON	RstCfgShadow	0
11	(if use UART boot) UART boot clock source : Not use UART boot	RstCfgUARTBootClk	meaningless
12	Not use UART boot	RstCfgBootDown	0
13	Not use ARM940 ICE debug option	ARM940TDEBUGEnable	0
14	Use buffer IC halfway between MMSP2 and NOR flash	RstCfgSR0Buf	1
15	Not use buffer IC halfway between MMSP2 and NAND flash	RstCfgNFBuf	0

Schematic



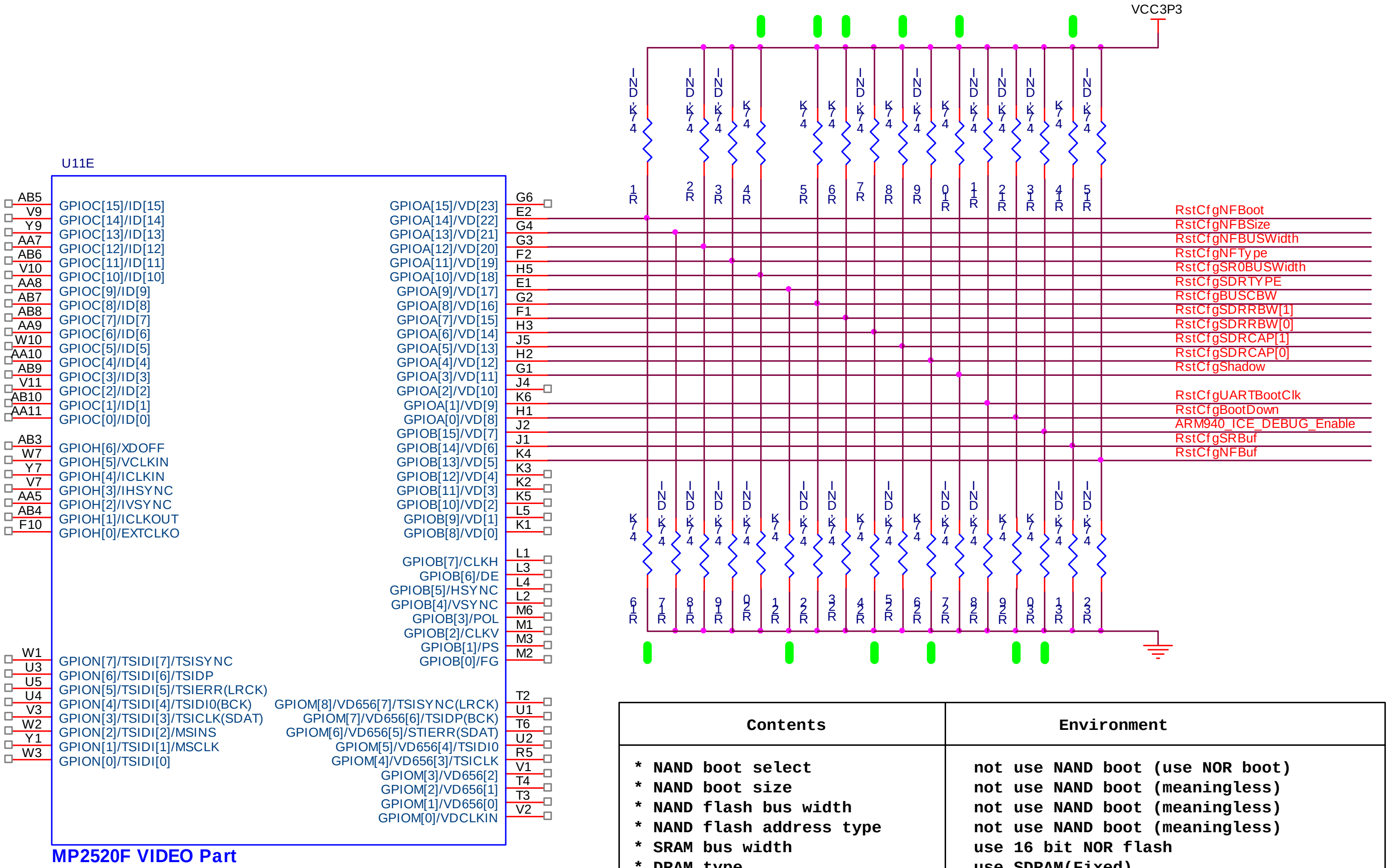
Contents	Environment
* NAND boot select	use NAND boot
* NAND boot size	512byte(Fixed)
* NAND flash bus width	use 8 bit bus NAND
* NAND flash address type	use 4 address type NAND
* SRAM bus width	not use NOR boot (meaningless)
* DRAM type	use SDRAM(Fixed)
* SDRAM bank total bus width	use 32 bit bus
* SDRAM bus width	use 16 bit SDRAM
* SDRAM capacity	use 256 Mbit SDRAM
* Shadow mode	use Shadow mode
* UART boot clock	not use UART boot (meaningless)
* UART boot setting	not use UART boot
* ARM940 ICE debug setting	use ARM940 ICE debug
* SRAM buffer setting	not use SRAM buffer
* NAND flash buffer setting	not use NAND buffer

<Figure 1-2. The example of System Configuration pin setting : Case 1>



Contents	Environment
* NAND boot select	use NAND boot
* NAND boot size	512byte(Fixed)
* NAND flash bus width	use 8 bit bus NAND
* NAND flash address type	use 3 address type NAND
* SRAM bus width	not use NOR boot (meaningless)
* DRAM type	use SDRAM(Fixed)
* SDRAM bank total bus width	use 32 bit bus
* SDRAM bus width	use 32 bit SDRAM
* SDRAM capacity	use 512 Mbit SDRAM
* Shadow mode	use Shadow mode
* UART boot clock	not use UART boot (meaningless)
* UART boot setting	not use UART boot
* ARM940 ICE debug setting	not use ARM940 ICE debug
* SRAM buffer setting	not use SRAM buffer
* NAND flash buffer setting	use NAND buffer

<Figure 1-3. The example of System Configuration pin setting : Case 2>



Contents	Environment
* NAND boot select	not use NAND boot (use NOR boot)
* NAND boot size	not use NAND boot (meaningless)
* NAND flash bus width	not use NAND boot (meaningless)
* NAND flash address type	not use NAND boot (meaningless)
* SRAM bus width	use 16 bit NOR flash
* DRAM type	use SDRAM(Fixed)
* SDRAM bank total bus width	use 32 bit bus
* SDRAM bus width	use 16 bit SDRAM
* SDRAM capacity	use 256 Mbit SDRAM
* Shadow mode	use Shadow mode
* UART boot clock	not use UART boot (meaningless)
* UART boot setting	not use UART boot
* ARM940 ICE debug setting	not use ARM940 ICE debug
* SRAM buffer setting	use SRAM buffer
* NAND flash buffer setting	not use NAND buffer

<Figure 1-4. The example of System Configuration pin setting : Case 3>

1.4.2. Test Mode Pins & Pull Down Pins

<Table 1-5> shows the MP2520F pins related to Test Mode. System developer of MP2520F should set Test Mode Pins disable. G12 and Y2 pins should be set to 0(zero) and can not be used for another service. For AA15, AB16, and V14 pins, we recommend that system developer should set these pins pulled-down. But unlike G12 & Y2 pins, these 3 pins can be used for another service, like as GPIO function or Alternate function.
<Table 1-6> shows the MP2520F pins which should be set to 0.

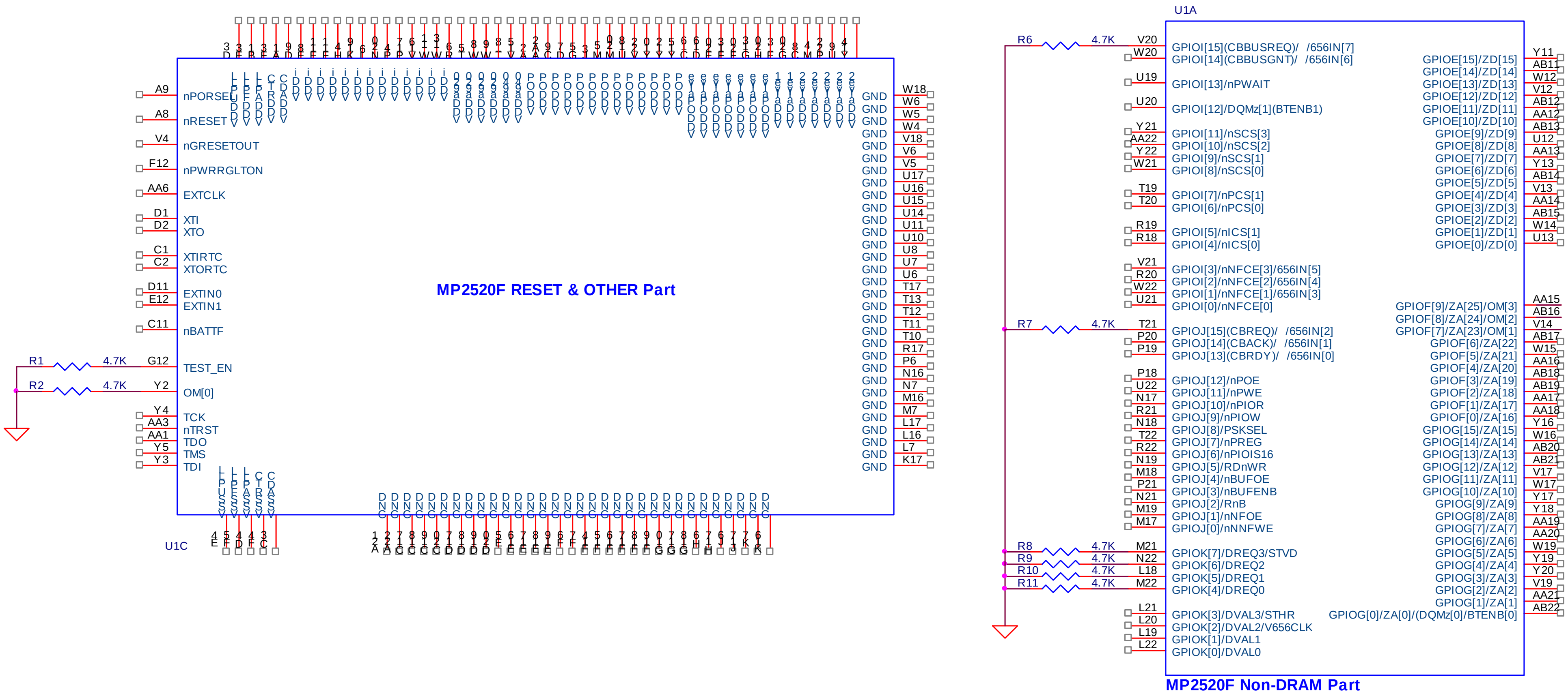
Pin Name	Pin No	Function Name	Descriptions
TEST_EN	G12	Test Enable	0 : Normal mode, 1 : Test mode Must be pulled-down (Do not set this pin to Test mode. This pin must be set to Normal mode)
OM0	Y2	External Clock Enable for FCLK	0 : PLL clock enable, 1 : External Clock enable Must be pulled-down (Do not set this pin to Test mode. This pin must be set to Normal mode)
GPIOF[9]/ZA[25]	AA15	OM[3]	If G12(TEST_EN) is set to high(Test mode), this signal is used as test pins Otherwise it can be used for another functions(GPIO or ALT)
GPIOF[8]/ZA[24]	AB16	OM[2]	If G12(TEST_EN) is set to high(Test mode), this signal is used as test pins Otherwise it can be used for another functions(GPIO or ALT)
GPIOF[7]/ZA[23]	V14	OM[1]	If G12(TEST_EN) is set to high(Test mode), this signal is used test pins Otherwise it can be used for another functions(GPIO or ALT)

<Table 1-5. MP2520F pins related to Test Mode>

Pin Name	Pin No	Function Name	Descriptions
GPIOI[15]/CBBUSREG	V20	GPIOI[15]/CBBUSREG	Must be pulled-down Can not be used for ALT function & GPIO function (N.C. Pin)
GPIOJ[15]/CBREQ	T21	GPIOJ[15]/CBREQ	Must be pulled-down Can not be used for ALT function & GPIO function (N.C. Pin)
GPIOK[7]/DREQ3/STVD	M21	GPIOK[7]/DREQ3/STVD	Must be pulled-down can be used for another functions(GPIO or ALT)
GPIOK[6]/DREQ2/656IN[1]	N22	GPIOK[6]/DREQ2/656IN[1]	Must be pulled-down can be used for another functions(GPIO or ALT).
GPIOK[5]/DREQ1	L18	GPIOK[5]/DREQ1	Must be pulled-down can be used for another functions(GPIO or ALT)
GPIOK[4]/DREQ0	M22	GPIOK[4]/DREQ0	Must be pulled-down can be used for another functions(GPIO or ALT)

<Table 1-6. MP2520F Pull-Down Pins>

Schematic



<Figure 1-4. Design guide of Test Mode & Pull-Down Pins>

1.4.3. Boot Mode

MP2520F supports three types of booting mode, NAND boot, NOR(or ROM) boot, and UART boot.

NAND Boot

MP2520F supports booting from NAND flash. When reset occurring, MP2520F reads 512byte data from NAND flash and writes them to address 0 region. The memory type of address 0 region can be SDRAM or Static Memory(SRAM) and it is configured by RstCfgShadow pin. (refer to <Table 1-4> and Chapter 4. MCU-A Bus>)
In NAND boot case, there is no need to use NOR flash(or ROM) for boot code.

NOR Boot

MP2520F supports the booting from NOR flash. <Table 1-4> shows the Configuration Pins related to NOR Boot and <Figure 1-4> shows the pin setting example of NOR Boot case.

UART Boot

MP2520F supports UART boot. When reset occurring, MP2520F downloads 16Kbyte data from the Host PC and writes them to address 0 region. The memory type of address 0 region can be SDRAM or Static Memory(SRAM) and it is configured by RstCfgShadow pin. (refer to <Table 1-4> and Chapter 4. MCU-A Bus>). <Table 1-7> shows the features related to UART Boot.

Contents	Descriptions	Notes
Transmit Speed	38,400 bps	Before UART Boot, user must set this feature on Host PC
Stop bit	No	Before UART Boot, user must set this feature on Host PC
Parity bit	No	Before UART Boot, user must set this feature on Host PC
Data bit	8 bit	Before UART Boot, user must set this feature on Host PC
UART Boot Setting	RstCfgBootDown : 1 (UART Boot Enable)	Refer to <Table 1-4>
UART Boot Clock	RstCfgUARTBootClk : 0 (PLL Clock)	Refer to <Table 1-4>
UART Channel	Channel **0** only	UART channel **0** of MP2520F and UART port of Host PC are connected by UART cable.

<Table 1-7. The features related to UART Boot>

2. Power

2.1. MP2520F Power

MP2520F has several power pins. <Table 2-1> shows the detail power source of MP2520F.

Power Name	Voltage level (V)	Description
VDDA920	2.0	Power for ARM920T
VDDA940	2.0	Power for ARM940T
VDDADC	3.3	Power for ADC block
VDDAlive1	2.0	Power for Alive1
VDDAlive2	2.0	Power for Alive2
VDDAPLL	2.0	Power for APLL
VDDFPLL	2.0	Power for FPLL
VDDUPLL	2.0	Power for UPLL
VDDI	2.0	Power for Internal Logic
VDDOP	3.3	Power for IO Pad
VDDOPAlive	3.3	Power for SDRAM I/O
VDDRTC	1.4 ~ 1.8	Power for RTC block

<Table 2-1. Power source of MP2520F>

Notice) RTC block of MP2520F needs VDDAlive1 & VDDAlive2 as well as VDDRTC. (refer to MP2520F Errata ver1.0)

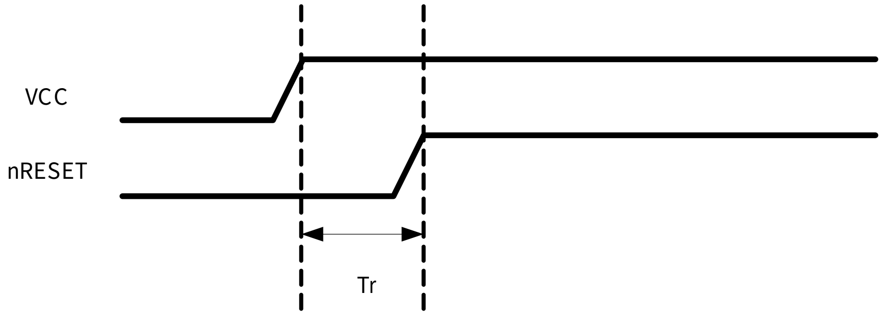
3. RESET

3.1. RESET

MP2520F has an internal Reset Generator. To use Power-On-Reset from internal Reset Generator, Power-On-Reset select pin(A9) of MP2520F must be set to Low(pulled-down). For using external Reset IC, Power-On-Reset select pin(A9) must be set to High(pulled-up). <Figure 3-3> shows the schematic design related to Reset of MP2520F. MagicEyes recommend using Reset IC (CASE 1 of <Figure 3-3>) to customers.

Reset-In Timing

<Figure 3-1> shows the reset transition timing of MP2520F. nRESET is a reset-in signal which is inputted to A8 pin of MP2520F.



<Figure 3-1. Reset transition timing of MP2520F>

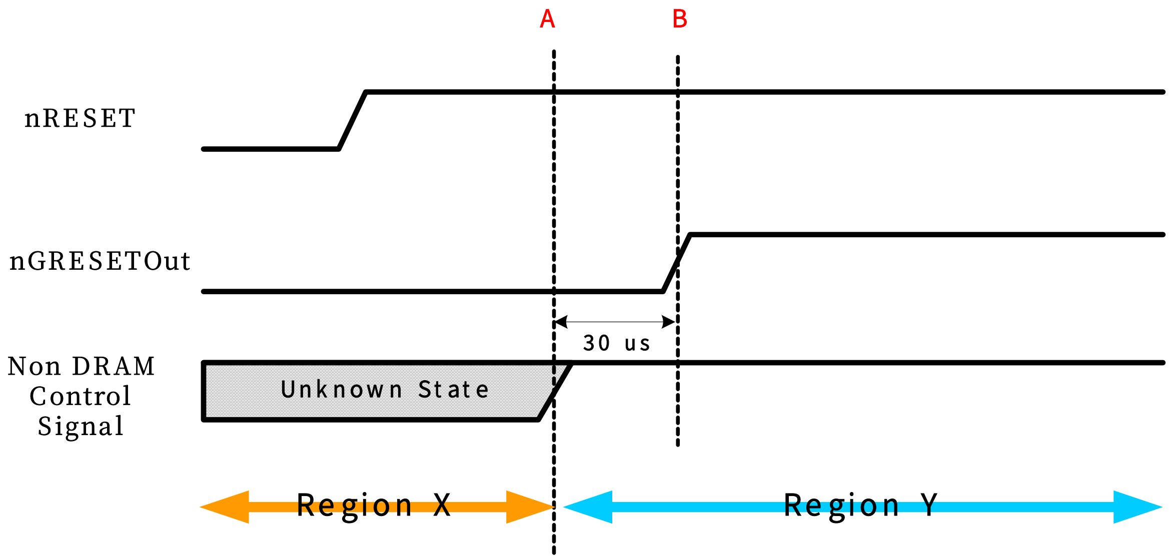
	MININUM	TYPICAL	MAXIMUM
Tr	**10** ms	**300** ms	-

'TYPICAL' is the measurement value in case of using MAX811 reset IC.

Reset-Out Timing

<Figure 3-2> shows the timing diagram of Reset-Out signal and MCU-C(or Non-DRAM) bus control signal. nRESET is a reset-in signal which is inputted to MP2520F and nGRESETOut is a global reset-out signal from MP2520F.

The Region X is an unknown state region and the region Y is a stable state region. So the value of Non-DRAM Control Signal becomes stable state from A point. Before A point, the state of Non-DRAM Control Signal is unknown and it has unknown value.
To make a system stable, Non-DRAM Control Signal must be used during Region Y. (Do not use Non-DRAM Control Signal during Resion X)



<Figure 3-2. Timing Diagram of Reset and MCU-C(or Non-DRAM) control signal>

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
MP2_nPORSEL	A9		MP2_nRESET	A8		MC_nRSTOUT	V4		-	-	-

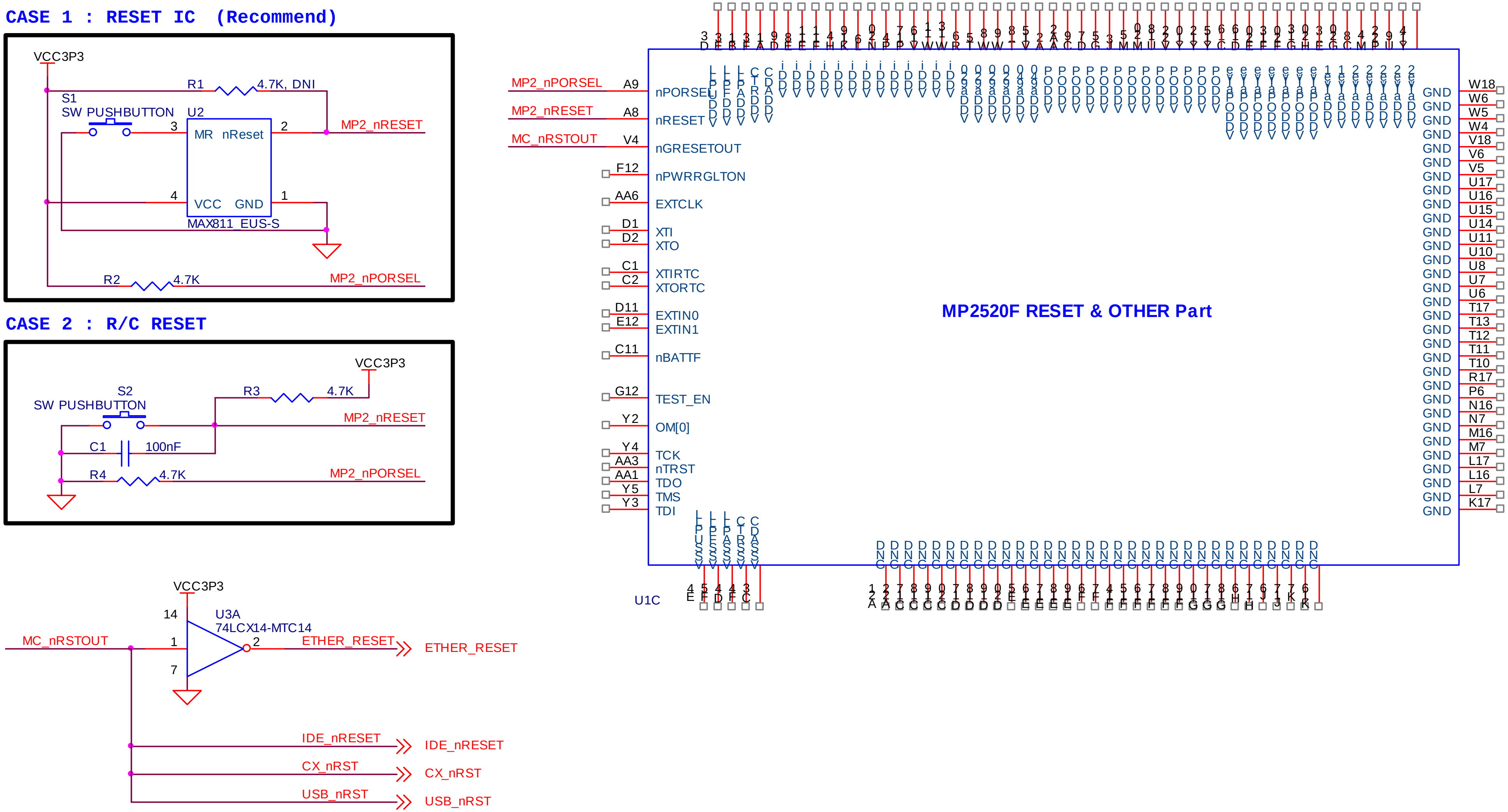
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
MP2_nPORSEL				POWER ON Reset Select signal. 0 : in case of using internal PowerOnReset 1 : in case of using external reset IC (recommended)
MP2_nRESET				Global Reset In
MC_nRSTOUT				Global Reset Out
ETHER_RESET				Ethernet controller Reset signal
IDE_nRESET				HDD Reset signal
CX_nRST				CX25874 Reset signal

- Component description

Component Number	Component Name	Descriptions
U1C	MP2520F	POWER & OTHER Part
U2	MAX811_EUS-S	Reset IC
U3	74LCX14 - MTC14	Inverter gate
S1,S2	Switches	Push Button Switch
R1	Registers	4.7K ohm (Do Not Install)
R2,R3,R4	Registers	4.7K ohm
C1	Capacitor	100nF

Schematic



<Figure 3-3. Design guide of Reset Part>

4. CLOCK

4.1. PLL Clock

<Figure 4-1> (A) shows the schematic design related to PLL clock of MP2520F. There are two PLL clock sources. One is OSC and the other is Crystal. The pin number of PLL clock pin is D1 & D2 and they are located nearby Video-out pins. Because the frequency of Video-out signal is so high, the PCB layout engineer has to pay attention to PCB pattern routing of PLL clock signal. Generally OSC is more stable than Crystal, so we recommend using OSC (CASE 1 of <Figure 4-1>) to customers.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
XTI	D1		XTO	D2		-	-	-	-	-	-

- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
XTI				7.3728 MHz OSC(or Crystal) In. (recommend)
XTO				7.3728 MHz Crystal Out. If OSC is used this signal is not used.

- Component description

Component Number	Component Name	Descriptions
OSC1	Oscillator	7.3728 MHz oscillator
Y1	Crystal	7.3728 MHz crystal
R1	Resistor	33 ohm
C1	Capacitor	100 nF
C2, C3	Capacitor	22 pF

4.2. RTC Clock

<Figure 4-1> (B) shows the schematic design related to RTC clock.

<Caution 4-1>
Although a system developer uses external RTC IC instead of internal RTC of MP2520F, he should supply RTC clock to MP2520F, so <Figure 4-1> (B) should be designed.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
XTIRTC	C1		XTORTC	C2		-	-	-	-	-	-

- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
XTIRTC				32.768KHz RTC crystal In
XTORTC				32.768KHz RTC crystal Out

- Component description

Component Number	Component Name	Descriptions
Y2	Crystal	32.768KHz crystal
C4, C5	Capacitor	22pF
R2	Resistor	10M ohm
R3	Resistor	2.2K ohm

4.3. External Video Clock

<Figure 4-1> (D) shows the schematic design related to video clock source.

<Caution 4-2> Video Encoder IC Selection
If a system developer uses a video encoder which does not support Pseudo Master mode, he should use external OSC and connect OSC clock signal to VCLKIN pin(GPIOH5) of MP2520F. If a system developer uses a video encoder which supports Pseudo Master mode, he should connect a video clock output signal of video encoder to VCLKIN pin(GPIOH5) of MP2520F and external OSC is not necessary for this application.

What is Pseudo Master mode ?
For Pseudo Master mode, MP2520F receives a video clock signal from external encoder IC and makes a video-out clock signal by using it, then outputs video-out clock signal, HSYNC signal, VSYNC signal, Data Enable signal and Video-out data.
In this mode, the external video encoder generates a clock reference signal, CLKO as an output. This signal informs the MP2520F the exact frequency at which the data must be sent to the external video encoder. Timing signals-HSYNC, VSYCN, and BLANK-are received by the external video encoder as inputs. That is to say, MP2520F receives a clock reference signal from external video encoder IC and makes a video-out clock signal by using it, then outputs video-out clock signal, HSYNC signal, VSYNC signal, BLANK signal, and Video-out data to external video encoder IC.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
VCLKIN	W7	H5	-	-	-	-	-	-	-	-	-

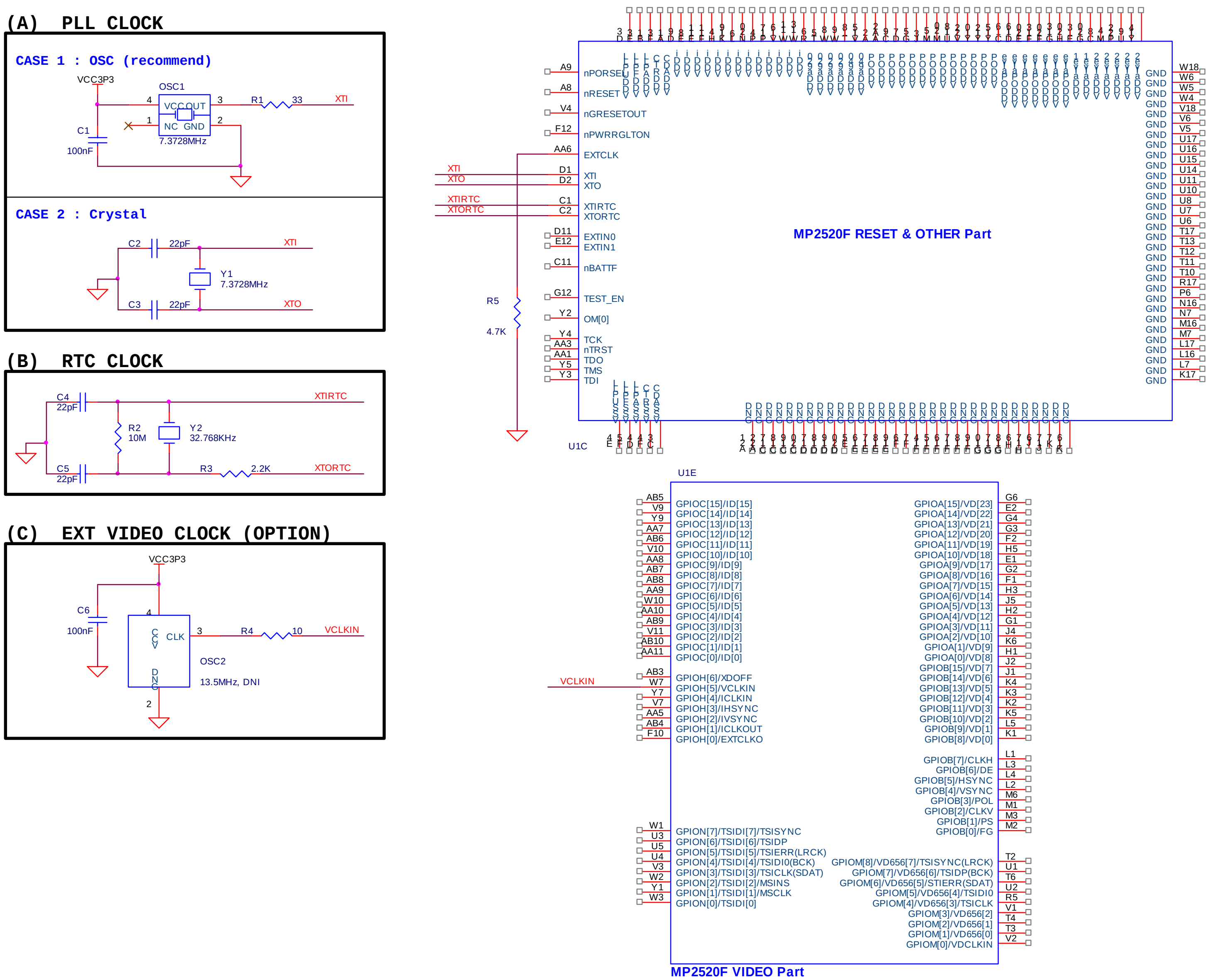
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
VCLKIN		✓		Video Clock In signal.

- Component description

Component Number	Component Name	Descriptions
OSC3	Oscillator	13.5MHz oscillator
C6	Capacitor	100nF
R4	Resistor	10 ohm

Schematic



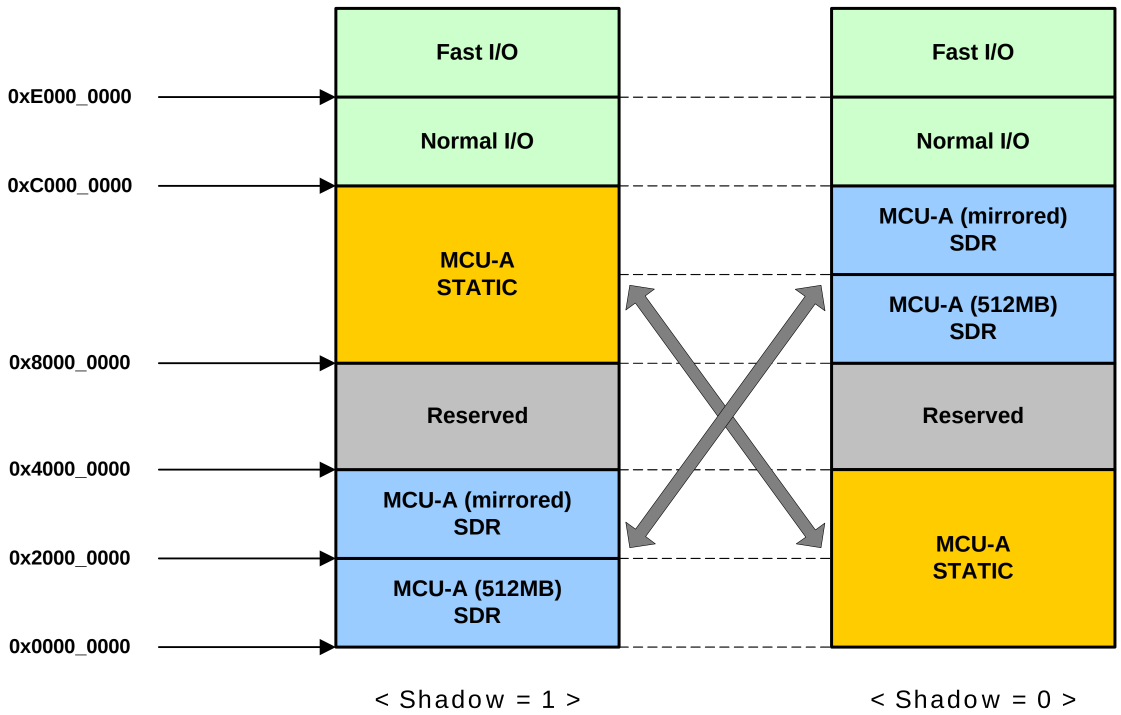
<Figure 4-1. Design guide of Clock>

5. MCU-A Bus

5.1. About MCU-A Bus

MCU-A Bus is a SDRAM memory bus of MP2520F and it has 32bit data bus width. It is possible to use 16bit data bus width but we don't recommend it to customers because of the decline of system performance. MP2520F has two SDRAM memory bank so the number of MCU-A Bus bank delect signals is two. (refer to <Caution 1-2>)

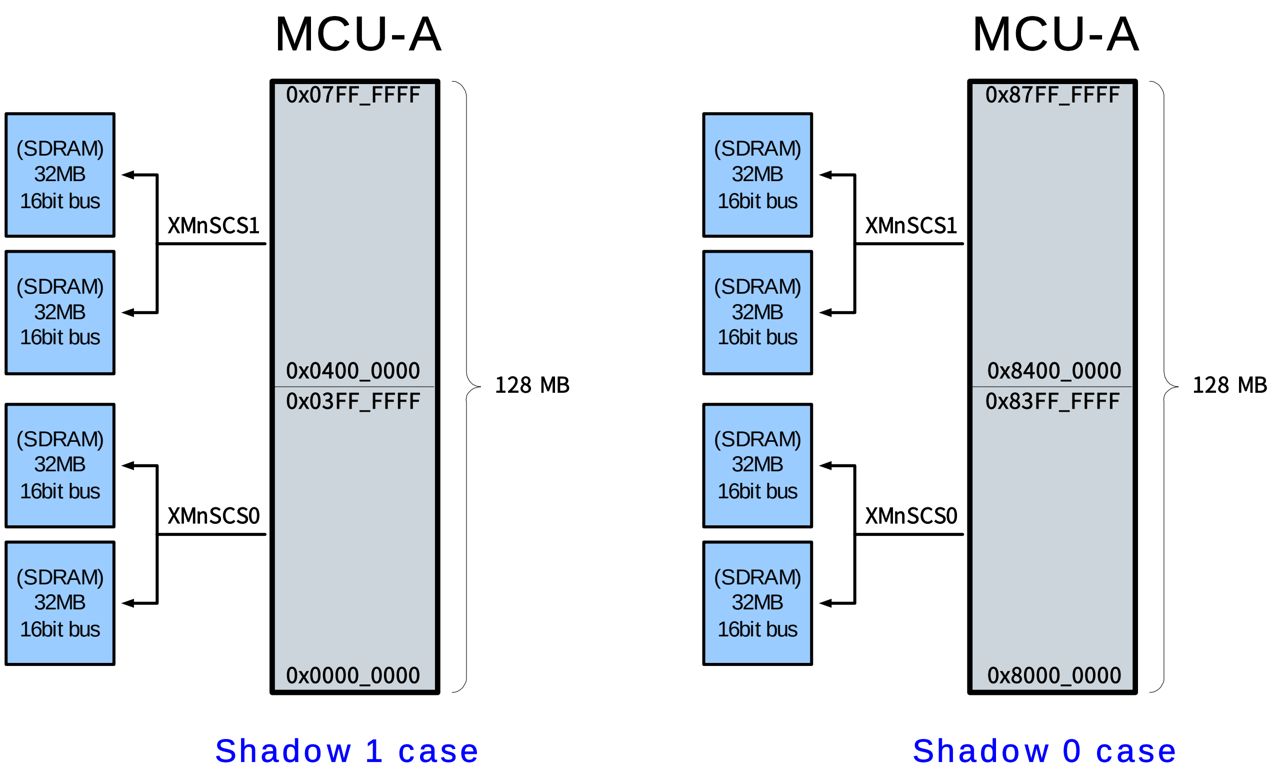
The SDRAM bank and the Static Memory bank can be exchanged by the setting value of RstCfgShadow pin. (refer to <Table 1-4>)



<Figure 5-1. SDRAM bank/Static Memory bank Exchange>

Example 5-1)

Supposing a system which uses four SDRAM ICs. Each SDRAM has 32MB capacity and 16bit data bus width. Two SDRAM banks are used and total memory capacity is 128MB.



<Figure 5-2. Example of MCU-A Memory Map change on Shadow mode setting>

<Table 5-1> shows combination examples of SDRAM bank.

Num	Case	Description
1	16bit SDRAM x 2	1 SDRAM memory bank system. Refer to <Figure 5-3>
2	16bit SDRAM x 2 x 2	2 SDRAM memory bank system. Refer to <Figure 5-4>
3	32bit SDRAM x 1	1 SDRAM memory bank system. Refer to <Figure 5-5>
4	32bit SDRAM x 2	2 SDRAM memory bank system
5	8bit SDRAM x 4	1 SDRAM memory bank system
6	8bit SDRAM x 4 x 2	2 SDRAM memory bank system

<Table 5-1. Union examples of SDRAM bank>

5.2. Schematic design examples

5.2.1. 16bit SDRAM case 1 : 16bit SDRAM x 2

<Figure 5-3> shows the schematic design example of MCU-A bus which is composed of two SDRAMs with 16bit data bus width. This architecture has one SDRAM memory bank.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
XMSBA1	C15		XMSBA0	D15		XMSA12	E15		XMSA11	E14	
XMSA10	D14		XMSA9	A12		XMSA8	C14		XMSA7	D13	
XMSA6	C13		XMSA5	B12		XMSA4	A11		XMSA3	D12	
XMSA2	C12		XMSA1	A10		XMSA0	B11		XSD31	A13	
XSD30	B13		XSD29	A14		XSD28	B14		XSD27	A15	
XSD26	B15		XSD25	A16		XSD24	B16		XSD23	A17	
XSD22	B17		XSD21	A18		XSD20	B18		XSD19	A19	
XSD18	B19		XSD17	A20		XSD16	B20		XSD15	B21	
XSD14	B22		XSD13	C21		XSD12	C22		XSD11	D21	
XSD10	D22		XSD9	E21		XSD8	E22		XSD7	F21	
XSD6	F22		XSD5	G21		XSD4	G22		XSD3	H21	
XSD2	H22		XSD1	J21		XSD0	J22		XMSCLK0	B10	
XMSCKE	G19		XMnSRAS	J19		XMnSCAS	J20		XMnSWE	J18	
XMnSDQM3	K20		XMnSDQM2	K22		XMnSDQM1	K21		XMnSDQM0	K18	
XMnSCS0	H19		-	-	-	-	-	-	-	-	-

- Signal description

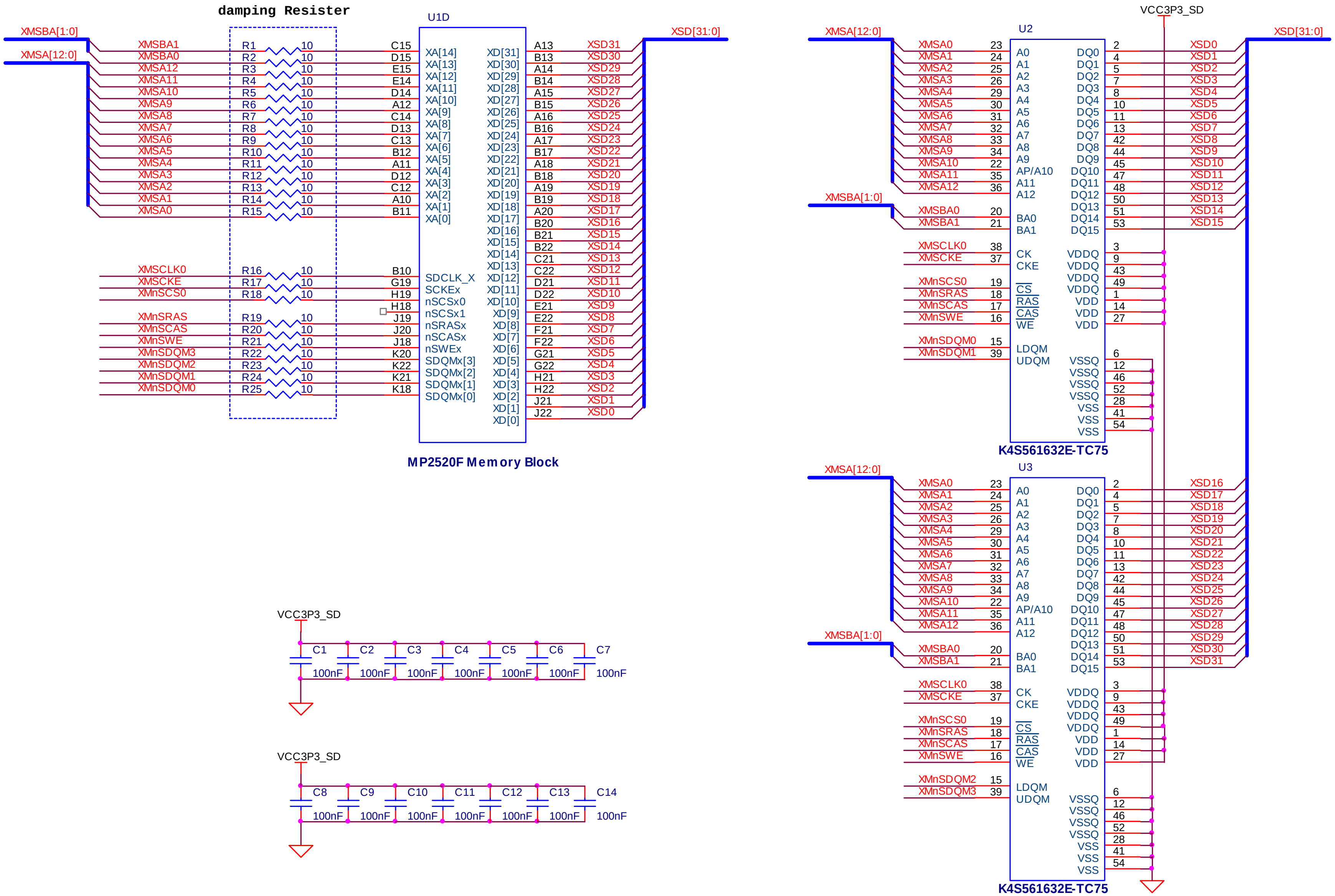
Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
XMSBA[1:0]				Bank select address 0 ~ 1 signal
XMSA[12:0]				Address 0 ~ 12 signal
XSD[31:0]				Data 0 ~ 31 signal
XMSCLK0				System clock signal
XMSCKE				Clock enable signal
XMnSCS0				Memory Bank select 0 signal. Connect to CS pin of SDRAM
XMnSRAS				Row address strobe signal
XMnSCAS				Column address strobe signal
XMnSWE				Write enable signal
XMnSDQM3				SDRAM DQM for data byte 3. ← [31:24]bit
XMnSDQM2				SDRAM DQM for data byte 2. ← [23:16]bit

XMnSDQM1				SDRAM DQM for data byte 1. ← [15:8]bit
XMnSDQM0				SDRAM DQM for data byte 0. ← [7:0]bit

- Component description

Component Number	Component Name	Descriptions
U1D	MP2520F	DRAM Part
U2, U3	SDRAM	SAMSUNG K4S561632E-TC75. 4M x 16bit x 4Banks
R1 ~ R25	Resistor	10 ohm. Damping resistors
C1 ~ C14	Capacitor	100nF. Bypass Capacitor

Schematic



<Figure 5-3. Design guide of 16bit SDRAM x 2 : 1 Memory Bank system>

5.2.2. 16bit SDRAM case 2 : 16bit SDRAM x 2 x 2

<Figure 5-4> shows the schematic design example of MCU-A bus which is composed of four SDRAMs with 16bit data bus width. This architecture has two SDRAM memory banks.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
XMSBA1	C15		XMSBA0	D15		XMSA12	E15		XMSA11	E14	
XMSA10	D14		XMSA9	A12		XMSA8	C14		XMSA7	D13	
XMSA6	C13		XMSA5	B12		XMSA4	A11		XMSA3	D12	
XMSA2	C12		XMSA1	A10		XMSA0	B11		XSD31	A13	
XSD30	B13		XSD29	A14		XSD28	B14		XSD27	A15	
XSD26	B15		XSD25	A16		XSD24	B16		XSD23	A17	
XSD22	B17		XSD21	A18		XSD20	B18		XSD19	A19	
XSD18	B19		XSD17	A20		XSD16	B20		XSD15	B21	
XSD14	B22		XSD13	C21		XSD12	C22		XSD11	D21	
XSD10	D22		XSD9	E21		XSD8	E22		XSD7	F21	
XSD6	F22		XSD5	G21		XSD4	G22		XSD3	H21	
XSD2	H22		XSD1	J21		XSD0	J22		XMSCLK0	B10	
XMSCKE	G19		XMnSRAS	J19		XMnSCAS	J20		XMnSWE	J18	
XMnSDQM3	K20		XMnSDQM2	K22		XMnSDQM1	K21		XMnSDQM0	K18	
XMnSCS0	H19		XMnSCS1	H18	-	-	-	-	-	-	-

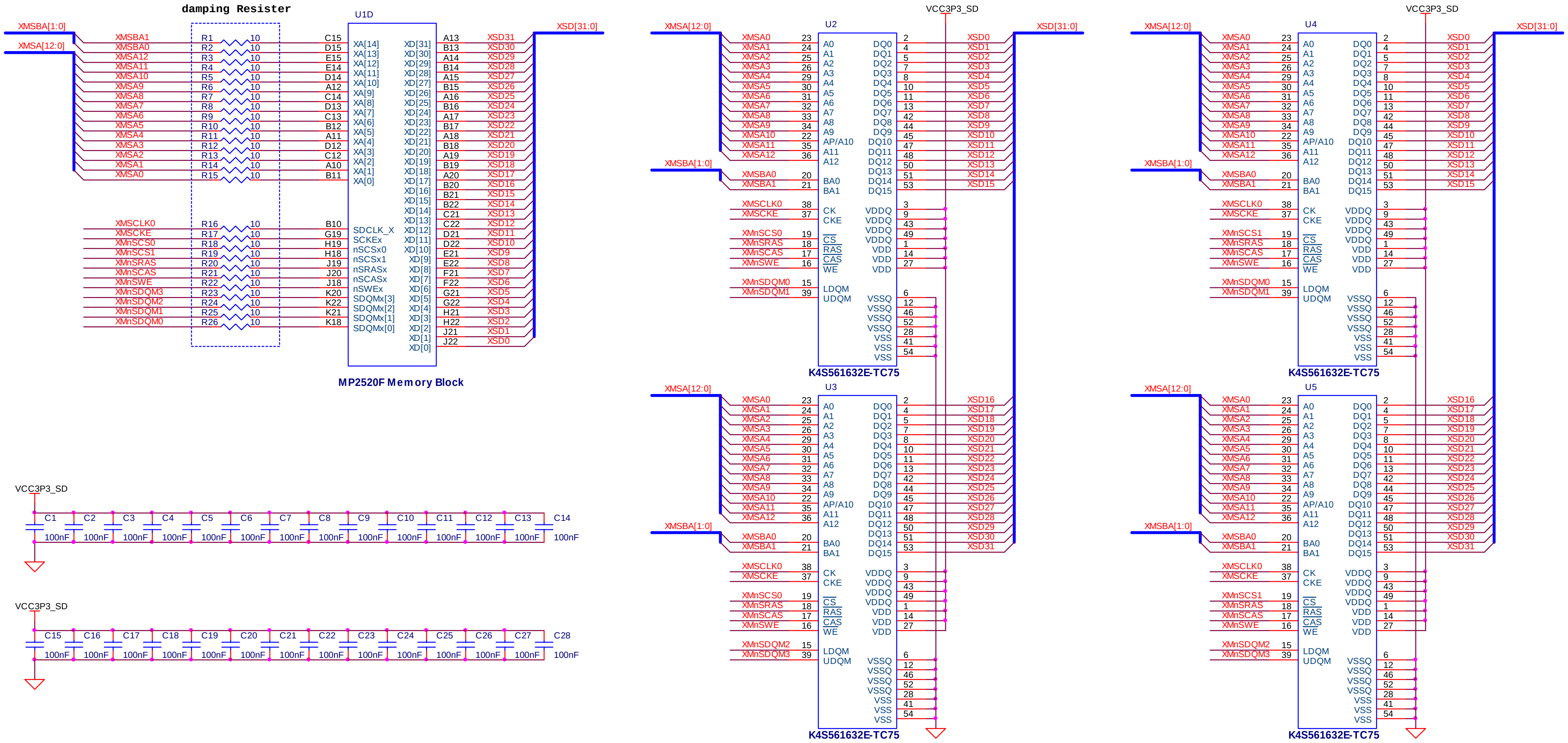
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
XMSBA[1:0]				Bank select address 0 ~ 1 signal
XMSA[12:0]				Address 0 ~ 12 signal
XSD[31:0]				Data 0 ~ 31 signal
XMSCLK0				System clock signal
XMSCKE				Clock enable signal
XMnSCS0				Memory Bank select 0 signal. Connect to CS pin of SDRAM
XMnSCS1				Memory Bank select 1 signal. Connect to CS pin of SDRAM
XMnSRAS				Row address strobe signal
XMnSCAS				Column address strobe signal
XMnSWE				Write enable signal
XMnSDQM3				SDRAM DQM for data byte 3. ← [31:24]bit
XMnSDQM2				SDRAM DQM for data byte 2. ← [23:16]bit
XMnSDQM1				SDRAM DQM for data byte 1. ← [15:8]bit
XMnSDQM0				SDRAM DQM for data byte 0. ← [7:0]bit

- Component description

Component Number	Component Name	Descriptions
U1D	MP2520F	DRAM Part
U2 ~ U5	SDRAM	SAMSUNG K4S561632E-TC75. 4M x 16bit x 4Banks
R1 ~ R26	Resistor	10 ohm. Damping resistors
C1 ~ C28	Capacitor	100nF. Bypass Capacitor

Schematic



<Figure 5-4. Design guide of 16bit SDRAM x 2 x 2 :   2 Memory Bank system>

5.2.3. 32bit SDRAM case : 32bit mobile SDRAM x 1

<Figure 5-5> shows the schematic design example of MCU-A bus which is composed of one Mobile SDRAM with 32bit data bus width. This architecture has one SDRAM memory bank.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
XMSBA1	C15		XMSBA0	D15		XMSA12	E15		XMSA11	E14	
XMSA10	D14		XMSA9	A12		XMSA8	C14		XMSA7	D13	
XMSA6	C13		XMSA5	B12		XMSA4	A11		XMSA3	D12	
XMSA2	C12		XMSA1	A10		XMSA0	B11		XSD31	A13	
XSD30	B13		XSD29	A14		XSD28	B14		XSD27	A15	
XSD26	B15		XSD25	A16		XSD24	B16		XSD23	A17	
XSD22	B17		XSD21	A18		XSD20	B18		XSD19	A19	
XSD18	B19		XSD17	A20		XSD16	B20		XSD15	B21	
XSD14	B22		XSD13	C21		XSD12	C22		XSD11	D21	
XSD10	D22		XSD9	E21		XSD8	E22		XSD7	F21	
XSD6	F22		XSD5	G21		XSD4	G22		XSD3	H21	
XSD2	H22		XSD1	J21		XSD0	J22		XMSCLK0	B10	
XMSCKE	G19		XMnSRAS	J19		XMnSCAS	J20		XMnSWE	J18	
XMnSDQM3	K20		XMnSDQM2	K22		XMnSDQM1	K21		XMnSDQM0	K18	
XMnSCS0	H19		-	-	-	-	-	-	-	-	-

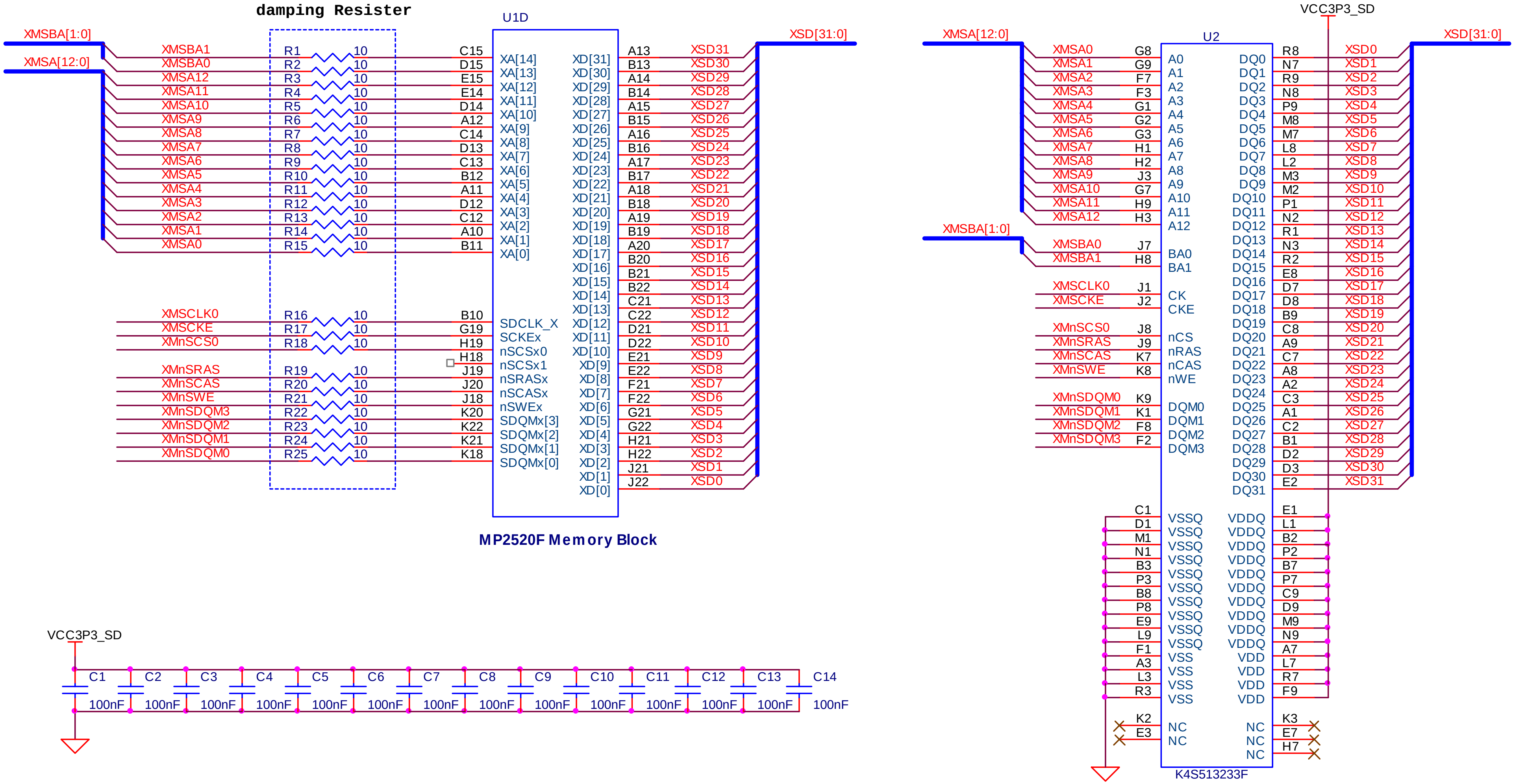
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
XMSBA[1:0]				Bank select address 0 ~ 1 signal
XMSA[12:0]				Address 0 ~ 12 signal
XSD[31:0]				Data 0 ~ 31 signal
XMSCLK0				System clock signal
XMSCKE				Clock enable signal
XMnSCS0				Memory Bank select 0 signal. Connect to CS pin of SDRAM
XMnSRAS				Row address strobe signal
XMnSCAS				Column address strobe signal
XMnSWE				Write enable signal
XMnSDQM3				SDRAM DQM for data byte 3. ← [31:24]bit
XMnSDQM2				SDRAM DQM for data byte 2. ← [23:16]bit
XMnSDQM1				SDRAM DQM for data byte 1. ← [15:8]bit
XMnSDQM0				SDRAM DQM for data byte 0. ← [7:0]bit

- Component description

Component Number	Component Name	Descriptions
U1D	MP2520F	DRAM Part
U2	SDRAM	SAMSUNG K4S513233F. 4M x 32bit x 4Banks
R1 ~ R25	Resistor	10 ohm. Damping resistors
C1 ~ C14	Capacitor	100nF. Bypass Capacitor

Schematic



<Figure 5-5. Design guide of 32bit mobile SDRAM x 1 : 1 Memory Bank system>

6. MCU-C Bus

6.1. Overview

MCU-C bus of MP2520F is a Static Bus with 16bit data bus width.

Supported memory(or device) types are as follows.

- Static Memory : Asynchronous SRAM, ROM, NOR flash, NAND flash
- Card interface : PCMCIA/CF
- IDE interface : HDD, CD-ROM, DVD-ROM

<Caution 6-1>
MP2520F has four Static Memory chip select signals, so if the number of external devices which are connected to Static Memory bus is more than four, a system developer should make another chip select signals by using the Address Decoding Method.

6.2. NAND Flash with Small Block size

MCU-C bus of MP2520F has four NAND flash chip select signal and it is possible to connect up to four NAND flash. <Figure 6-1> shows the schematic design example about NAND flash with 8bit data bus width.

<Caution 6-2>
The RnB pin of NAND flash is Open-Drain pin, so this pin must be set to High by using pull-up Resistor. If the system developer wants to know the proper value of pull-up Resistor, please refer to the datasheet of NAND flash IC. (Normally, 2Kohm ~ 3Kohm is used)

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
MC_DATA7	AA13	E 7	MC_DATA6	Y13	E 6	MC_DATA5	AB14	E 5	MC_DATA4	V13	E 4
MC_DATA3	AA14	E 3	MC_DATA2	AB15	E 2	MC_DATA1	W14	E 1	MC_DATA0	U13	E 0
MC_nNFCS0	U21	I 0	MC_nPIOR	N17	J 10	MC_nPIOW	R21	J 9	MC_NFRnB	N21	J 2
MC_nNFOE	M19	J 1	MC_nNFWE	M17	J 0	MC_WP	reserved	reserved	-	-	-

- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	

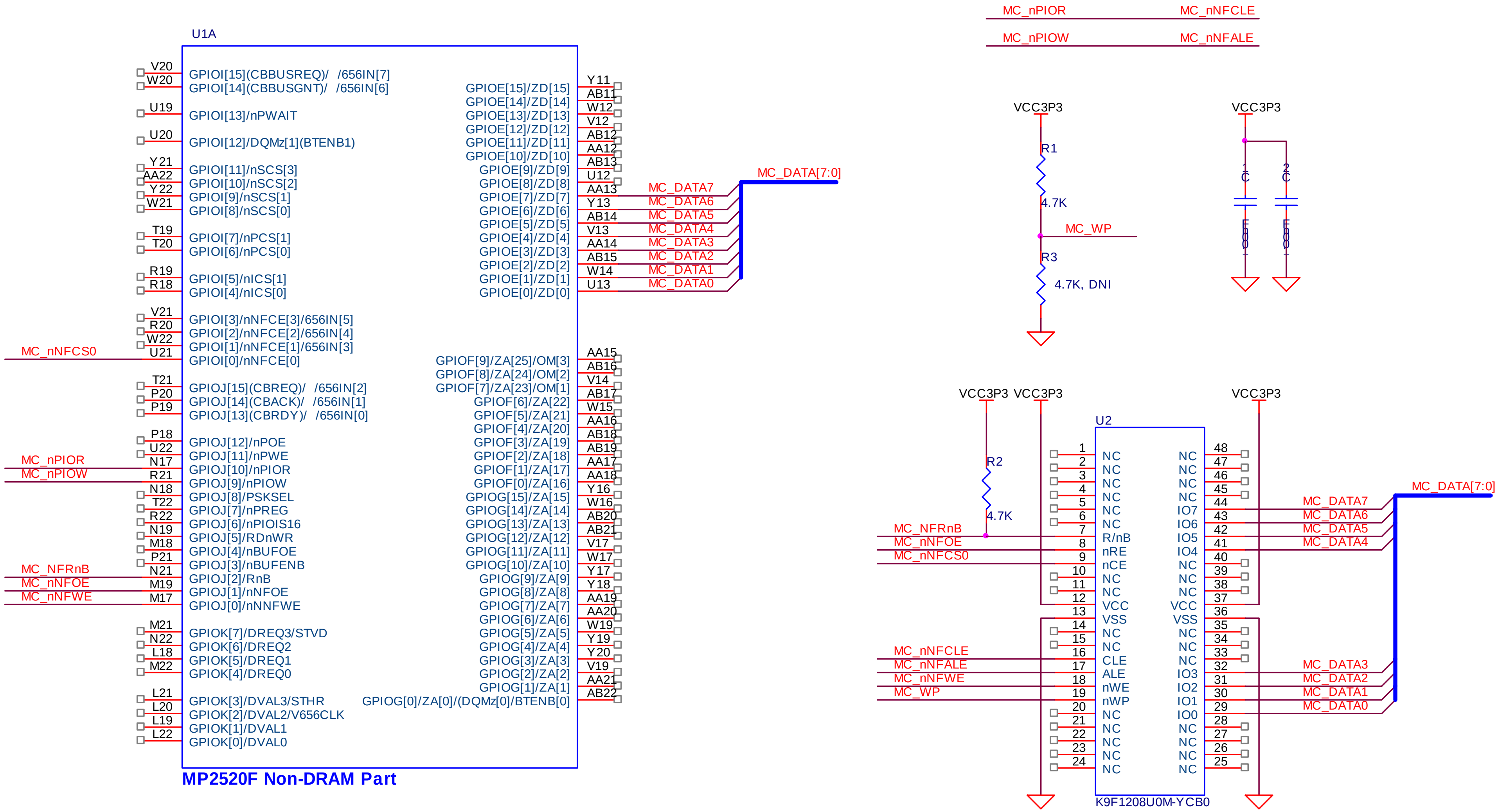
MC_DATA[7:0]		✓		MCU-C bus data 0 ~ 7 signal
MC_nNFCS0		✓		NAND flash Chip Select 0 signal
MC_nPIOR		✓		NAND flash Command Latch Enable signal
MC_nPIOW		✓		NAND flash Address Latch Enable signal
MC_NFRnB		✓		NAND flash Ready/Busy signal
MC_nNFOE		✓		NAND flash Read Enable signal
MC_nNFWE		✓		NAND flash Write Enable signal
[1] MC_WP	✓			NAND flash Write Protect signal

[1] : MC_WP is a Write Protect signal of NAND flash. This signal can be controlled by hardware or software. If a system developer wants to control this signal by software, he should connect it to GPIO pin and then control it as GPIO_OUT function.

- Component description

Component Number	Component Name	Descriptions
U1A	MP2520F	Non-DRAM Part
U2	NAND flash	SAMSUNG K9F1208U0M. 64Mbyte NAND flash
R1,R2	Resistor	4.7K ohm
R3	Resistor	4.7K ohm (Do Not Install)
C1 ~ 2	Capacitor	100nF. Bypass Capacitor

Schematic



<Figure 6-1. Design guide of NAND flash with small block size>

6.3. NAND Flash with Large Block size

Originally, MP2520F does not support large block NAND flash for NAND boot. But if a system developer wants to use NAND boot from Large Block NAND flash, he can do that by changing his schematic design like as <Figure 6-2>.

For normal data access, there is no difference of schematic design between Large Block NAND flash and Small Block NAND flash.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
MC_DATA7	AA13	E 7	MC_DATA6	Y13	E 6	MC_DATA5	AB14	E 5	MC_DATA4	V13	E 4
MC_DATA3	AA14	E 3	MC_DATA2	AB15	E 2	MC_DATA1	W14	E 1	MC_DATA0	U13	E 0
MC_nNFCS0	U21	I 0	MC_nPIOR	N17	J 10	MC_nPIOW	R21	J 9	MC_NFRnB	N21	J 2
MC_nNFOE	M19	J 1	MC_nNFWE	M17	J 0	NAND_nBOOT	L21	K3	MC_WP	reserved	reserved

- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
MC_DATA[7:0]		√		MCU-C bus data 0 ~ 7 signal
MC_nNFCS0		√		NAND flash Chip Select 0 signal
MC_nPIOR		√		NAND flash Command Latch Enable signal
MC_nPIOW		√		NAND flash Address Latch Enable signal
MC_NFRnB		√		NAND flash Ready/Busy signal
MC_nNFOE		√		NAND flash Read Enable signal
MC_nNFWE		√		NAND flash Write Enable signal
[1] NAND_nBOOT	√			Large Block NAND flash Enable signal
[2] MC_WP	√			NAND flash Write Protect signal
[3] XTI				7.3728 MHz OSC(or Crystal) In.
[4] MC_nRSTOUT				Global Reset Out

[1] : This signal should be changed to 0(Low) value to access NAND flash after NAND boot.

[2] : MC_WP is a Write Protect signal of NAND flash. This signal can be controlled by hardware or software. If a system developer wants to control this signal by software, he should connect it to GPIO pin and then control it as GPIO_OUT function.

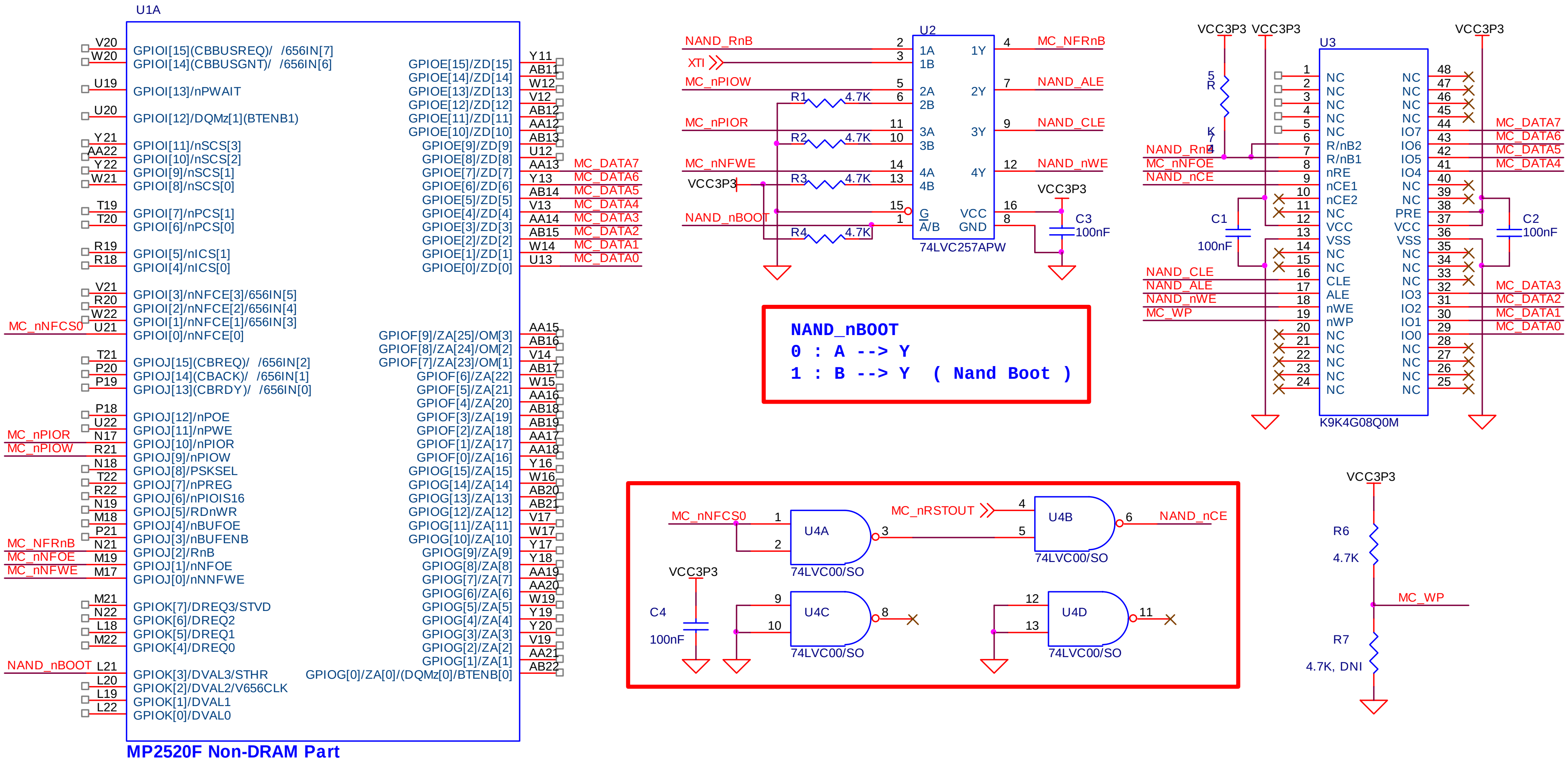
[3] : XTI is 7.3728MHz OSC clock in signal. Refer to <Figure 4-1>

[4] : MC_nRSTOUT is global reset out signal. Refer to <Figure 3-1>

- Component description

Component Number	Component Name	Descriptions
U1A	MP2520F	Non-DRAM Part
U2	Quad 2-input multiplexer	74LVC257APW
U3	NAND flash	K9K4G08Q0M. 512Mbyte NAND flash
U4	Quad 2-Input NAND Gate	74LVC00/SO
R1 ~ 6	Resistor	4.7K ohm
R7	Resistor	4.7K ohm (Do Not Install)
C1 ~ 4	Capacitor	100nF. Bypass Capacitor

Schematic



<Figure 6-2. Design guide of NAND flash with Large Block size>

6.4. NOR Flash

For using NOR boot, NOR flash is connected to Static Memory region 0 of MCU-C bus so it is controlled by static bus chip select 0(MC_nSCS0). <Figure 6-3> shows the schematic design example of NOR flash.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
MC_DATA15	Y11	E 15	MC_DATA14	AB11	E 14	MC_DATA13	W12	E 13	MC_DATA12	V12	E 12
MC_DATA11	AB12	E 11	MC_DATA10	AA12	E 10	MC_DATA9	AB13	E 9	MC_DATA8	U12	E 8
MC_DATA7	AA13	E 7	MC_DATA6	Y13	E 6	MC_DATA5	AB14	E 5	MC_DATA4	V13	E 4
MC_DATA3	AA14	E 3	MC_DATA2	AB15	E 2	MC_DATA1	W14	E 1	MC_DATA0	U13	E 0
MC_ADDR20	AA16	F 4	MC_ADDR19	AB18	F 3	MC_ADDR18	AB19	F 2	MC_ADDR17	AA17	F 1
MC_ADDR16	AA18	F 0	MC_ADDR15	Y16	G 15	MC_ADDR14	W16	G 14	MC_ADDR13	AB20	G 13
MC_ADDR12	AB21	G 12	MC_ADDR11	V17	G 11	MC_ADDR10	W17	G 10	MC_ADDR9	Y17	G 9
MC_ADDR8	Y18	G 8	MC_ADDR7	AA19	G 7	MC_ADDR6	AA20	G 6	MC_ADDR5	W19	G 5
MC_ADDR4	Y19	G 4	MC_ADDR3	Y20	G 3	MC_ADDR2	V19	G 2	MC_ADDR1	AA21	G 1
MC_nSCS0	W21	I 8	MC_nPOE	P18	J 12	MC_nPWE	U22	J 11	NOR_RnB	reserved	reserved

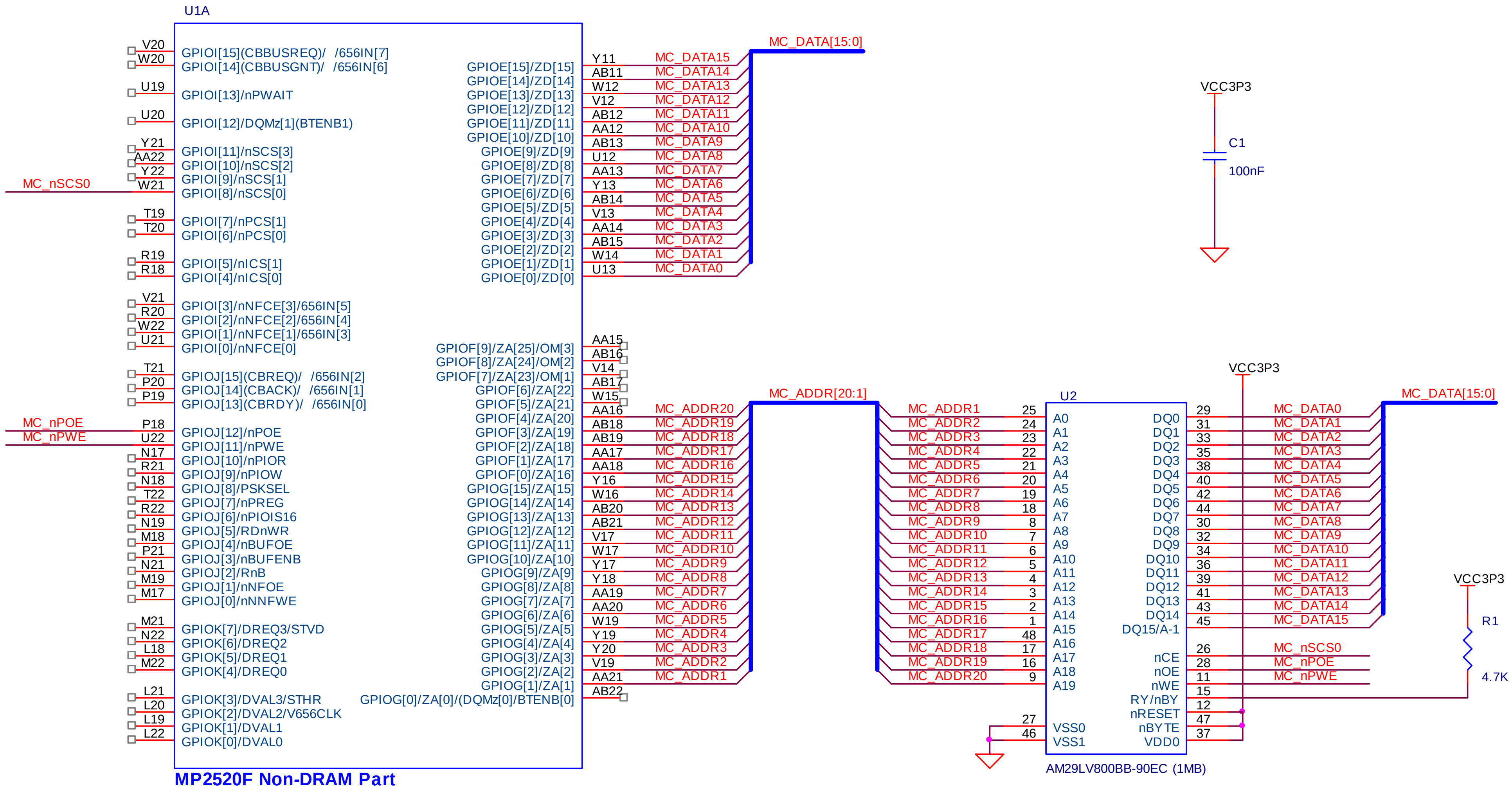
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
MC_DATA[15:0]		✓		MCU-C bus data 0 ~ 15 signal
MC_ADDR[20:1]		✓		MCU-C bus address 1 ~ 20 signal
MC_nSCS0		✓		Static Memory 0 chip select signal
MC_nPOE		✓		PCMCIA/CF/IDE/StaticMemory read enable signal
MC_nPWE		✓		PCMCIA/CF/IDE/StaticMemory write enable signal

- Component description

Component Number	Component Name	Descriptions
U1A	MP2520F	Non-DRAM part
U2	NOR Flash	AM29LV800B
R1	Resistor	4.7K ohm
C1	Capacitor	100nF. Bypass Capacitor

Schematic



<Figure 6-3. Design guide of NOR flash>

6.5. SRAM

SRAM is connected to Static Memory Region of MCU-C bus. <Figure 6-4> shows the schematic design example of SRAM with 16bit data bus width.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
MC_DATA15	Y11	E 15	MC_DATA14	AB11	E 14	MC_DATA13	W12	E 13	MC_DATA12	V12	E 12
MC_DATA11	AB12	E 11	MC_DATA10	AA12	E 10	MC_DATA9	AB13	E 9	MC_DATA8	U12	E 8
MC_DATA7	AA13	E 7	MC_DATA6	Y13	E 6	MC_DATA5	AB14	E 5	MC_DATA4	V13	E 4
MC_DATA3	AA14	E 3	MC_DATA2	AB15	E 2	MC_DATA1	W14	E 1	MC_DATA0	U13	E 0
MC_ADDR16	AA18	F 0	MC_ADDR15	Y16	G 15	MC_ADDR14	W16	G 14	MC_ADDR13	AB20	G 13
MC_ADDR12	AB21	G 12	MC_ADDR11	V17	G 11	MC_ADDR10	W17	G 10	MC_ADDR9	Y17	G 9
MC_ADDR8	Y18	G 8	MC_ADDR7	AA19	G 7	MC_ADDR6	AA20	G 6	MC_ADDR5	W19	G 5
MC_ADDR4	Y19	G 4	MC_ADDR3	Y20	G 3	MC_ADDR2	V19	G 2	MC_ADDR1	AA21	G 1
MC_nSCS0	W21	I 8	MC_nPOE	P18	J 12	MC_nPWE	U22	J 11	-	-	-

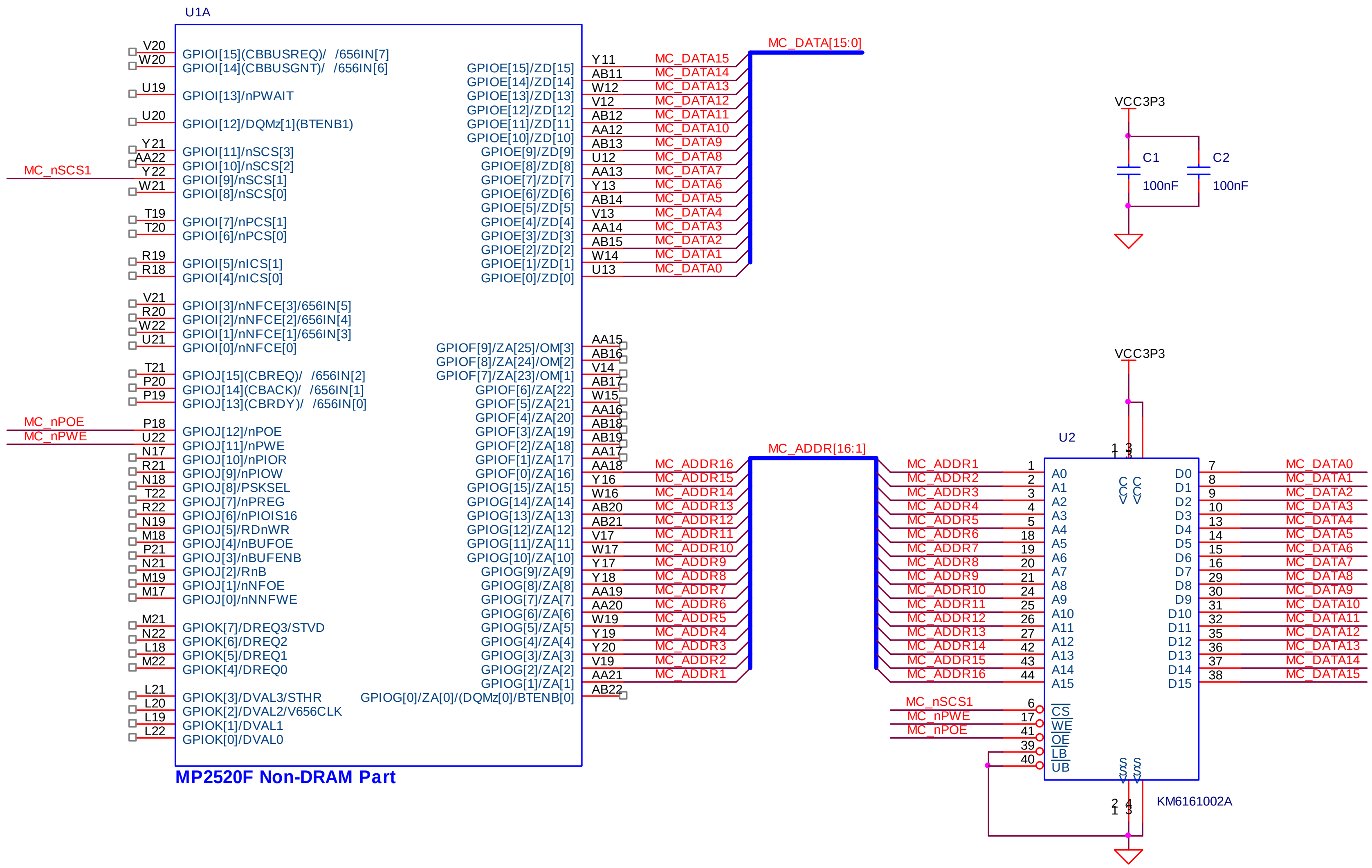
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
MC_DATA[15:0]		✓		MCU-C bus data 0 ~ 15 signal
MC_ADDR[16:1]		✓		MCU-C bus address 1 ~ 16 signal
MC_nSCS1		✓		Static Memory 1 chip select signal
MC_nPOE		✓		PCMCIA/CF/IDE/StaticMemory read enable signal
MC_nPWE		✓		PCMCIA/CF/IDE/StaticMemory write enable signal

- Component description

Component Number	Component Name	Descriptions
U1A	MP2520F	Non-DRAM part
U2	SRAM	KM6161002A
C1 ~ C2	Capacitor	100nF. Bypass Capacitor

Schematic



<Figure 6-4. Design guide of SRAM>

6.6. IDE

Originally, HDD is connected to IDE Interface of MCU-C bus and controlled by IDE controller. But in this case, only PIO mode is usable so there is a restriction of data-transfer performance. We recommend another solution which is called Static IDE to increase the data-transfer performance of IDE.

6.6.1. The method using IDE controller

This is an original method to connect HDD with MP2520F. <Figure 6-5> (A) shows the schematic design example of 3.5" HDD case and <Figure 6-5> (B) shows 1.8" & 2.5" case. For 3.5" HDD, it uses 40pin connector (20 x 2), on the other side 1.8" & 2.5" HDD uses 44pin connector(22 x 2). <Table 6-1> describes the HDD Connector Pin Description.

Pin No	Description	Pin No	Description
1	nRESET	2	GND
3	DATA7	4	DATA8
5	DATA6	6	DATA9
7	DATA5	8	DATA10
9	DATA4	10	DATA11
11	DATA3	12	DATA12
13	DATA2	14	DATA13
15	DATA1	16	DATA14
17	DATA0	18	DATA15
19	GND	20	N.C
21	DMA Request	22	GND
23	Write Enable	24	GND
25	Read Enable	26	GND
27	IORDY	28	GND
29	DMA Acknowledge	30	GND
31	IDE IRQ	32	Obsolete (see note)
33	ADDRESS1	34	N.C
35	ADDRESS0	36	ADDRESS2
37	IDE CS0	38	IDE CS1
39	-	40	GND
41	VCC5	42	VCC5
43	GND	44	N.C

<Table 6-1. HDD Connector Pin Descriptions>

Note)

- Pin 32 is defined as IOCS16 in ATA-2 Standard but IOCS16 signal is not used recent ATA Standard (ATA/ATAPI-5). MP2520F supports ATA/ATAPI-5 type HDD as well as ATA-2 type HDD.
- Pin 41, 42, 43, and 44 are only used for 1.8" & 2.5" HDD.
- Pin to Pin distance of 44 Pin connector is 2 mm, on the other side 40 Pin connector is 2.54 mm.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
MC_DATA15	Y11	E 15	MC_DATA14	AB11	E 14	MC_DATA13	W12	E 13	MC_DATA12	V12	E 12
MC_DATA11	AB12	E 11	MC_DATA10	AA12	E 10	MC_DATA9	AB13	E 9	MC_DATA8	U12	E 8
MC_DATA7	AA13	E 7	MC_DATA6	Y13	E 6	MC_DATA5	AB14	E 5	MC_DATA4	V13	E 4
MC_DATA3	AA14	E 3	MC_DATA2	AB15	E 2	MC_DATA1	W14	E 1	MC_DATA0	U13	E 0
MC_ADDR2	V19	G 2	MC_ADDR1	AA21	G 1	MC_ADDR0	AB22	G 0	MC_nPWAIT	U19	I 13
MC_nICS0	R18	I 4	MC_nICS1	R19	I 5	MC_nPIOR	N17	J 10	MC_nPIOW	R21	J 9
MC_nIOCS16	R22	J 6	MC_DREQ0	M22	K 4	MC_DACK0	L22	K 0	ATA_IRQ	V3	N 3

- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
MC_DATA[15:0]		✓		MCU-C bus data 0 ~ 15 signal
MC_ADDR[2:0]		✓		MCU-C bus address 0 ~ 2 signal
[1]MC_nPWAIT		✓		MCU-C bus WAIT signal for PCMCIA/CF/IDE/StaticMemory
MC_nICS0		✓		IDE chip select 0 signal
MC_nICS1		✓		IDE chip select 1 signal
MC_nPIOR		✓		PCMCIA/CF/IDE/StaticMemory IO Read enable signal
MC_nPIOW		✓		PCMCIA/CF/IDE/StaticMemory IO Write enable signal
MC_nIOCS16		✓		PCMCIA/CF/IDE/StaticMemory 16bit access enable signal
MC_DREQ0		✓		DMA Request 0 signal
MC_DACK0		✓		DMA Acknowledge 0 signal
ATA_IRQ	✓			IDE interrupt signal
[2]ATA_IORDY				IDE WAIT signal
[3]ETHER_IORDY				Ethernet Controller WAIT signal
[4]CF0_nWAIT				PCMCIA/CF WAIT signal
IDE_nIOCS16				IDE nIOCS16 signal
CF_nIOCS16				CF nIOCS16 signal
[5]IDE_nRESET				IDE reset signal

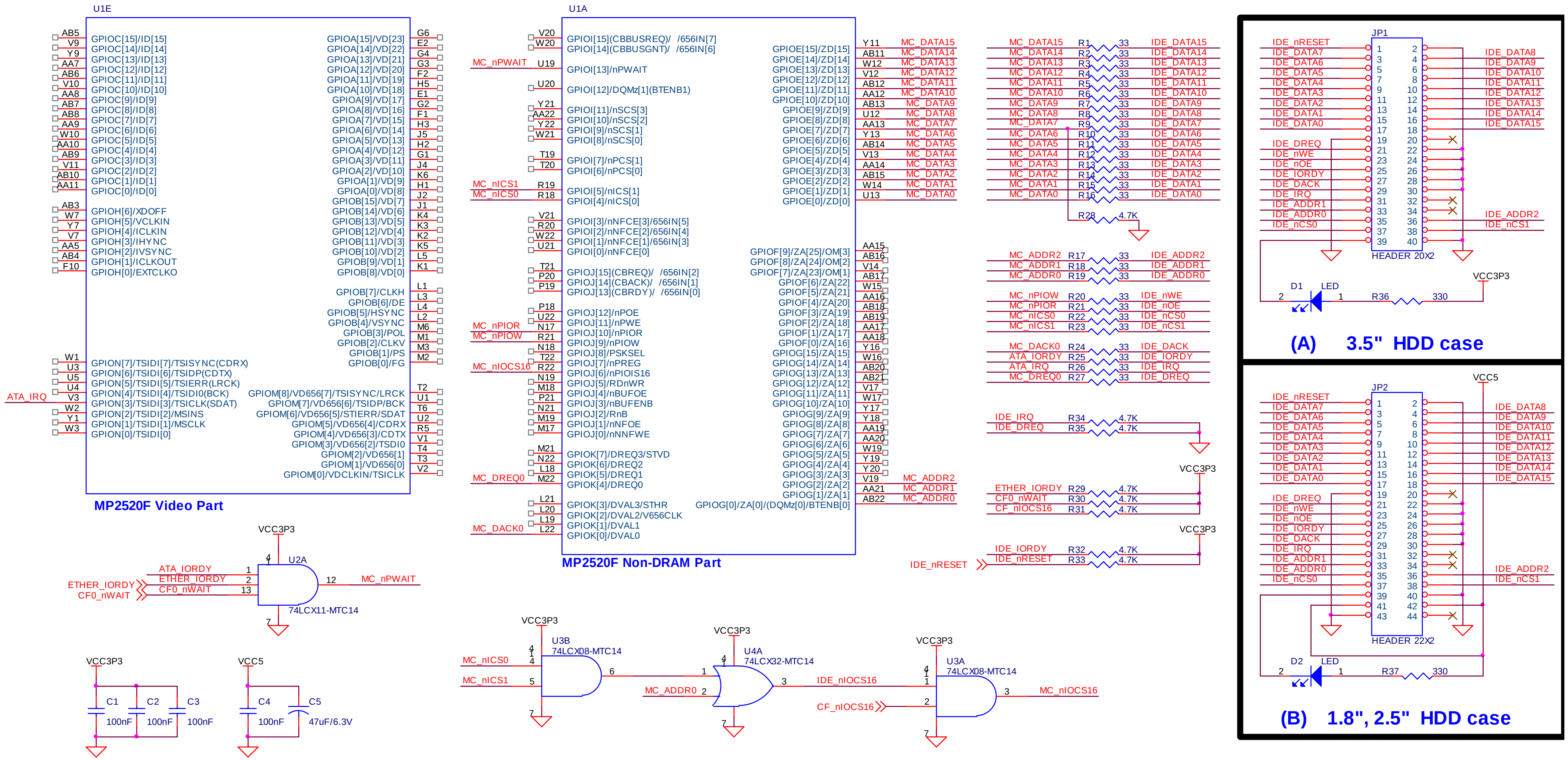
[1, 2, 3, 4] : ATA_IORDY is a wait signal from HDD, ETHER_IORDY is a wait signal from Ethernet controller, and CF0_nWAIT is a wait signal from PCMCIA/CF. These three signals are inputted to 3-Input AND Gate IC(74LCX11-MTC14) and the IC outputs MC_nPWAIT signal to MP2520F.

[5] : IDE_nRESET is a reset signal which is inputted to HDD. Basically this signal is generated from MC_nRSTOUT signal which is shown in <Figure 3-3>. **If a system developer wants to control IDE_nRESET signal by software, connect it to GPIO pin of MP2520F and then control it as GPIO_OUT function.**

- Component description

Component Number	Component Name	Descriptions
U1A	MP2520F	Non-DRAM part
U2A	74LCX11-MTC14	3 input AND gate
U3A, U3B	74LCX08-MTC14	2 input AND gate
U4A	74LCX32-MTC14	2 input OR gate
JP1	Header 20 x 2	2.54 mm. Connector for 3.5" HDD
JP2	Header 22 x 2	2 mm. Connector for 1.8" or 2.5" HDD
R1 ~ R27	Resistor	33 ohm. Damping resistors
R28 ~ R35	Resistor	4.7K ohm
R36 ~ R37	Resistor	330 ohm
D1, D2	LED	Chip LED
C1,C2,C3,C4	Capacitor	100nF. Bypass Capacitor
C5	Capacitor	47uF/6.3V

Schematic



<Figure 6-5. Design guide of HDD>

6.6.2. WORKAROUND : The method using Static Controller

For Static IDE, HDD is not connected to IDE controller but connected to Static controller of MCU-C bus. The data transfer between HDD and main memory(SDRAM) is achieved by **memory to memory DMA(not Multi-word DMA or UDMA)**, and as a result, the data transfer performance is better than using IDE controller.
To use this method, some of signals in <Figure 6-5> should be modified. <Figure 6-5> shows the modified schematic design.

HDD side	MP2520F side	
Signal List	Before changing	After changing
IDE_nWE	MC_nPIOW	MC_nPWE
IDE_nOE	MC_nPIOR	MC_nPOE
IDE_nCS0	MC_nICS0	nICS0
IDE_nCS1	MC_nICS1	nICS1
IDE_ADDR0	MC_ADDR0	MC_ADDR2
IDE_ADDR1	MC_ADDR1	MC_ADDR3
IDE_ADDR2	MC_ADDR2	MC_ADDR4

<Table 6-2. Modified Pin list>

- IDE_nWE signal is a Write Enable signal inputted to HDD. MC_nPIOW signal is a IDE/PCMCIA/CF Write Enable signal of MP2520F and originally it is connected to IDE_nWE signal. But for Static IDE, MC_nPWE signal which is generated from Static controller of MP2520F is a Write Enable signal and it is connected to IDE_nWE signal instead of MC_nPIOW signal.
- IDE_nOE signal is a Read Enable signal inputted to HDD. MC_nPIOR signal is a IDE/PCMCIA/CF Read Enable signal of MP2520F and originally it is connected to IDE_nOE signal. But for Static IDE, MC_nPOE signal which is generated from Static controller of MP2520F is a Read Enable signal and it is connected to IDE_nOE signal instead of MC_nPIOR signal.
- IDE_nCS0 signal is a IDE Chip Select 0 signal inputted to HDD. MC_nICS0 signal is a IDE Chip Select 0 signal of MP2520F and originally it is connected to IDE_nCS0 signal. But for Static IDE, nICS0 signal which is generated from the combination of Static Chip Select signal & MC_ADDR8 signal of MP2520F is a IDE Chip Select 0 signal and it is connected to IDE_nCS0 signal instead of MC_nICS0 signal.
- IDE_nCS1 signal is a IDE Chip Select 1 signal inputted to HDD. MC_nICS1 signal is a IDE Chip Select 1 signal of MP2520F and originally it is connected to IDE_nCS1 signal. But for Static IDE, nICS1 signal which is generated from the combination of Static Chip Select signal & MC_ADDR8 signal of MP2520F is a IDE Chip Select 1 signal and it is connected to IDE_nCS1 signal instead of MC_nICS1 signal.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
MC_DATA15	Y11	E 15	MC_DATA14	AB11	E 14	MC_DATA13	W12	E 13	MC_DATA12	V12	E 12
MC_DATA11	AB12	E 11	MC_DATA10	AA12	E 10	MC_DATA9	AB13	E 9	MC_DATA8	U12	E 8
MC_DATA7	AA13	E 7	MC_DATA6	Y13	E 6	MC_DATA5	AB14	E 5	MC_DATA4	V13	E 4
MC_DATA3	AA14	E 3	MC_DATA2	AB15	E 2	MC_DATA1	W14	E 1	MC_DATA0	U13	E 0
MC_ADDR8	Y18	G 8	MC_ADDR4	Y19	G 4	MC_ADDR3	Y20	G 3	MC_ADDR2	V19	G 2
MC_nSCS0	W21	I 8	MC_nPOE	P18	J 12	MC_nPWE	U22	J 11	MC_DREQ0	M22	K 4
MC_DACK0	L22	K 0	ATA_IRQ	V3	N 3	MC_nPWAIT	U19	I 13	-	-	-

- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
MC_DATA[15:0]		✓		MCU-C bus data 0 ~ 15 signal
MC_ADDR[4:2]		✓		MCU-C bus address 2 ~ 4 signal
MC_ADDR8		✓		MCU-C bus address 8 signal
[1]MC_nPWAIT		✓		MCU-C bus WAIT signal for PCMCIA/CF/IDE/StaticMemory
MC_nSCS0		✓		Static bus chip select 0 signal
MC_nICS1		✓		IDE chip select 1 signal
MC_nPOE		✓		Static bus Read enable signal
MC_nPWE		✓		Static bus Write enable signal
MC_DREQ0		✓		DMA Request 0 signal
MC_DACK0		✓		DMA Acknowledge 0 signal
ATA_IRQ	✓			IDE interrupt signal
[2]ATA_IORDY				IDE WAIT signal
[3]ETHER_IORDY				Ethernet Controller WAIT signal
[4]CF0_nWAIT				PCMCIA/CF WAIT signal
[5]IDE_nRESET				IDE reset signal

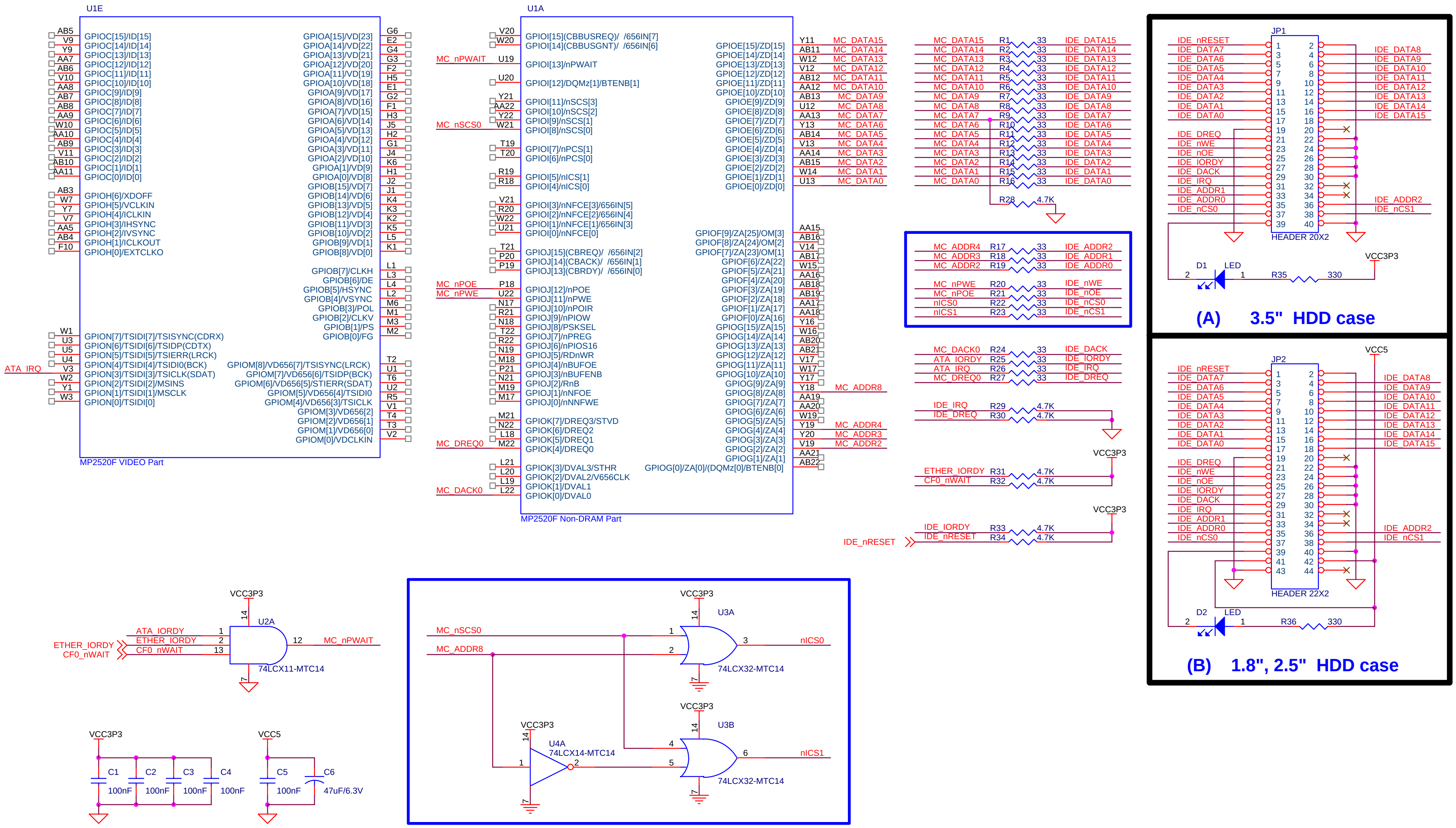
[1, 2, 3, 4] : ATA_IORDY is a wait signal from HDD. ETHER_IORDY is a wait signal from Ethernet controller. And CF0_nWAIT is a wait signal from PCMCIA/CF. If CF & Ethernet controller as well as HDD are used for user' system, these three signals are inputted to 3-Input AND Gate IC(74LCX11-MTC14) and the IC outputs MC_nPWAIT signal to MP2520F.

[5] : IDE_nRESET is a reset signal which is inputted to HDD. Basically this signal is generated from MC_nRSTOUT signal(this signal is generated from V4 pin of MP2520F). **If a system developer wants to control IDE_nRESET signal by software, connect it to GPIO pin of MP2520F and then control it as GPIO_OUT function.**

- Component description

Component Number	Component Name	Descriptions
U1A	MP2520F	Non-DRAM part
U1E	MP2520F	Video part
U2A	74LCX11-MTC14	3 input AND gate
U3A, U3B	74LCX32-MTC14	2 input OR gate
U4A	74LCX14-MTC14	Inverter
JP1	Header 20 x 2	2.54 mm. Connector for 3.5" HDD
JP2	Header 22 x 2	2 mm. Connector for 1.8" or 2.5" HDD
R1 ~ R27	Resistor	33 ohm. Damping resistors
R28 ~ R34	Resistor	4.7K ohm
R35 ~ R36	Resistor	330 ohm
D1, D2	LED	Chip LED
C1,C2,C3,C4,C5	Capacitor	100nF. Bypass Capacitor
C6	Capacitor	47uF/6.3V

Schematic



<Figure -6-6. Design example of IDE interface in case of Static controller>

6.7. PCMCIA/CF

PCMCIA/CF is connected to Card Interface of MCU-C bus and <Figure 6-6> shows the schematic design example of CF. CF has 11bit address, on the other side PCMCIA has 26bit address and one more power source VPP while the other signals are the same. PCMCIA/CF needs many GPIO-controlled signals besides of Card Interface control signals. The lists of GPIO-controlled signals are as follows :
GPIO_P0EN, GPIO_P1EN, GPIO_P0VCC5EN, GPIO_P0VCC3EN, CF0_nRST, CF0_RDYnBSY1, CF0_S1_VS1, CF0_S1_VS2, CF0_nCD, CF0_FAIL

<Figure 6-7> shows that signals are connected to SPDIF, SSP, and UART1 pin. If a system developer wants to use SPDIF(or SSP or UART1) to his system board, he should change these GPIO-controlled signals' pin position to another GPIO pins which are not used in his system.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
MC_DATA15	Y11	E 15	MC_DATA14	AB11	E 14	MC_DATA13	W12	E 13	MC_DATA12	V12	E 12
MC_DATA11	AB12	E 11	MC_DATA10	AA12	E 10	MC_DATA9	AB13	E 9	MC_DATA8	U12	E 8
MC_DATA7	AA13	E 7	MC_DATA6	Y13	E 6	MC_DATA5	AB14	E 5	MC_DATA4	V13	E 4
MC_DATA3	AA14	E 3	MC_DATA2	AB15	E 2	MC_DATA1	W14	E 1	MC_DATA0	U13	E 0
MC_ADDR10	W17	G 10	MC_ADDR9	Y17	G 9	MC_ADDR8	Y18	G 8	MC_ADDR7	AA19	G 7
MC_ADDR6	AA20	G 6	MC_ADDR5	W19	G 5	MC_ADDR4	Y19	G 4	MC_ADDR3	Y20	G 3
MC_ADDR2	V19	G 2	MC_ADDR1	AA21	G 1	MC_ADDR0	AB22	G 0	MC_nPWAIT	U19	I 13
MC_nPCS1	T19	I 7	MC_nPCS0	T20	I 6	MC_nPOE	P18	J 12	MC_nPWE	U22	J 11
MC_nPIOR	N17	J 10	MC_nPIOW	R21	J 9	MC_PSKSEL	N18	J 8	MC_nPREG	T22	J 7
MC_nIOCS16	R22	J 6	MC_RDnWR	N19	J 5	MC_nBUFENB	P21	J 3	GPIO_P0EN	AA17	F 1
GPIO_P1EN	AA18	F 0	GPIO_P0VCC5EN	A5	O 5	GPIO_P0VCC3EN	A4	O 4	CF0_nRST	B8	D 9
CF0_RDYnBSY1	E10	D 8	CF0_S1_VS1	A7	D 7	CF0_S1_VS2	B7	D 6	CF0_nCD	A6	D 5
CF0_FAIL	N1	L 14	-	-	-	-	-	-	-	-	-

- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
MC_DATA[15:0]		✓		MCU-C bus data 0 ~ 15 signal
MC_ADDR[10:0]		✓		MCU-C bus address 0 ~ 10 signal
[1]MC_nPWAIT		✓		MCU-C bus WAIT signal for PCMCIA/CF/IDE/StaticMemory
MC_nPCS1		✓		PCMCIA/CF Chip Select signal. This is used to enable the low byte.
MC_nPCS0		✓		PCMCIA/CF Chip Select signal. This is used to enable the high byte.
MC_nPOE		✓		PCMCIA/CF/IDE/StaticMemory Read enable signal
MC_nPWE		✓		PCMCIA/CF/IDE/StaticMemory Write enable signal
MC_nPIOR		✓		PCMCIA/CF/IDE/StaticMemory IO Read enable signal
MC_nPIOW		✓		PCMCIA/CF/IDE/StaticMemory IO Write enable signal

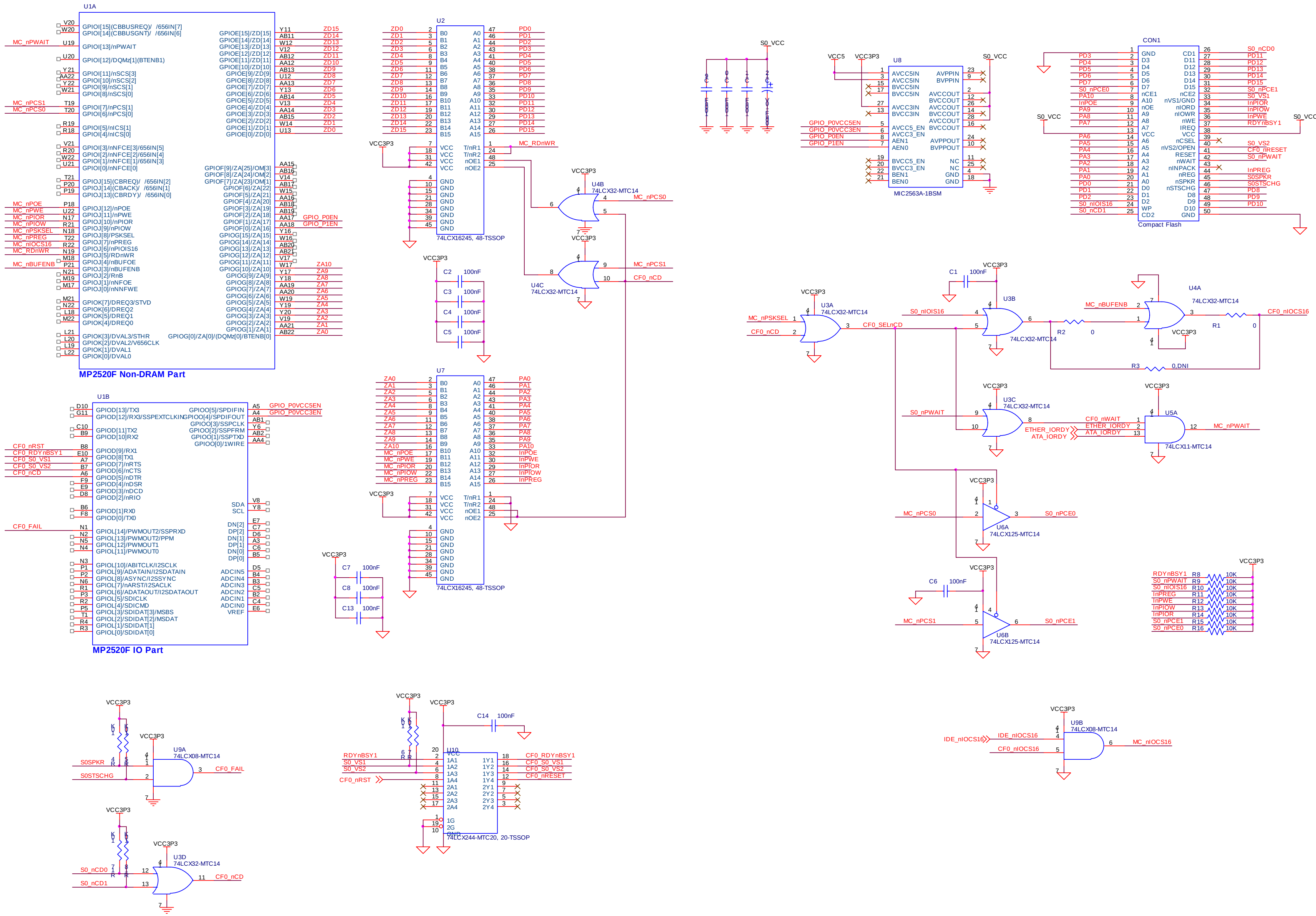
MC_PSKSEL		✓		PCMCIA/CF Socket Select signal. 0 : PCMCIA/CF region 0, 1 : PCMCIA/CF region 1
MC_nPREG		✓		PCMCIA/CF Register Select signal.
MC_nIOCS16		✓		PCMCIA/CF/IDE/StaticMemory 16bit access enable signal
MC_RDnWR		✓		Buffer direction control signal. 0 : Write, 1 : Read
MC_nBUFENB		✓		Buffer enable signal. Intended for use as a buffer enable for data bus buffering logic.
GPIO_P0EN	✓			Enable(EN0) signal for PCMCIA/CF Power control IC
GPIO_P1EN	✓			Enable(EN1) signal for PCMCIA/CF Power control IC
GPIO_P0VCC5EN	✓			Enable(VCC5EN) signal for PCMCIA/CF Power control IC
GPIO_P0VCC3EN	✓			Enable(VCC3EN) signal for PCMCIA/CF Power control IC
CF0_nRST	✓			PCMCIA/CF Reset signal
CF0_RDYnBSY1	✓			PCMCIA/CF Ready & Busy signal
CF0_S1_VS1	✓			PCMCIA/CF Voltage Sense 1 signal
CF0_S1_VS2	✓			PCMCIA/CF Voltage Sense 2 signal
CF0_nCD	✓			PCMCIA/CF Card detect signal
CF0_FAIL	✓			PCMCIA/CF Fail signal
[2]ATA_IORDY				IDE WAIT signal
[3]ETHER_IORDY				Ethernet Controller WAIT signal
[4]CF0_nWAIT				PCMCIA/CF WAIT signal
IDE_nIOCS16				IDE nIOCS16 signal

[1, 2, 3, 4] : ATA_IORDY is a wait signal from HDD. ETHER_IORDY is a wait signal from Ethernet controller. And CF0_nWAIT is a wait signal from PCMCIA/CF. These three signals are inputted to 3-Input AND Gate IC(74LCX11-MTC14) and the IC outputs MC_nPWAIT signal to MP2520F.

- Component description

Component Number	Component Name	Descriptions
U1A	MP2520F	Non-DRAM Part
U1B	MP2520F	IO Part
U2, U7	74LCX16245	16-bit bidirectional Transceiver
U3, U4	74LCX32	Quad 2-input OR Gate
U9	74LCX08	Quad 2-input AND Gate
U10	74LCX244	Buffer
U6	74LCX125	Quad Buffer
U8	MIC2563A-1BSM	Dual Slot PCMCIA/CF Power control IC
U5	74LCX11	Triple 3-input AND Gate
CON1	Connector	Connector for CF card
C1 ~ C11,C13, C14	Capacitor	100nF. Bypass capacitor
C12	Capacitor	100uF. Tantalum capacitor. 10V
R1, R2	Resistor	0 ohm
R3	Resistor	0 ohm (Do Not Install)
R4 ~ R18	Resistor	10K ohm. Pull-up resistor

Schematic



<Figure 6-7. Design guide of CF>

7. Video OUT

The display controller of MP2520F creates video signals and data to connect to external display devices. Such external devices are TFT LCD, Video Encoder, DAC, etc. The connection to external devices requires various kinds of Synchronized Signals and Data Formats. The supported formats are as follows :

- Supports Composite Sync
- Supports CCIR601 video format
- Supports CCIR656 video format
- Supports YCbCr 3x8/3x10 mode (CCUR656 like)
- Supports RGB
- Supports Multiplexed RGB(MRGB)

For more information about the Video Encoder interface, refer to the application note for the Video Encoder interface.

7.1. Connection example of 18bit LCD module

LCD module is connected to Video out port of MP2520F and <Figure 7-1> shows the schematic design example of LCD module interface.

- LCD module part name : TX18D11VM1CAA [HITACHI]
- Video data format : RGB666

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
VD17	E1	A[9]	VD16	G2	A[8]	VD15	F1	A[7]	VD14	H3	A[6]
VD13	J5	A[5]	VD12	H2	A[4]	VD11	G1	A[3]	VD10	J4	A[2]
VD9	K6	A[1]	VD8	H1	A[0]	VD7	J2	B[15]	VD6	J1	B[14]
VD5	K4	B[13]	VD4	K3	B[12]	VD3	K2	B[11]	VD2	K5	B[10]
VD1	L5	B[9]	VD0	K1	B[8]	HSYNC	L4	B[5]	VSYNC	L2	B[4]
CLKHO	L1	B[7]	ACTIVE	L3	B[6]	BL_ON_OFF	W3	N[0]	LCD_BR_CON	N5	L[12]

- Signal description

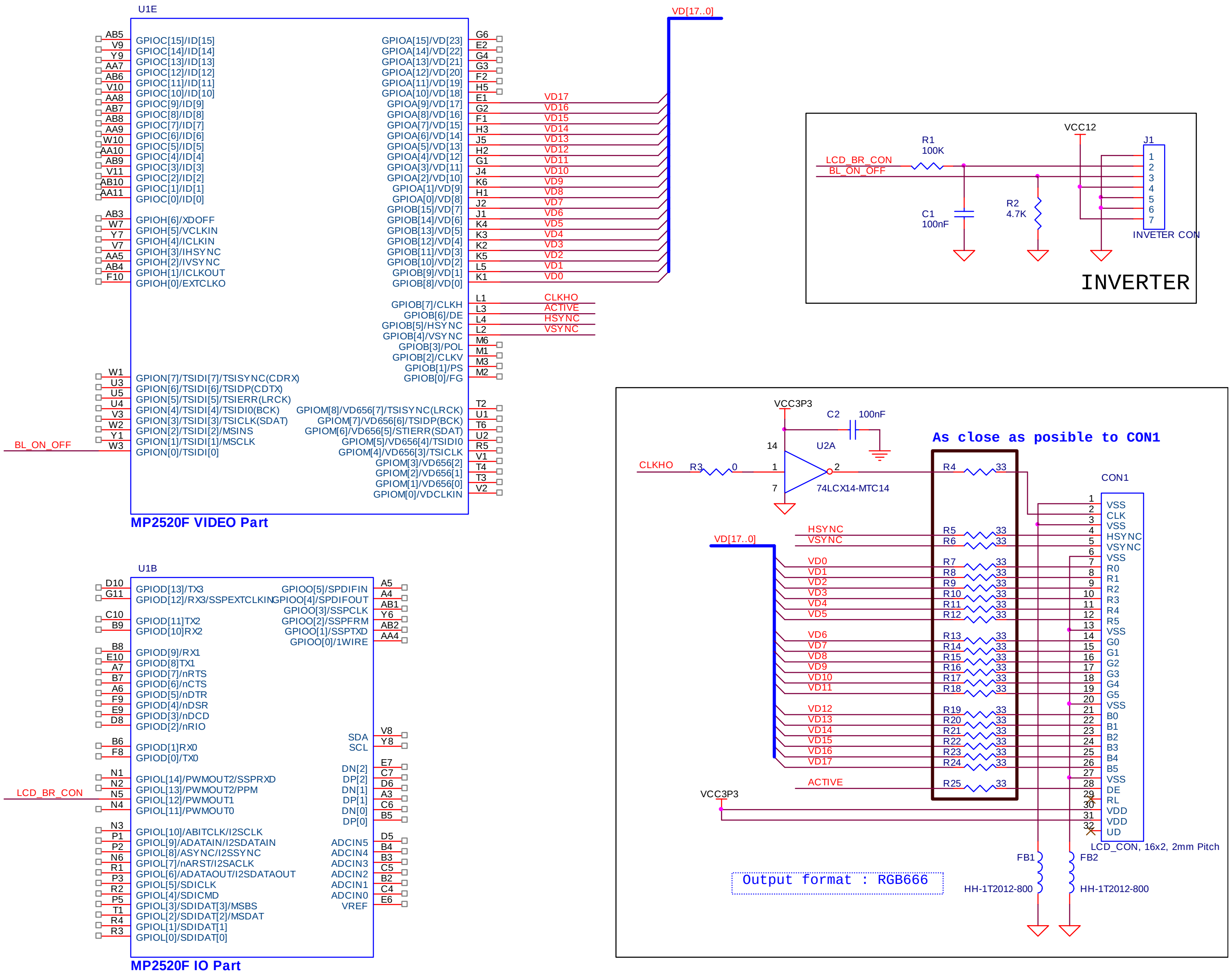
Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
VD[17:0]		✓	-	Video output data
HSYNC		✓	-	Horizontal Sync
VSYNC		✓	-	Vertical Sync

CLKHO		✓		Pixel clock output
ACTIVE		✓		Data enable
BL_ON_OFF	✓			Back Light ON-OFF
LCD_BR_CON		✓		LCD Bright Control (PWMOUT1)

- Component description

Component Number	Component Name	Descriptions
U1E	MP2520F	VIDEO Part
U1B	MP2520F	IO Part
U2	74LCX14-MTC14	Low Voltage Hex Inverter with 5V Tolerant Schmitt Trigger Inputs
CON1	Connector	LCD Connector
C1,C2	Capacitor	100nF
FB1,FB2	Bead	HH-1T2012-800
J1	Connector	INVETER CON
R1	Resistor	100K ohm
R2	Resistor	4.7K ohm
R3	Resistor	0 ohm
R4 ~ R25	Resistor	33 ohm

Schematic



<Figure 7-1. Design guide of LCD Module interface>

8. Video IN

MP2520F can receive 6 types of Input format and <Table 8-1> shows the input format list.

Video Input format
YCrCb 4:2:2 16bit format (601)
ITU-R BT 656(CCIR656) 8bit
ITU-R BT 656(CCIR656) Like 8bit
8/10 bit Bayer RGB
8/10 bit Serial RGB
8 bit parallel RGB format

<Table 8-1. Video Input format list>

There are two video input ports in MP2520F. GPIOC port is a normal video input port and GPIOM port is a CCIR656-only port. MP2520F uses GPIOC port as an input port from Camera and GPIOM port as an input port from Video Decoder.

For more information about Camera interface & Video decoder interface, refer to the application note for the Camera interface & Video decoder interface

9. RTC

MP2520F has a RTC block but there is power-supply problem. To operate the RTC block of MP2520F, VDDAlive1 & VDDAlive2 power as well as VDDRTC power must be supplied. So it consumes more power than external RTC. MagicEyes recommend the using external RTC to customers.

<Figure 9-1> shows the schematic design example of External RTC which is controlled by I2C. MagicEyes recommends GPIO I2C to customers. (refer to <Caution 14-1>)
The part name of External RTC is DS1672S of Maxim semiconductor.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
GPIO_SCL	M3	B[1]	GPIO_SDA	M2	B[0]	SCL	V8	-	SDA	Y8	-

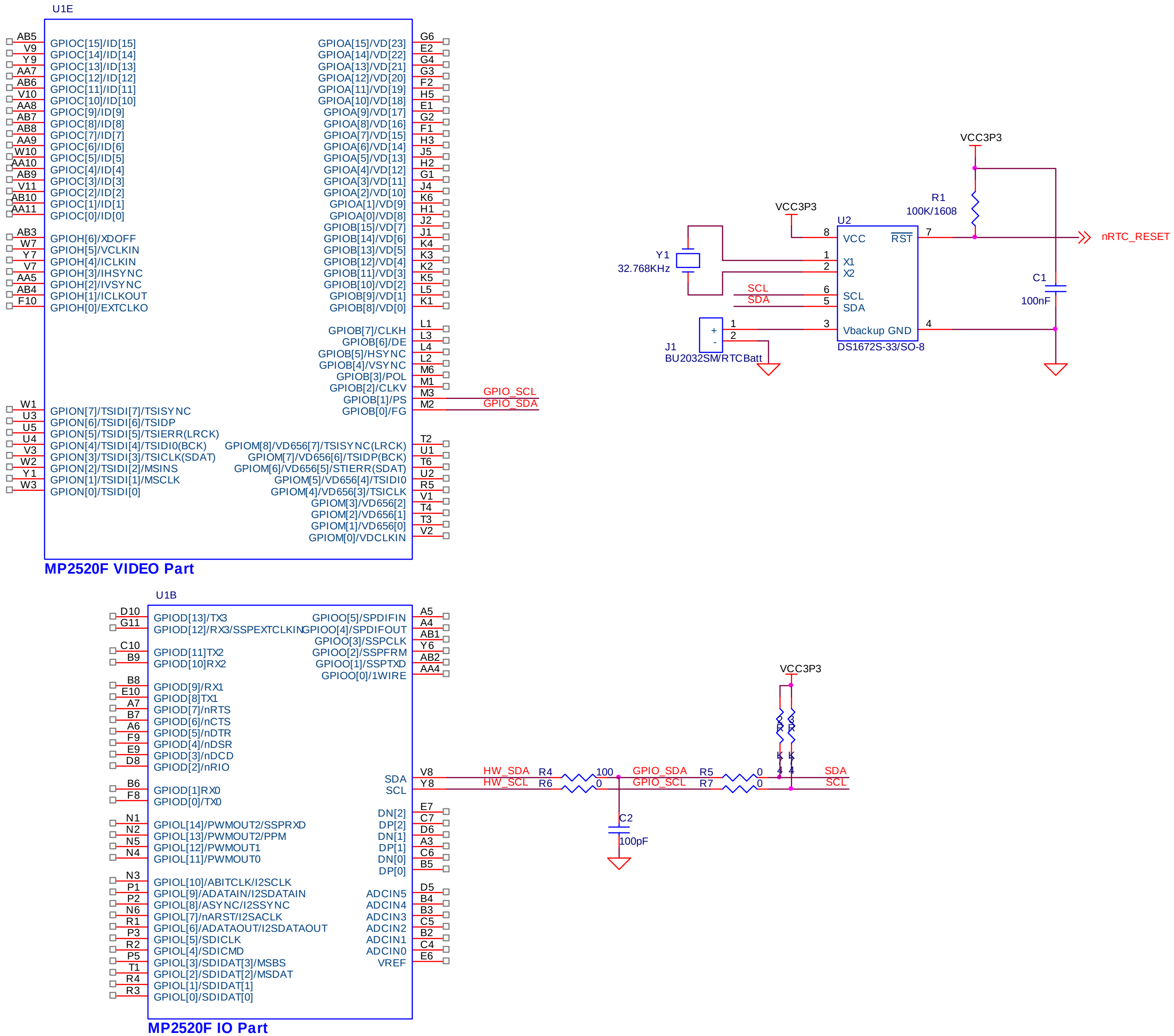
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
GPIO_SCL	✓			GPIO I2C Serial clock line
GPIO_SDA	✓			GPIO I2C Serial data line
SCL				HW I2C Serial clock line
SDA				HW I2C Serial data line
nRTC_RESET	✓			RTC Reset out signal

- Component description

Component Number	Component Name	Descriptions
U1E	MP2520F	VIDEO Part
U1B	MP2520F	IO Part
U2	DS1672S-33/SO-8	Low-Voltage Serial Timekeeping Chip
C1	Capacitor	100nF
C2	Capacitor	100pF
J1	Connector	BU2032SM/RTCBatt
R2,R3	Resistor	4.7K ohm
R1	Resistor	100K ohm
R4,R6,R7	Resistor	0 ohm
R5	Resistor	100 ohm
Y2	Crystal	32.768KHz

Schematic



<Figure 9-1. Design guide of External RTC>

10. UART

10.1. Normal Function (Tx/Rx only)

MP2520F has four UART channels and each channel operates independently. <Figure 10-1> shows the schematic design example of UART channel 0 case.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
UART_TX	F8	D[0]	UART_RX	B6	D[1]						

- Signal description

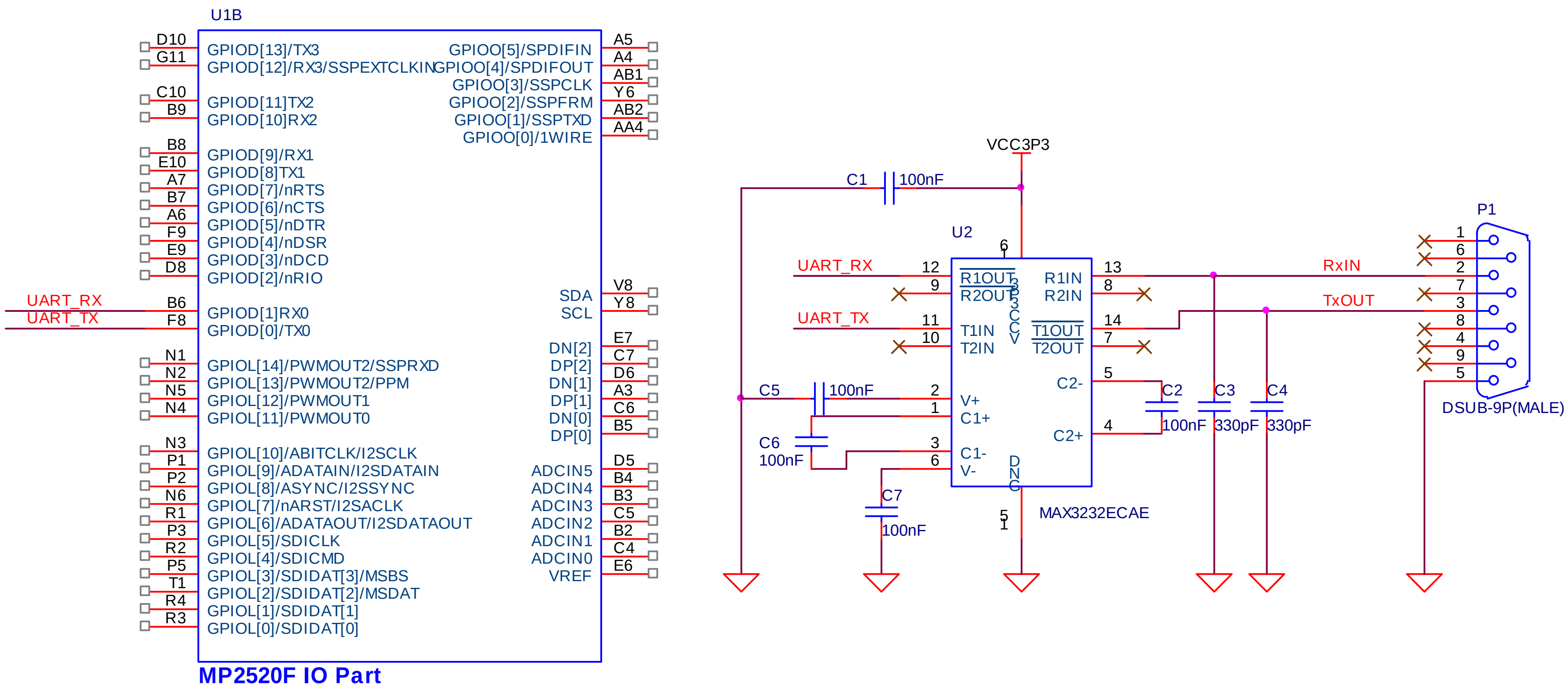
Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
[1]UART_RX		✓		UART Receive Pin
[2]UART_TX		✓		UART Transmit Pin

[1, 2] : UART_RX & UART_TX signal can be connected to UART channel 0 port, 1 port, 2 port, and 3 port.

- Component description

Component Number	Component Name	Descriptions
U1B	MP2520F	I/O Part
U2	Transceiver	MAX3232(Maxim)
P1	D-SUB 9Pin(MALE)	DS02-09-M-R
C1,C2,C5,C6,C7	Capacitors	100nF
C3,C4	Capacitors	330pF

Schematic



<Figure10-1.Desing guide of UART Normal>

10.2. Full Modem Function

UART ch1 and ch2 has full modem control function but it is not possible to use it at the same time, so full modem control function should be used at only one port at a time. <Figure 10-2> shows the schematic design example of UART channel 1 case.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
UART_RTS	A7	D[7]	UART_CTS	B7	D[6]	UART_DTR	A6	D[5]	UART_DSR	F9	D[4]
UART_CD	E9	D[3]	UART_RI	D8	D[2]	UART_RX	B8	D[9]	UART_TX	E10	D[8]

- Signal description

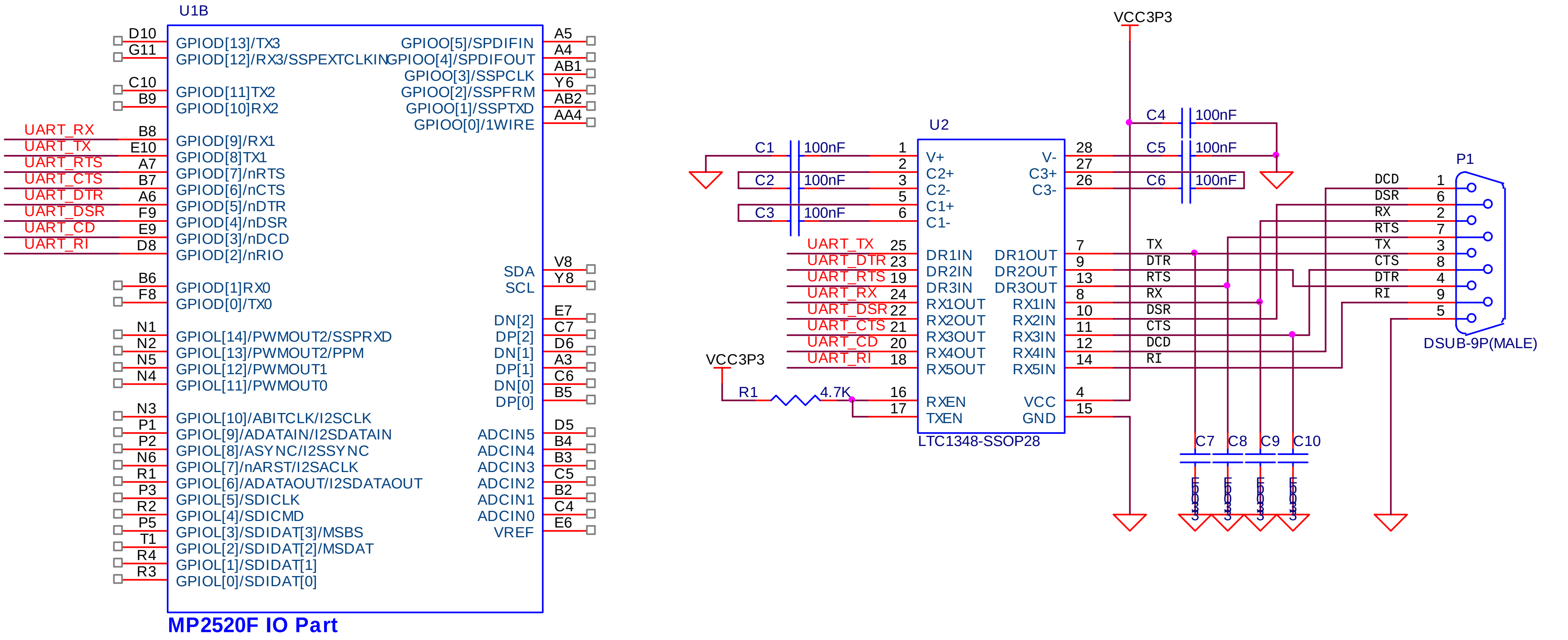
Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
[1]UART_RTS		✓		UART1 Ready-To-Send Pin
[2]UART_CTS		✓		UART1 Clear-To-Send Pin
[3]UART_DTR		✓		UART1 Data-Terminal-Ready Pin
[4]UART_DSR		✓		UART1 Data-Set-Ready Pin
[5]UART_CD		✓		UART1 Data-Carrier-Detect Pin
[6]UART_RI		✓		UART1 Ring Indication Pin
[7]UART_RX		✓		UART1 Receive Pin
[8]UART_TX		✓		UART1 Transmit Pin

1, 2, 3, 4, 5, 6, 7, 8 : UART_RTS, UART_CTS, UART_DTR, UART_DSR, UART_CD, UART_RI, UART_RX and UART_TX signal can be connected to UART channel 2 port as well as channel 1 port.

- Component description

Component Number	Component Name	Descriptions
U1	MP2520F	I/O Part
U2	Transceiver	LTC1348-SSOP28(Linear)
P1	D-SUB 9Pin(MALE)	DS02-09-M-R
R1	Resistor	4.7K ohm
C1,C2,C3,C4,C5,C6	Capacitors	100nF
C7,C8,C9,C10	Capacitors	330pF

Schematic



<Figure10-2.Desing guide of UART Full Modem>

11. PWM

MP2520F has four Pulse Width Modulator(PWM) ports. Each PWM operates respectively and is controlled by its own register set. <Figure 11-1> shows the schematic design example of PWM connection.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
LCD_BR_CON	N5	L[12]									

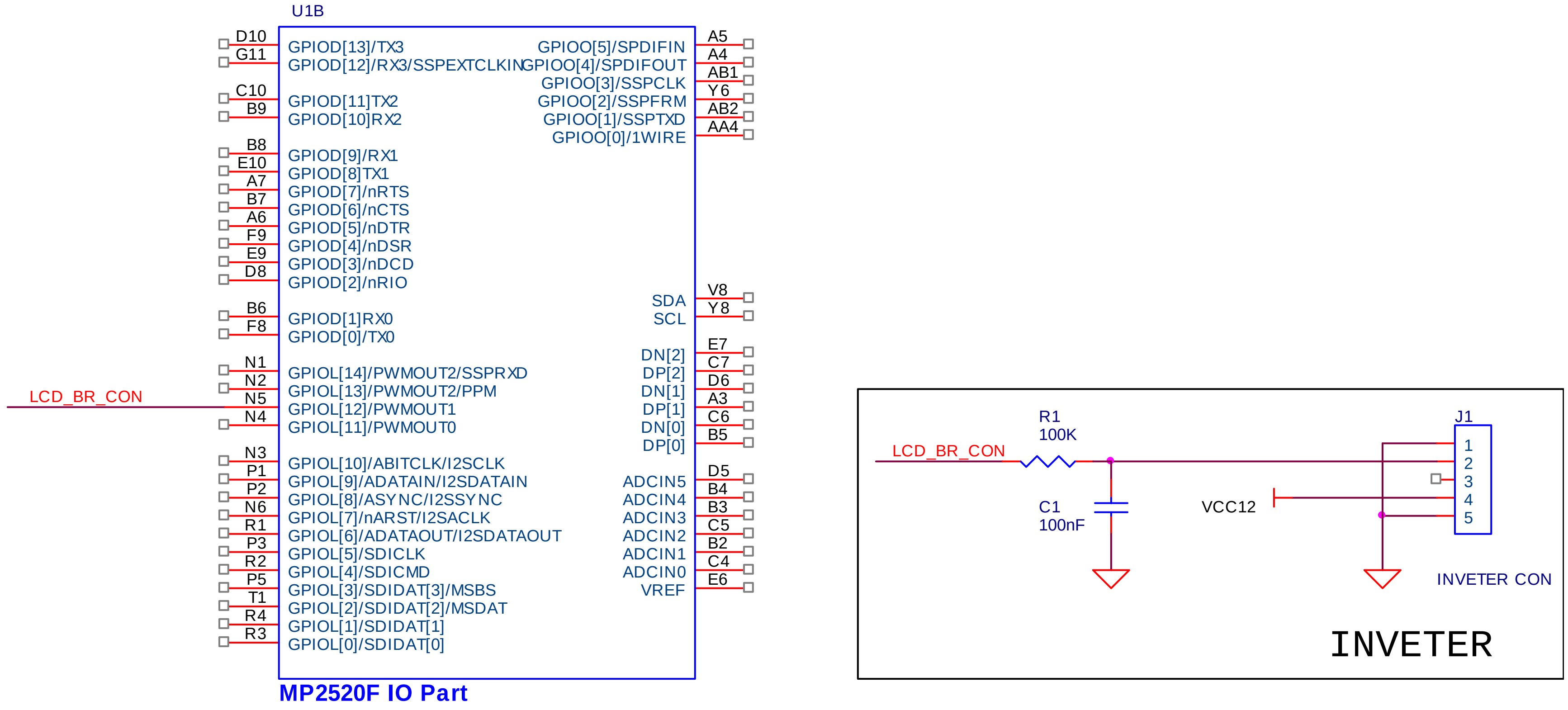
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
LCD_BR_CON		✓		LCD Bright Control (PWMOUT1)

- Component description

Component Number	Component Name	Descriptions
U1B	MP2520F	IO Part
C1	Capacitor	100nF
J1	Connector	INVERTER CON
R1	Resistor	100K

Schematic



<Figure 11-1. Design guide of PWM>

12. PPM

The Pulse Period Measurement (PPM) measures the period of the high level and the low level of a 1-bit Signal entered from the outside. By using this PPM block, user can receive the input signal from remote controller. <Figure 12-1> shows the schematic design example of PPM connection.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
IR_REMO	N2	L[13]									

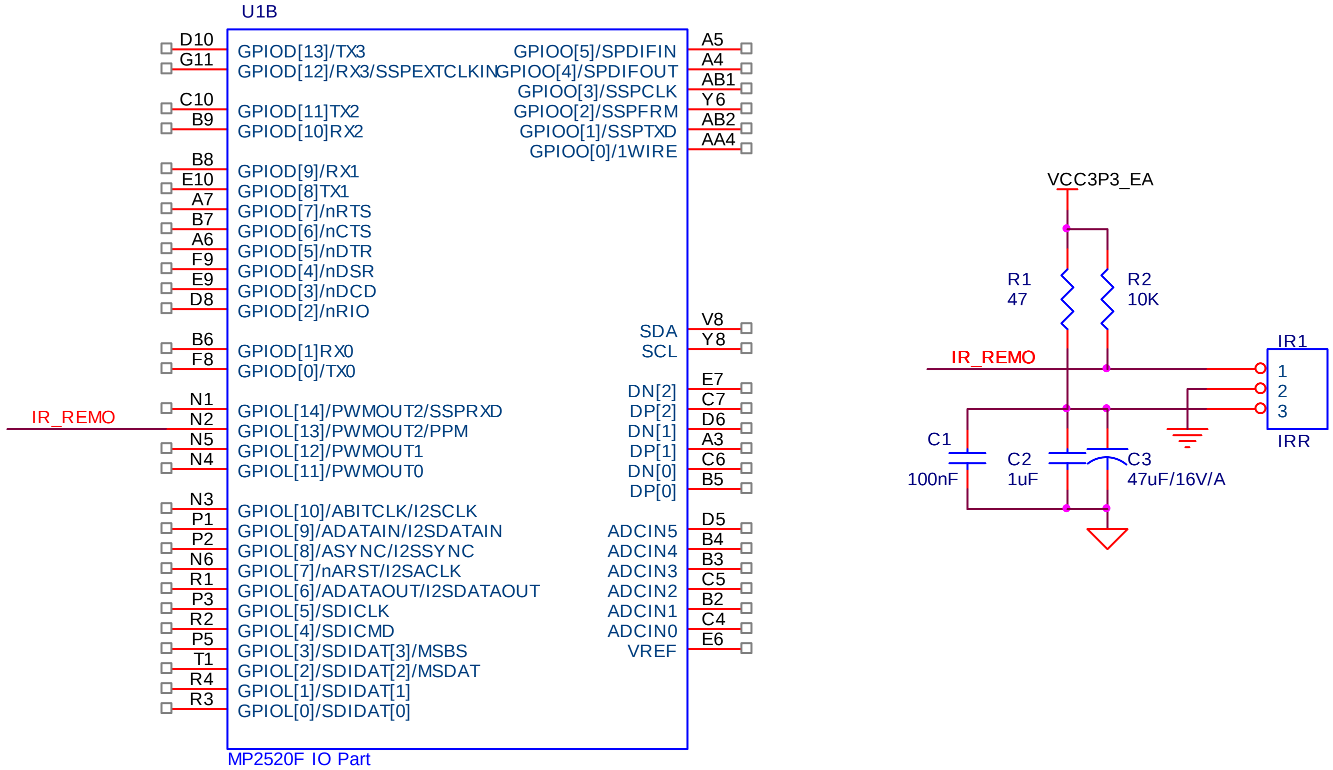
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
IR_REMO			✓	Pulse Period Measurement (PPM) interface Pin

- Component description

Component Number	Component Name	Descriptions
U1B	MP2520F	IO Part
IR1	ROM-N338LM	Receiver Module for infrared remote control
C1	Capacitor	100nF
C2	Capacitor	1uF
C3	Capacitor	47uF/16V/A
R1	Resistor	47 ohm
R2	Resistor	10K ohm

Schematic



<Figure 12-1. Design guide of PPM>

13. USB

13.1. USB 1.1 Host & Device

MP2520F supports three ports USB 1.1 host interface. Two of them have USB Device function as well as USB Host function. <Figure 13-1> shows the schematic design example of USB 1.1 Host & Device connection.

- USB port 1 only has USB host function
- USB port 0 and 2 have USB device function as well as USB host function

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
USB_HD1_M	E7		USB_HD1_P	C7		USB_H_M	D6		USB_H_P	A3	
USB_HD0_M	C6		USB_HD0_P	B5		USB_DETECT	AB3	H6	USBD_CON	E9	D3
USB_P0_EN	P19	J13									

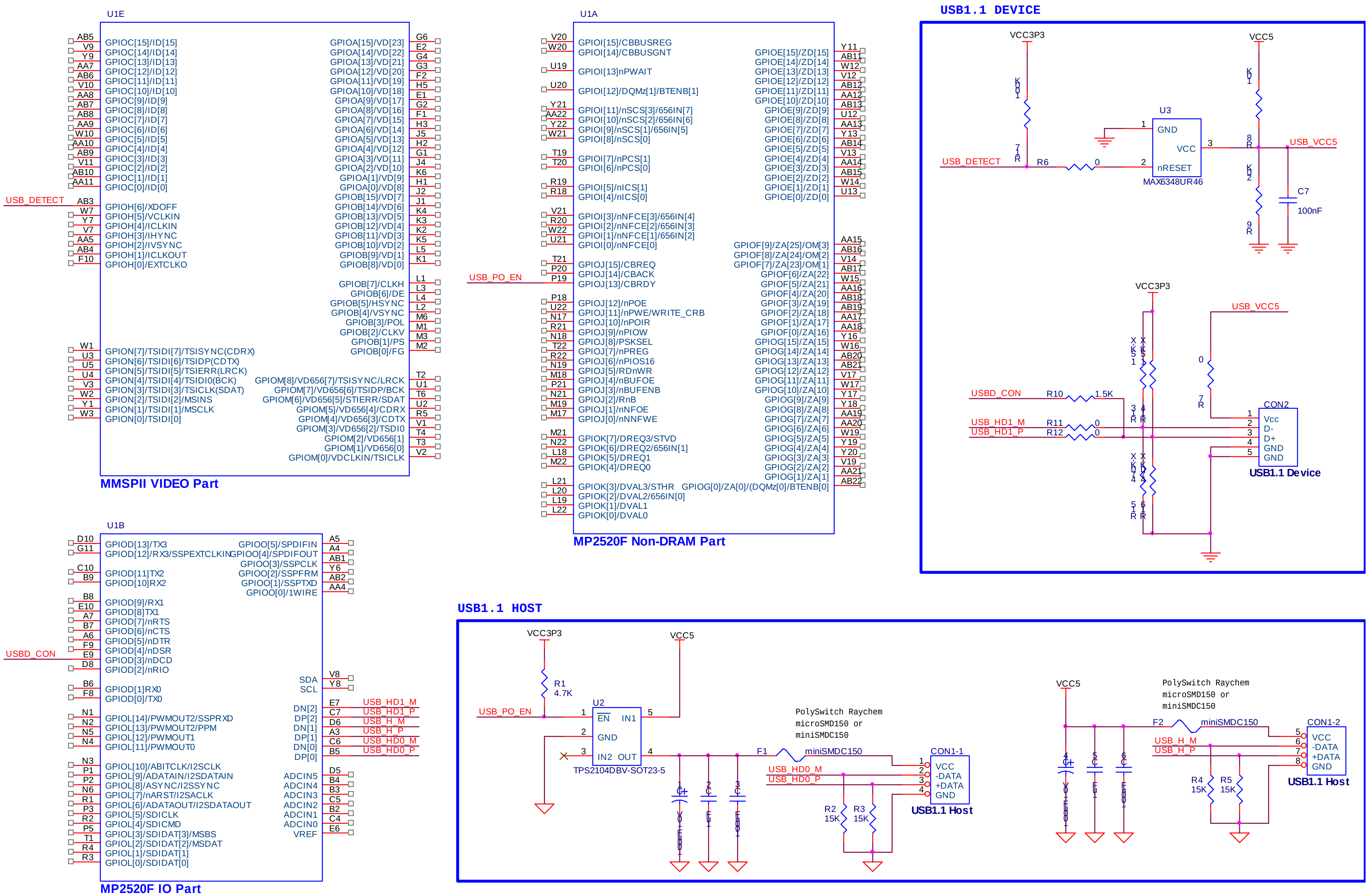
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
USB_HD1_M				**USB Host/Device (-)**
USB_HD1_P				**USB Host/Device (+)**
USB_H_M				**USB only Host (-)**
USB_H_P				**USB only Host (+)**
USB_HD0_M				**USB Host/Device (-)**
USB_HD0_P				**USB Host/Device (+)**
USB_DETECT	✔			**PC(USB1.1 Host) detection signal. This signal is used to detect whether PC is connected or not. Normal status(not connected status) value is LOW. If PC is connected to USB1.1 Device port of MP2520F, this signal value is changed from LOW to HIGH. This is a GPIO IN signal**
USBD_CON	✔			**Control the USB1.1 Device Plus signal. This signal is used to control the Pull-up & Pull-down value of USB1.1 Device Plus signal. If PC is not connected to USB1.1 Device port of MP2520F, program should make this signal LOW. If PC is connected, program initializes the device driver of USB1.1 Device port and then should make this signal HIGH. This is a GPIO OUT signal.**
USB_P0_EN	✔			Control the power of USB1.1 Host port. 0 : enable the power of USB1.1 Host 1 : disable the power of USB1.1 Host This is a GPIO OUT signal

- Component description

Component Number	Component Name	Descriptions
U1B	MP2520F	IO Part
U2	TPS2104DBV	Power Distribution Switch
CON1	Connector	USB Port A type
CON2	Connector	USB Port B type
C1,C4	Capacitor	100uF/10V
C2,C5,C7	Capacitor	1uF
C3,C6,C8	Capacitor	100nF
F1,F2	Fuse	miniSMDC150
R1	Resistor	4.7K ohm
R2,R3,R4,R5	Resistor	15K ohm
R6	Resistor	0 ohm
R7	Resistor	0 ohm
R8	Resistor	10K ohm
R9	Resistor	20K ohm
R10	Resistor	1.5K ohm
R11, R12	Resistor	0 ohm
R13, R14	Resistor	1.5K ohm, DNI
R15, R16	Resistor	470K ohm, DNI
R17	Resistor	100K ohm
C7	Capacitor	100nF
U3	MAX6348UR46	Reset generator with voltage comparator

Schematic



<Figure 13-1. Design guide of USB1.1 Host & Device>

14. I2C

<Figure 14-1> shows the schematic design example of I2C interface.

<Caution 14-1>
System developer can use GPIO I2C as well as Hardware I2C. **We recommend GPIO I2C to system developer, because Hardware I2C needs extra Resistor & Capacitor circuit to set the exact setup hold time.**

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
GPIO_SCL	M3	B[1]	GPIO_SDA	M2	B[0]	SCL	V8	-	SDA	Y8	-

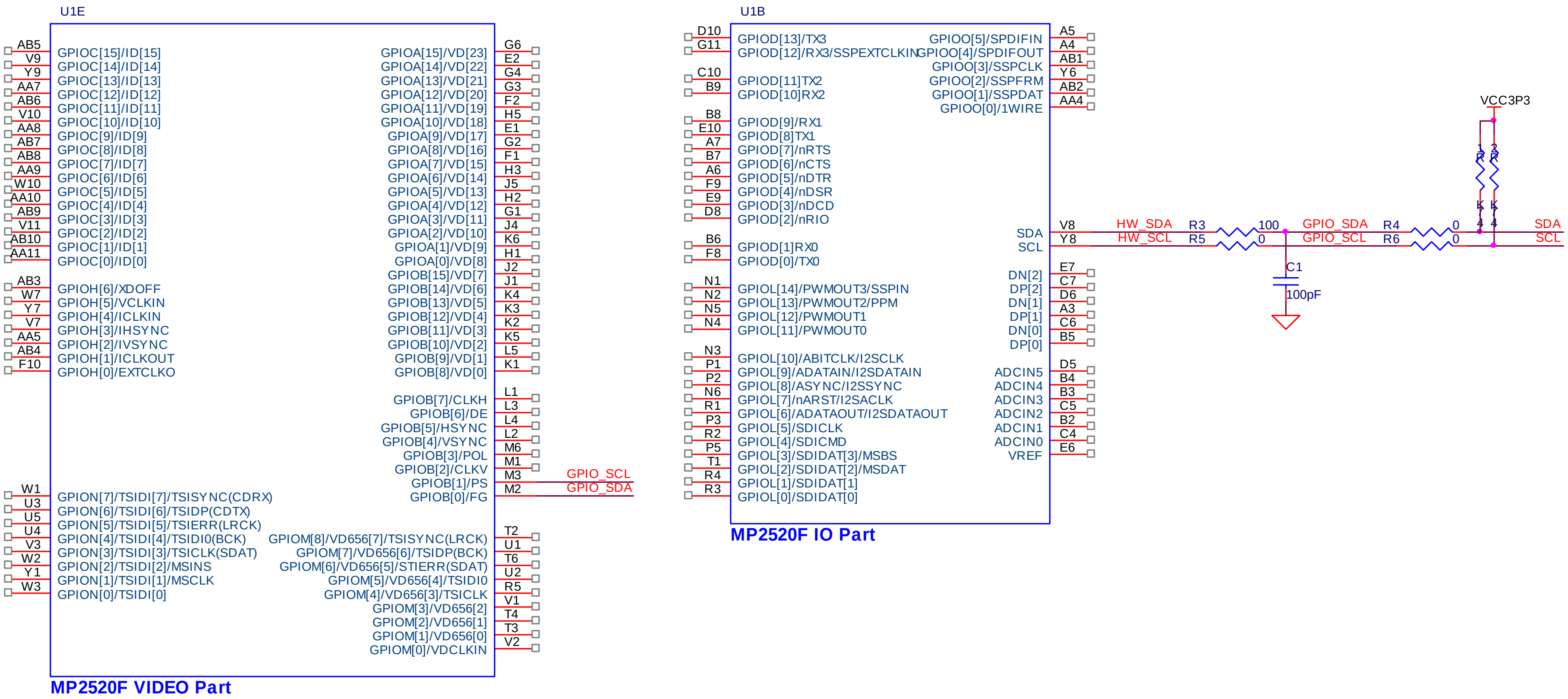
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
GPIO_SCL	√			GPIO I2C Serial clock line
GPIO_SDA	√			GPIO I2C Serial data line
SCL				HW I2C Serial clock line
SDA				HW I2C Serial data line

- Component description

Component Number	Component Name	Descriptions
U1E	MP2520F	VIDEO Part
U1B	MP2520F	IO Part
C1	Capacitor	100pF
R1,R2	Resistor	4.7K ohm
R4,R5,R6	Resistor	0 ohm
R3	Resistor	100 ohm

Schematic



<Figure 14-1 Design guide of I2C>

15. SPDIF

The SPDIF controller supports Standard IEC958 Digital audio interface and Input/Output Mode. <Figure 15-1> shows the schematic design example of SPDIF interface.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
SPDIF_RX	A5	O[5]	SPDIF_TX	A4	O[4]						

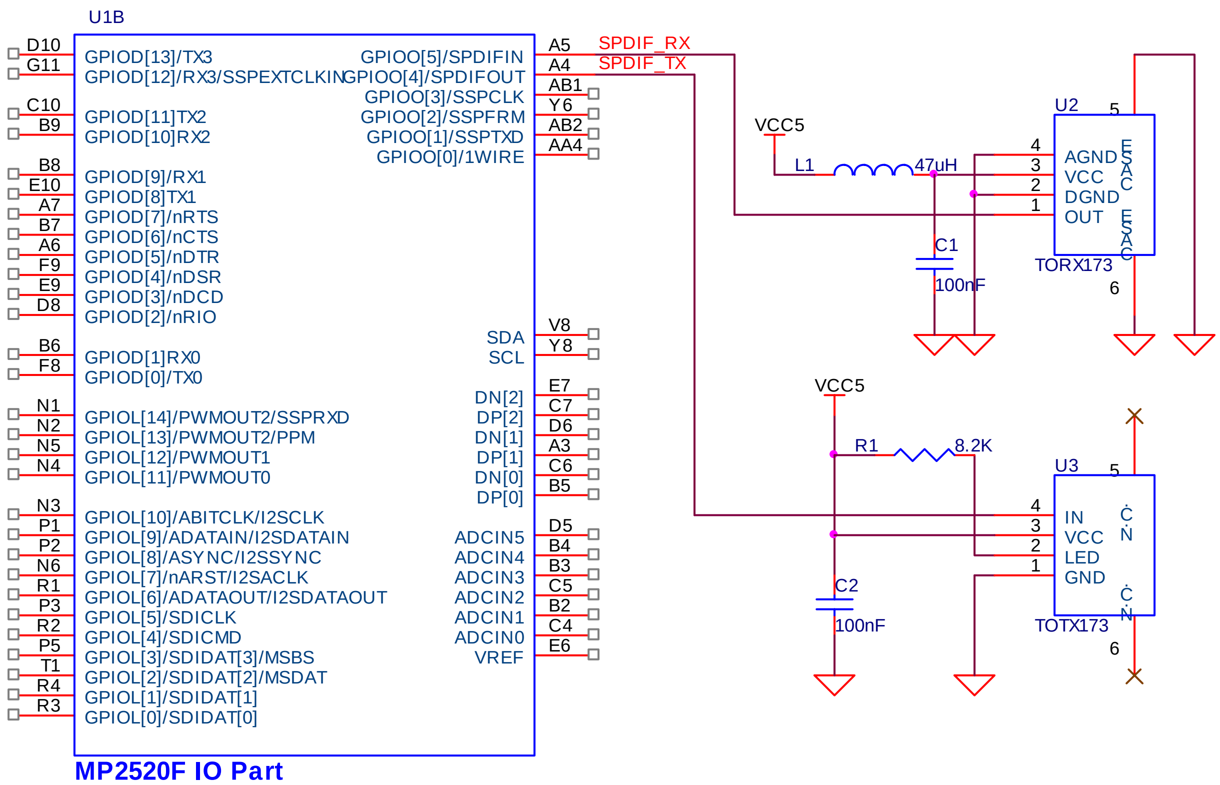
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
SPDIF_RX		√		SPDIF Input
SPDIF_TX		√		SPDIF Output

- Component description

Component Number	Component Name	Descriptions
U1B	MP2520F	I/O Part
U2	Fiber Optic Receiving Module	TORX173(Toshiba)
U3	Fiber Optic Transmitting Module	TOTX173(Toshiba)
L1	Inductor	47uH
R1	Resistor	8.2K ohm
C1,C2	Capacitors	100nF

Schematic



<Figure15-1.Desing guide of SPDIF>

16. SSP/SPI

SSP/SPI is a full-duplex synchronous serial interface. It supports Synchronous Serial Protocol(SSP) and Serial Peripheral Interface(SPI) protocol. <Figure 16-1> shows the schematic design example of Serial NAND flash.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
SPI_CLK	AB1	O[3]	SPI_CS	Y6	O[2]	SPI_DO	AB2	O[1]	SPI_DI	N1	L[14]

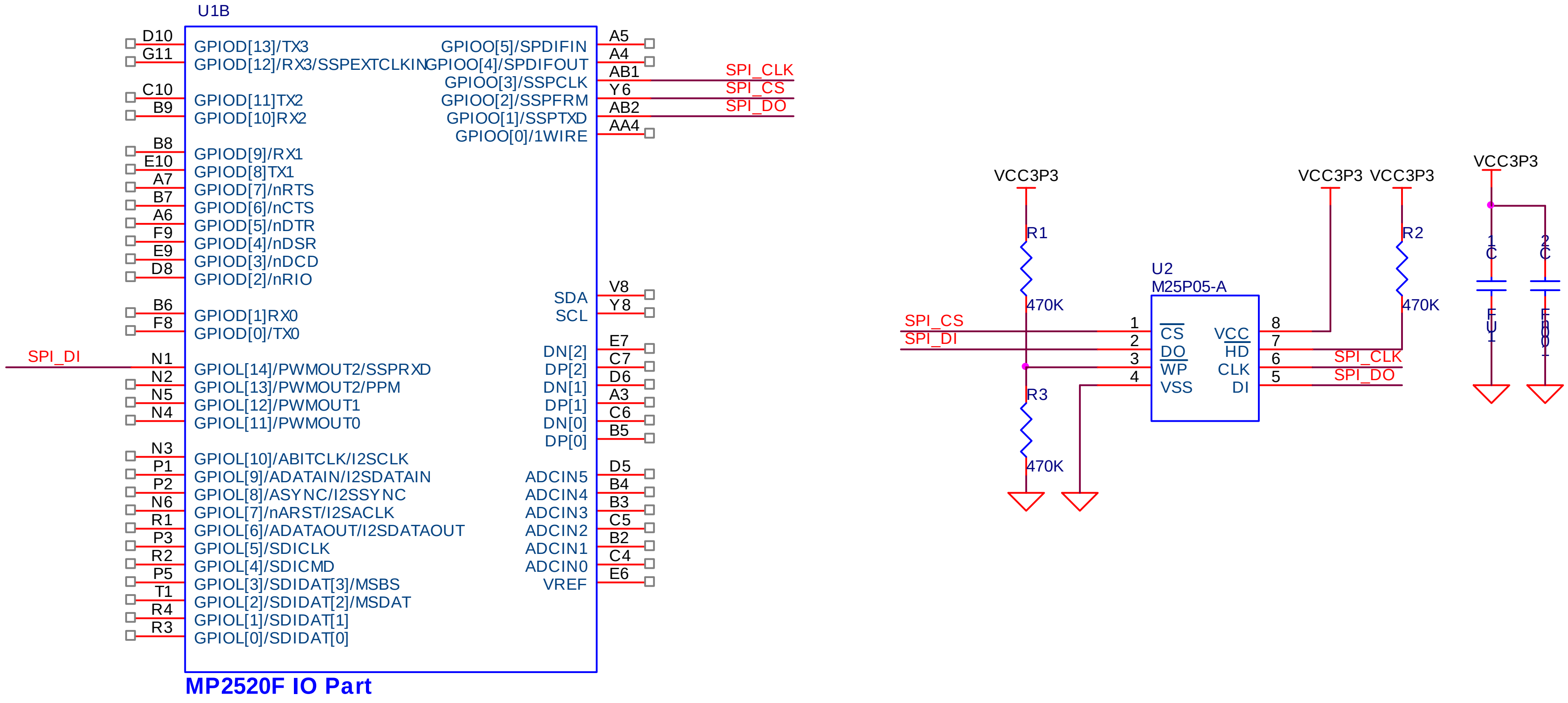
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
SPI_CLK		√		Synchronous serial port clock
SPI_CS		√		Synchronous serial port frame signal
SPI_DO		√		Synchronous serial port transmit data
SPI_DI			√	Synchronous serial port receive data

- Component description

Component Number	Component Name	Descriptions
U1B	MP2520F	IO Part
U2	M25P05-A	1 Mbit, Low Voltage, Serial Flash Memory
C1	Capacitor	1uF
C2	Capacitor	100nF
R1,R2,R3	Resistor	470K ohm

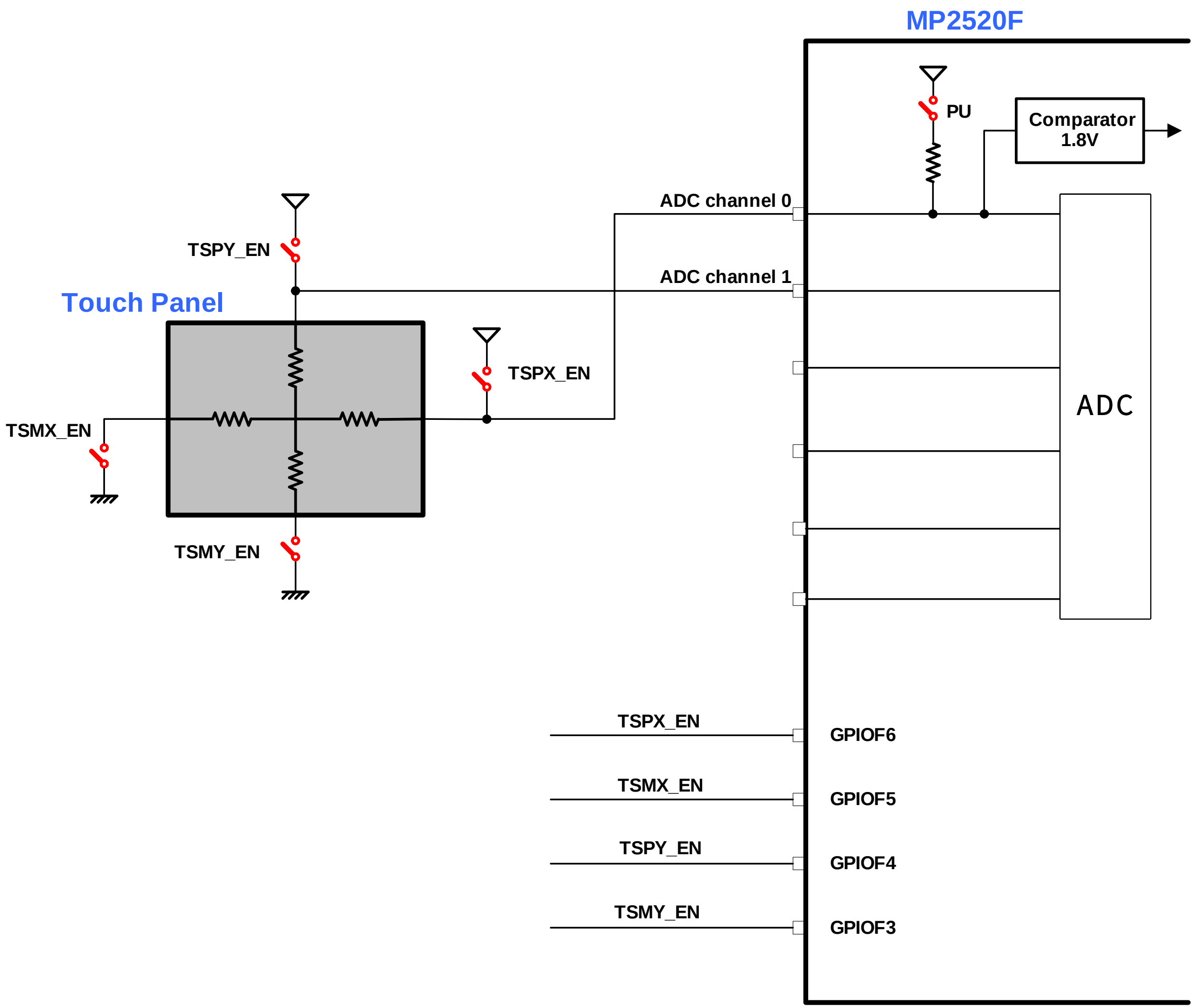
Schematic



<Figure 16-1. Design guide of SSP/SPI>

17. ADC

ADC controller reads the data from six ADC channels. By using internal Comparator of MP2520F, it is possible to connect Touch Panel. <Figure 17-1> shows the connection diagram of Touch panel.



<Figure 17-1. Touch Panel connection diagram>

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
TSPX_EN	AB17	F[6]	TSMX_EN	W15	F[5]	TSPY_EN	AA16	F[4]	TSMY_EN	AB18	F[3]
TSPY	B2		TSPX	C4							

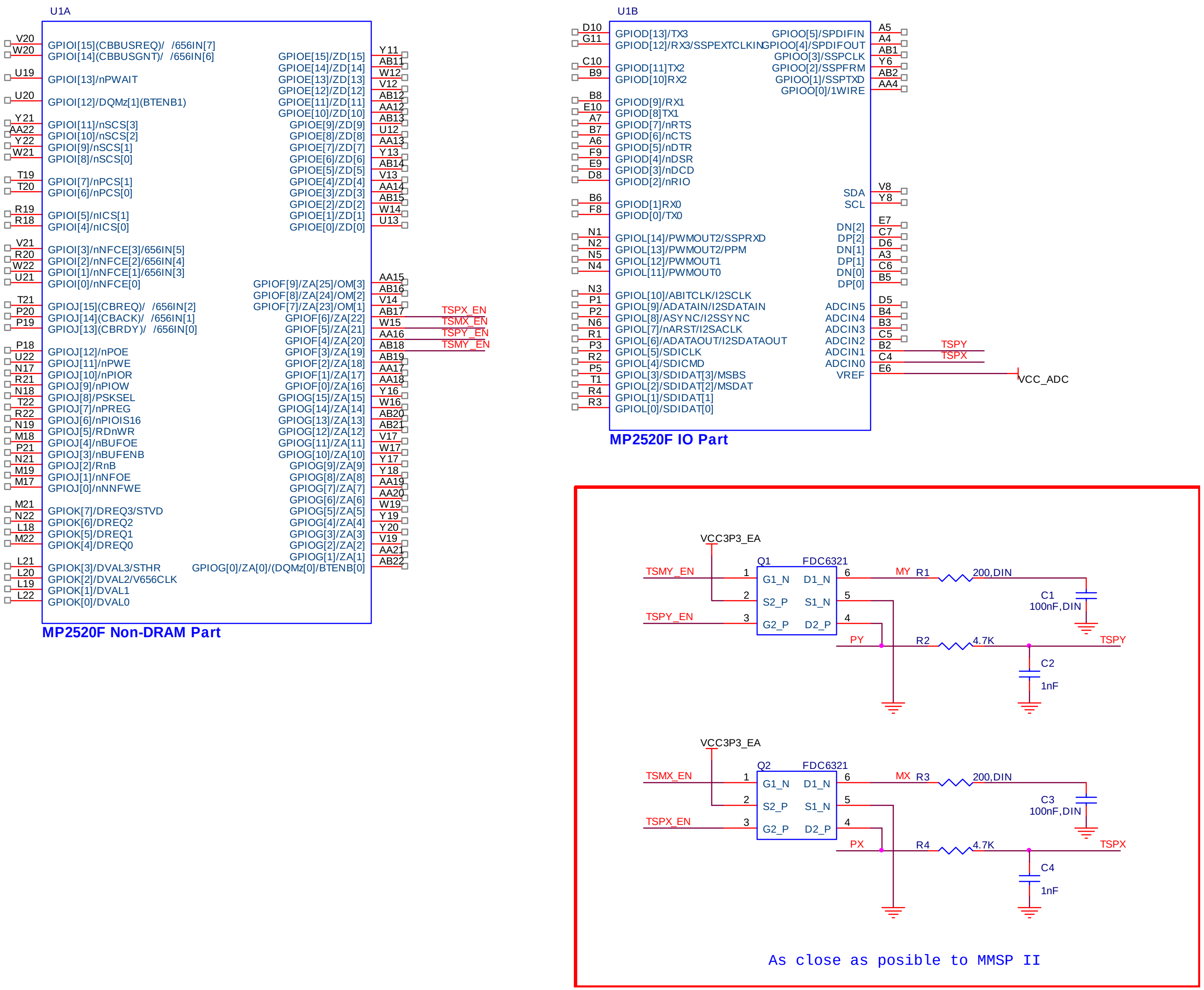
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
TSPX_EN		✓		Image Capture Data Input
TSMX_EN		✓		Image Capture Data Input
TSPY_EN		✓		Image Capture Data Input
TSMY_EN		✓		Image Capture Data Input
TSPY				ADC Analog Input
TSPX				ADC Analog Input

- Component description

Component Number	Component Name	Descriptions
U1B	MP2520F	Non DRAM Bus Part
U1A	MP2520F	I/O Part
Q1,Q2	Dual N & P Channel , Digital FET	FDC6321(Fairchild)
R1,R3	Resistors	200 ohm(Do not Install)
R2,R4	Resistors	4.7K ohm
C1,C3	Capacitors	100nF(Do not Install)
C2,C4	Capacitors	1nF

Schematic



<Figure17-2. Design guide of ADC>

18. SD/MMC

<Figure 18-1> shows the schematic design example of SD/MMC.

- Pin assignment

Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No	Signal Name	Pin No	GPIO No
SD_DAT[3]	P5	L3	SD_DAT[2]	T1	L2	SD_DAT[1]	R4	L1	SD_DAT[0]	R3	L0
SD_CMD	R2	L4	SD_CLK	P3	L5	SD_WP	D8	D2	SD_CD	E9	D3

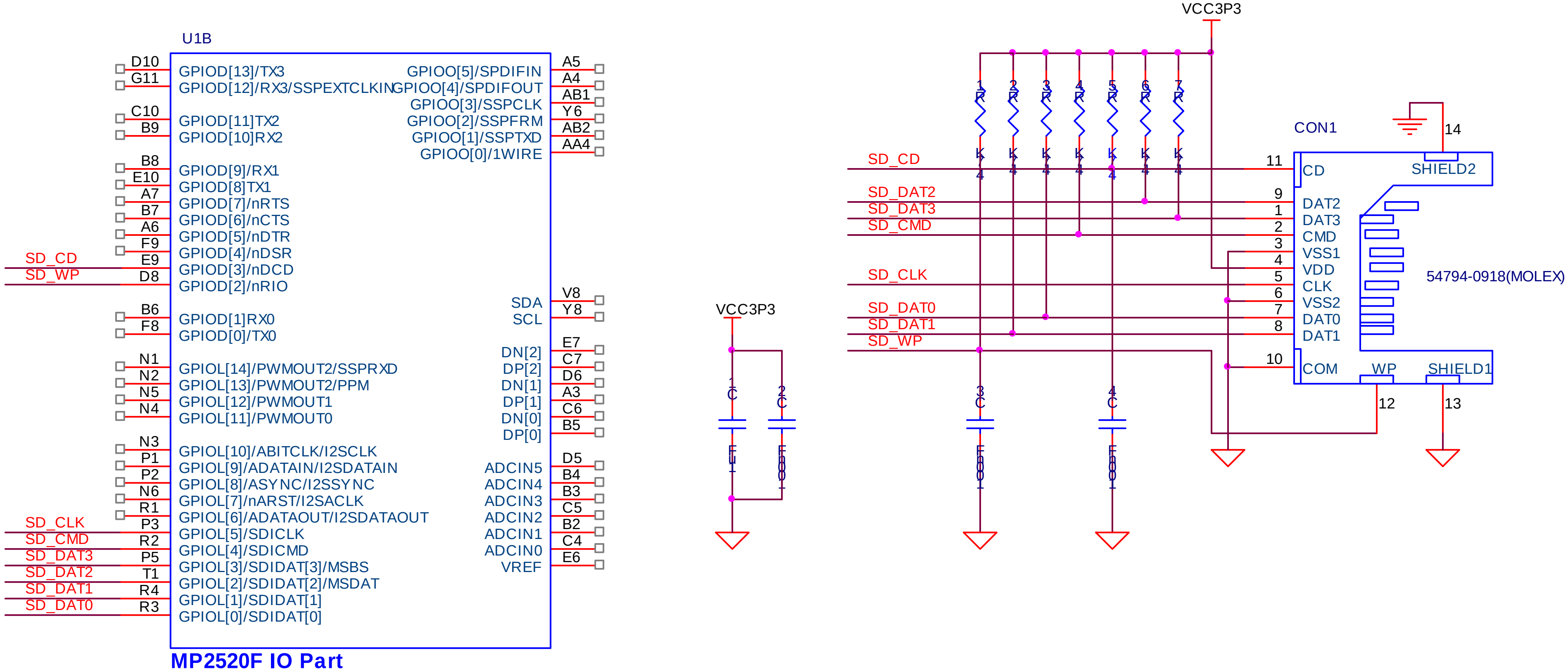
- Signal description

Signal Name	Function			Descriptions
	GPIO	ALT 1	ALT 2	
SD_DAT[3:0]		√		SD/MMC Data
SD_CMD		√		SD/MMC Command
SD_CLK		√		SD/MMC Clock
SD_WP	√			SD/MMC Write Protect
SD_CD	√			SD/MMC Card Detect

- Component description

Component Number	Component Name	Descriptions
U1B	MP2520F	IO Part
R2,R3,R4,R6,R7	Resistor	47K ohm
R1,R5	Resistor	4.7K ohm
C2,C3,C4	Capacitor	100nF
C1	Capacitor	1uF
CON1	Connector	54794-0918(MOLEX)

Schematic



<Figure 18-1. Design guide of SD/MMC>